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GROSS DRY MATTER PRODUCTIVITY (GDMP)
NET PRIMARY PRODUCTION (NPP)
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List of Acronyms

A	Accuracy
ACD	Algorithm Change Document
AD	Applicable Document
APAR	Absorbed Photosynthetically Active Radiation
APU	Accuracy, precision and uncertainty
ATBD	Algorithm Theoretical Basis Document
B	Mean Bias
BA	Barren and sparsely vegetated
C	Carbon
CAL/VAL	Calibration / Validation
CCI	Climate Change Initiative
CEOS	Committee on Earth Observation
CGLOPS	Copernicus Global Land Operations
CLMS	Copernicus Land Monitoring Service
COG	Cloud Optimized Geotiff
CRO	Cropland
DBF	Deciduous Broadleaved Forest
DM	Dry Matter
DMP	Dry Matter Productivity
EBF	Broadleaved Evergreen Forest
EC	Eddy Covariance
ECV	Essential Climate Variable
EO	Earth Observation
EOS	Earth Observation System
EPSG	European Petroleum Survey Group
ESA	European Space Agency
ETC	Ecosystem Thematic Centre
FAPAR	Fraction of Absorbed Photosynthetically Active Radiation
FLUXNET	A global network of micrometeorological flux measurement sites that measure the exchanges of carbon dioxide, water vapor, and energy between the biosphere and atmosphere
GBOV	Ground-Based Observations for Validation
GCOS	Global Climate Observing System
GDMP	Gross Dry Matter Productivity
GM	Geometric Mean
GPP	Gross Primary Production
GMR	Geometric Mean Regression

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GRA	Grassland
HER	Herbaceous
ICOS	Integrated Carbon Observation System
INSPIRE	Infrastructure for Spatial Information in Europe
IPCC	Intergovernmental Panel on Climate Change
JCGM	Joint Committee for Guides in Metrology
JRC	Joint Research Centre
LAI	Leaf Area Index
LANDVAL	Validation sites for land
LC100	Land Cover at 100 m
LPV	Land Product Validation
LUE	Light Use Efficiency
MAD	Mean Absolute Difference
MBE	Mean Bias Error
MD	Median Deviation
MODIS	Moderate Resolution Imaging Spectroradiometer
MXF	Mixed forest
NEE	Net Ecosystem Exchange
NetCDF	Network Common Data Form
NLF	Needleleaved Forest
NPP	Net Primary Production
NRT	Near Real Time
OLCI	Ocean and Land Colour Instrument
PQAD	Product Quality Assessment Document
P	Precision
PROBA-V	Project for On-Board Autonomy - Vegetation
PSN	Net Photosynthesis
PUM	Product User Manual
QAR	Quality Assessment Report
QC	Quality Control
QFLAG	Quality Flag
R	Pearson correlation coefficient
Reco	Ecosystem Respiration
RMSD	Root Mean Squared Difference
RT	Real Time
RTx	Consolidated mode after x dekads
S1	1-day synthesis
S3	Sentinel-3

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SHR	Shrubland
SPOT	Satellite Pour l'Observation de la Terre
STD	Standard deviation of the bias. A measure of precision
SZA	Solar Zenith Angle
TOC	Top-of-Canopy
U	Uncertainty
UTM	Universal Transverse Mercator
VGT	Vegetation instrument
VITO	Vlaams Instituut Technologisch Onderzoek
VR	Validation Report
WGS	World Geodetic System

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EXECUTIVE SUMMARY

The Copernicus Land Monitoring Service (CLMS) produces a series of qualified bio-geophysical products on the status and evolution of the land surface. The products are used to monitor vegetation, crops, water cycle, energy budget and terrestrial cryosphere. Production and delivery of the parameters take place in a timely manner and are complemented by the constitution of long-term time series.

This document presents the results of validation of the 300m Dry Matter Productivity (DMP), Gross Dry Matter Productivity (GDMP), Net Primary Production (NPP), and Gross Primary Production (GPP) Version 2 (V2).

The timeseries of DMP/GDMP/NPP/GPP 300m V2 is based on Fraction of Absorbed Photosynthetically Active Radiation (FAPAR) 300m V2 derived from Project for On-Board Autonomy - Vegetation (PROBA-V) for 2014 – 2018 and from the Ocean and Land Colour Imager (OLCI) onboard Sentinel-3 platform for 2019 – present.

The quality assessment of the DMP/GDMP/NPP/GPP 300m V2 product is based on the periods 2019 (for comparison with PROBA-V DMP/GDMP 300m V1.0, and consistency between V2 derived from PROBA-V and from Sentinel-3/OLCI) and 2023 (for comparison with OLCI DMP/GDMP 300m V1.1) for consistency analysis, and on the full time series 2014 – 2024 for analysis of temporal stability and consistency.

Each of the four DMP/GDMP/NPP/GPP 300m V2 products contains a layer with the respective product data and a quality flag layer (QFLAG) describing the quality of the input FAPAR and meteorological data.

Intercomparison with DMP/GDMP 300m V1.0, DMP/GDMP 300m V1.1, DMP/GDMP 1km V2, and MOD17 V6.1 GPP were performed with a selection of full resolution tiles, global sub-sampled images, and extractions over the LANDVAL V2 sites (Martínez-Sánchez et al., 2024).

A comparison with in-situ data from the Integrated Carbon Observation System (ICOS) and Ameriflux network is made based on temporal profiles and statistical consistency checks.

From the analyses, it is concluded that the DMP/GDMP/NPP/GPP 300m V2 dataset shows high product completeness and good temporal stability. The statistical consistency with the reference datasets is large, although some inconsistencies are present. These can be attributed to differences in the FAPAR input and to changes in the surface reflectance collections on which the products are based. The agreement between PROBA-V and Sentinel-3/OLCI is very good in DMP/GDMP/NPP/GPP 300m V2, confirming the high product consistency from 2014 up until present.

1 BACKGROUND OF THE DOCUMENT

1.1 SCOPE AND OBJECTIVES

This document presents the results of the quality assessment of the Copernicus Land Monitoring Service (CLMS) 300m Dry Matter Productivity (DMP), Gross Dry Matter Productivity (GDMP), Net Primary Production (NPP), and Gross Primary Production (GPP) Version 2 (V2) product, available to users from 2014 onwards. DMP/GDMP/NPP/GPP 300m V2 was designed to provide a continuous time series integrating input data from both PROBA-V (2014 – 2018) and Sentinel-3/OLCI (2019 – onwards).

The validation encompasses a robust and scientific analysis of the new version of the DMP/GDMP/NPP/GPP 300m product, with the objective to evaluate the product accuracy, stability, and performance. The DMP/GDMP/NPP/GPP 300m V2 product is compared to remote sensing-based reference datasets DMP/GDMP 300m V1.0, DMP/GDMP 300m V1.1, DMP/GDMP 1km V2, and MOD17 V6.1 GPP, and to flux tower-based estimates from FLUXNET.

1.2 CONTENT OF THE DOCUMENT

This document is structured as follows:

- Chapter 2 recalls the requirements and the expected performance.
- Chapter 3 briefly describes the DMP/GDMP/NPP/GPP 300m V2 product.
- Chapter 4 lists the scientific questions to be answered in the quality assessment.
- Chapter 5 describes the methodology for quality assessment, the metrics, the evaluation criteria, and the reference datasets.
- Chapter 6 presents the results of the scientific analyses.
- Chapter 7 summarises the main conclusions, perspectives, and recommendations of the study.

1.3 RELATED DOCUMENTS

1.3.1 Applicable documents

AD1: Part 2: Technical specifications of Framework Service Contract – Operation of the bio-geophysical variables systematic monitoring of the Global Land Component of the Copernicus Land Service 'CGLOPS' JRC/2023/OP/0273, 19th April 2023.

Available at https://etendering.ted.europa.eu/cft/cft-display.html?cftId=13795

1.3.2 Input

Document ID	Descriptor
CGLOPS1_PQAD	Product Quality Assurance Document for Vegetation and Energy products of CLMS
CGLOPS1_ATBD_DMP1km-V2	Algorithm Theoretical Basis Document of the DMP 1km V2 product
CGLOPS1_ATBD_DMP300m-V1	Algorithm Theoretical Basis Document of the DMP 300m V1 product
CGLOPS1_ATBD_DMP300m-V1.1	Algorithm Theoretical Basis Document of the DMP 300m V1.1 product
CGLOPS1_ATBD_DMP300m-V2	Algorithm Theoretical Basis Document of the DMP 300m V2 product
CGLOPS1_ATBD_FAPAR1km-V2	Algorithm Theoretical Basis Document of the FAPAR 1km V2 product
CGLOPS1_ATBD_FAPAR300m-V1	Algorithm Theoretical Basis Document of the FAPAR 300m V1 product
CGLOPS1_ATBD_FAPAR300m-V1.1	Algorithm Theoretical Basis Document of the FAPAR 300m V1.1 product
CGLOPS1_ATBD_FAPAR300m-V2	Algorithm Theoretical Basis Document of the FAPAR 300m V2 product
CGLOPS1_PUM_FAPAR300m-V2	Product User Manual of the FAPAR 300m V2 product
CGLOPS1_VR_FAPAR300m-V2	Validation report of the FAPAR 300m V2 product
CGLOPS1_VR_FAPAR300m-V1.1	Validation report of the FAPAR 300m V1.1 product
CGLOPS1_VR_TOC-Reflectance300m-V2.3	Validation report of the surface reflectance V2.3 product
CGLOPS1_ATBD_LC100mV3	Algorithm Theoretical Basis Document of the global land cover 100m map

Available at https://land.copernicus.eu/en/technical-library

1.3.3 Output

Document ID	Descriptor

CGLOPS1_PUM_DMP-NPP300m-V2.0	Product User Manual describing all information about the DMP/GDMP/NPP/GPP 300m V2 products

Available at https://land.copernicus.eu/en/technical-library

1.3.4 External documents

GCOS#245	The 2022 GCOS ECVs requirements, World Meteorological Organisation, Geneva, Switzerland (2022)

Available at https://library.wmo.int/records/item/58111-the-2022-gcos-ecvs-requirements-gcos-245

JCGM, 2014	JCGM, 2014. International Vocabulary of Metrology–Basic and General Concepts and Associated Terms, Chemistry International -- Newsmagazine for IUPAC. Walter de Gruyter GmbH. https://doi.org/10.1515/ci.2008.30.6.21
PROBA-V C2 ACD	Algorithm change document of PROBA-V Collection 2 data

Available at https://proba-v.vgt.vito.be/sites/probavvgt/files/downloads/PROBA-V_C2_Algorithm_Change_Document.pdf

2 PRODUCT REQUIREMENTS

According to the applicable document [AD1], the requirements relevant for DMP and NPP are described below.

2.1 SPECIFIC TECHNICAL DETAILS AND REQUIREMENTS

GENERAL PRODUCT SPECIFICATION	
Geometric Properties:	
Baseline dataset pixel resolutions	300m
Target baseline location accuracy	Better than 0.5 pixels
Coordinate position	Centre of the pixel
Geodetical datum	WGS84
Geographic projection	Regular latitude/longitude grid
Geographic Coverage	Global
Temporal resolution	10-day period (dekad: days 1–10, 11–20, 21–end of month)
Timeliness	Within 2 days (optimally 1 day) after the end of each dekad

DMP – DRY MATTER PRODUCTIVITY	
Amount (weight) of dry matter (DM) produced per surface unit and per time unit expressed in kilograms of dry matter per hectare per day (kgDM/ha/day). The accumulation over time leads to the assessment of the biomass (in this context, phytomass), and can be defined as above-ground mass per unit area of living plant material (GCOS#245)	
Uncertainty (2-sigma)	Threshold: 20% (Can be biome specific: <10% in crop areas; 20% for grassland) Goal: 10% (Can be biome specific: <10% in crop areas; 20% for grassland)
Stability (per decade, i.e. 10 years)	Threshold: 10% Goal: 5%

NPP – ABOVE GROUND NET PRIMARY PRODUCTION	
Net primary production is the rate at which plants produce net useful chemical energy expressed in mass of carbon per unit area per day ($gC/m^2/day$)	
Uncertainty (2-sigma)	Threshold: 20% (Can be biome specific: <10% in crop areas; 20% for grassland) Goal: 10% (Can be biome specific: <10% in crop areas; 20% for grassland);
Stability (per decade, i.e. 10 years)	Threshold: 10% Goal: 5%

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2.2 FURTHER REQUIREMENTS

2.2.1 Output product composition

Products may contain various information layers and ancillary information – the base reference for product packages are the operational products as on 01.03.2023.

2.2.2 Data Structure

Data coding[1] shall be compatible with the Global Land products as on 01.03.2023 and/or follow the INSPIRE specifications, where applicable.
Ancillary information shall be as currently used and include at least the following:

- The number of measurements per pixel used to generate any synthesis product
- The per-pixel date of the individual measurement or the start-end dates of the period covered
- Quality indicators, with explicit per-pixel identification of the cause of anomalous parameter result.

The product naming and filename conventions that are used in the Copernicus Global Land component production as on 01.03.2023 shall be followed. This may be adapted for complete product collections upon agreement with the contracting authority during the Framework.

2.2.3 Data format

To ensure interoperability with the current Global Land component (operational product data formats and archive data formats) and other Copernicus services, all datasets shall be available in NETCDF. Additional format such as Cloud Optimized Geotiff (COG) or ZAR format can be proposed for production or could be requested by the Contracting Authority during the Framework Contract.

2.2.4 Uncertainties and Validation

Uncertainties indicated in the product specifications above follow the threshold and goal proposed by GCOS#245.

Uncertainty estimates should account for the error propagation uncertainty coming from input data though the retrieval algorithms as during the contract period, ESA plans to imbed uncertainties in the Sentinel-3 ground segment products and, for Sentinel-2, there is an offline tool to determine Level 1 uncertainties; these can be used for propagation in the production chain.

Validation of the products shall conform to at least the CEOS LPV standards. Wherever appropriate the bio-geophysical variables shall be validated and compared to CEOS CAL/VAL datasets and/or

[1] Data coding is the provision for the number of bits, range of values, usage of reserved values, content of status map, conversion to physical values formulas.

Ground-Based Observations for Validation (GBOV) of Copernicus Global Land component when biophysical parameters are available.

2.2.5 Input data

Copernicus sentinel data are available from https://dataspace.copernicus.eu

Bio-geophysical variable products should be based on common base reflectance data:

- Sentinel 3: Derived Top-of-Canopy reflectance may be brokered or produced as it is the case on 01.03.2023 under the CGLOPS contracts;
- Sentinel 2: the Global Land Sentinel-2 Global Mosaic (S2GM)[2] component provides temporal mosaic of surface spectral bands that can be brokered and/or directly used.

Up to and including 2019, products archives of the Global Land component have been based on SPOT VGT, Proba-V, ENVISAT, MODIS, TOPEX/Poseidon, Jason-1, Jason-2, Jason-3, datasets, which are available through the https://dataspace.copernicus.eu.

Ancillary satellite data that is purchased through Copernicus and put at disposal of the Services is available through the Data Warehouse and will become available on the Copernicus Dataspace Ecosystem.

Ancillary datasets, other than satellite imagery described above, that might be required shall be the responsibility of the contractor.

2.2.6 Product delivery

Products shall be delivered to the Copernicus Land Component dissemination.

The Copernicus Data Space Ecosystem infrastructure will be used to provide access to the final map products.

[2] https://land.copernicus.eu/en/products/global-image-mosaic

3 DESCRIPTION OF THE DMP/GDMP/NPP/GPP PRODUCTS

This chapter provides a summarized description of the products under evaluation in this QAR. More details are provided in the PUM [CGLOPS1_PUM_DMP-NPP300m-V2.0] and the ATBD [CGLOPS1_ATBD_DMP300m-V2].

The DMP product of CLMS is an indicator for vegetation productivity. It represents the amount (weight) of dry matter (DM) produced per surface unit and per time unit expressed in kilograms of dry matter per hectare per day (kgDM/ha/day). It is based on the Light Use Efficiency (LUE) approach, first formulated by (Monteith, 1972): vegetation growth is completely defined as the part of the incoming solar radiance that is used for photosynthesis and which is absorbed by the plants (APAR, [$kJ_{AP}/m^2/d$]), and a number of conversion efficiency factors.

Table 1: Parameters of the Monteith formula used for CLMS DMP/GDMP.

Parameter	Meaning	Value	Unit
GDMP	Gross Dry Matter Productivity	0 – 640	kgDM/ha/day
DMP	Dry Matter Productivity	0 – 320	kgDM/ha/day
R	Total shortwave incoming radiation (0.2 – 3.0µm)	0 – 320	GJ_T/ha/day
ε_c	Fraction of Photosynthetically Active Radiation (0.4 – 0.7µm) in total shortwave	0.48	J_P/J_T
FAPAR	Fraction of Absorbed Photosynthetically Active Radiation by green vegetation	0.0 – 1.0	J_{PA}/J_P
ε_{LUEc}	Light use efficiency at optimum	Biome-specific	kgDM/GJ_{PA}
ε_T	Normalized temperature effect	0.0 – 1.0	-
ε_{CO2}	Normalized CO_2 fertilization effect	0.0 – 1.0	-
ε_{AR}	Fraction kept after autotrophic respiration	0.5	-
ε_{RES}	Fraction remaining after omitted effects (drought, pests ...)	1.0	-

The global CLMS DMP and GDMP products, based on FAPAR 300m V2 [CGLOPS1_ATBD_FAPAR300m-V2, CGLOPS1_PUM_FAPAR300m-V2], are computed with the following Monteith variant (see Table 1 for a description of the symbols and their values):

$$\mathbf{DMP\ [kgDM/ha/day]} = R \times \varepsilon_c \times \mathit{FAPAR} \times \varepsilon_{LUEc} \times \varepsilon_T \times \varepsilon_{CO2} \times \varepsilon_{AR} [\times \varepsilon_{RES}] \quad \textbf{(Eq. 1)}$$

$$\mathbf{GDMP\ [kgDM/ha/day]} = R \times \varepsilon_c \times \mathit{FAPAR} \times \varepsilon_{LUEc} \times \varepsilon_T \times \varepsilon_{CO2} [\times \varepsilon_{RES}] \quad \textbf{(Eq. 2)}$$

DMP and GDMP are directly related to the more commonly used NPP and GPP, respectively, via a conversion factor between dry matter and carbon, and scaling according to the units. According to

the Intergovernmental Panel on Climate Change (IPCC) (Penman, 2003), the efficiency of the conversion between carbon and dry matter is on average 0.5 gC/gDM. So, NPP (GPP) and DMP (GDMP) only differ by a constant factor. In practice, to scale DMP to NPP and GDMP to GPP the following formulas are used:

$$\textbf{NPP [gC/m}^2\textbf{/day] = DMP [kgDM/ha/day]} \times \textbf{0.5} \times \textbf{0.1} \quad \textbf{(Eq. 3)}$$

$$\textbf{GPP [gC/m}^2\textbf{/day] = GDMP [kgDM/ha/day]} \times \textbf{0.5} \times \textbf{0.1} \quad \textbf{(Eq. 4)}$$

Since DMP, GDMP, NPP, and GPP differ only by a constant factor, the validation results for one of the products is also valid for the other three products. Therefore, **the quality assessment is only performed for GPP**. For the DMP 300m V1.0, DMP 300m V1.1, and DMP 1km V2 datasets no GPP product is available, but it is obtained by multiplying the DMP product by 0.1 (or the GDMP product by a factor of 0.05, see Eq. 4).

Each of the products contains two output layers:

- the DMP, GDMP, NPP, or GPP layer,
- and a quality flag (QFLAG) derived from the quality layer of the input FAPAR 300m V2 product (see Table 2).

In the first scientific research question (see Section 6.1), we investigate the occurrences of the various quality flags in the QFLAG layer of DMP/GDMP/NPP/GPP 300m V2 and define which flags are used in the subsequent analysis.

Table 2: Quality flags of DMP, GDMP, NPP, and GPP

Bit	Qualitative indicator	Meaning	Mapping on FAPAR 300m V2
1	FAPAR missing or invalid	0 = Available FAPAR 300m V2 value 1 = Missing FAPAR 300m V2 value	FAPAR = 255 (missing value)
2	Meteorological data missing or invalid	0 = Available meteorological data values 1 = Missing meteorological data values	
3	Not used		
4	Evergreen Broadleaf Forest (EBF)	1 = EBF 0 = no EBF	FAPAR QFLAG bit 2
5	Not used		
6	Low FAPAR correction	1 = FAPAR < 0.15 and rescaled 0 = FAPAR is unchanged	
7	FAPAR computation for EBF	0 = FAPAR from daily observations 1 = FAPAR filled from previous dekad	FAPAR QFLAG bit 5
8	FAPAR gap filling by interpolation	0 = FAPAR not interpolated 1 = FAPAR interpolated	FAPAR QFLAG Bit 3

The production of DMP/GDMP/NPP/GPP is managed similarly to the production of FAPAR 300m V2.0. A brief description is given below but the reader is also referred to the product user manual of FAPAR [CGLOPS1_PUM_FAPAR300m-V2]. The compositing for real time estimates is achieved according to the scheme described in Figure 1. We consider a particular dekadal date, *D*.

- In real time mode, RT0, (when the actual date corresponds to *D*, i.e. the first line in Figure 1), the compositing is achieved using only the past observations, with a maximum compositing window spanning c_- days in the past. The value of the product at dekadal date *D* is therefore relatively inaccurate.
- Then, the observations accumulate after dekadal date *D* when time passes. The value at dekadal date *D* is updated using the observations available after dekadal date *D* within the consolidation period. The accuracy of the product value progressively improves during this consolidation period.
- At the end of the consolidation period that lasts c_+ days, the value of the product at dekadal date *D* has converged towards the 'historical' value, i.e. when there is at least c_- days (maximum length of the compositing window) after dekadal date *D*.

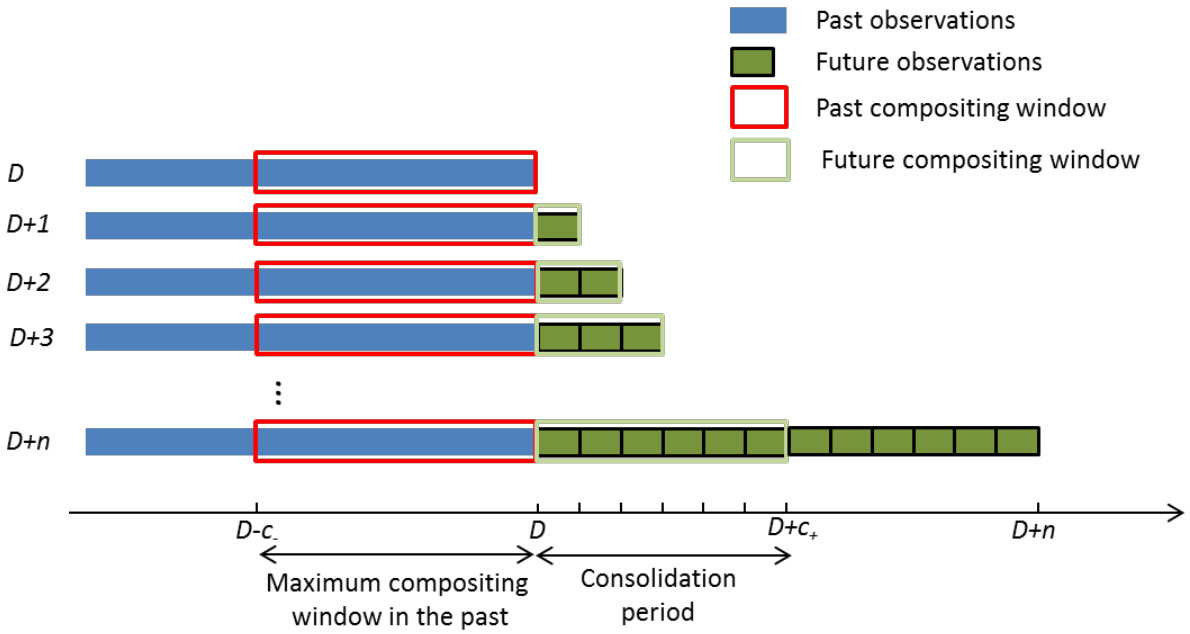

Figure 1: Compositing scheme used for real time estimates

Note that the compositing period is not symmetric: the maximum compositing window in the past is longer than the maximum compositing window in the future (after date *D*) that is limited by the length of the consolidation period. The length of the maximum compositing window is set to 60 days;

however more daily observations are used to classify the pixel as EBF or no-EBF. If insufficient daily observations are available to estimate the dekadal value, either an interpolation with the surrounding dekads is performed or, in case of EBF, the value will be based on the previous dekadal value.

Once a new dekad is available after dekadal date D, the value of the product at dekadal date D is updated and replaces the previous update. This step enables significant saving in storage. The reliability of the estimate increases with the consolidations. The product is provided in the following consolidations: RT0 (at dekad D), RT1 (at dekad D+1), RT2 (at dekad D+2) and RT6 (at dekad D+6).

In this report, we focus on the validation of the RT6 mode, which is considered as the consolidated product, meaning that the product is not updated after this update. The consistency of RT6 with RT2, RT1, and RT0 is assessed as well (see Section 6.3).

4 SCIENTIFIC QUESTIONS

The product under evaluation in the quality assessment report is the DMP/GDMP/NPP/GPP 300m V2 dataset. This dataset is created based on input data from two different sensors. From 2014 to 2018, DMP/GDMP/NPP/GPP 300m V2 is based on the FAPAR derived from Top-Of-Canopy (TOC) reflectances from PROBA-V Collection 2 (C2) daily (S1) synthesis 300m data. From 2019 onwards, the product is based on the FAPAR derived from TOC reflectances V2.3 from the OLCI sensor onboard Sentinel-3A and -3B (with future continuity ensured by the planned Sentinel-3C and -3D missions). For validation purposes, a test dataset of DMP/GDMP/NPP/GPP 300m V2 based on PROBA-V FAPAR covering the year 2019 is available. This period is used for quality assessment and for comparison with the Sentinel-3/OLCI DMP/GDMP/NPP/GPP 300m V2 dataset, but this test dataset is not distributed to users.

To assess the quality of the DMP/GDMP/NPP/GPP 300m V2 dataset, it is compared to previous DMP/GDMP products. An overview of the CLMS 300m and 1km global DMP/GDMP product versions is provided in Table 3 and the datasets are discussed in more detail in §5.2.2.

Table 3: Algorithmic differences between successive versions of DMP/GDMP products at 1km and 300m resolution.

Spatial resolution	DMP version	Algorithm summary	Time series available
1km	V1	Products originated from MARSOP, using FAPAR of MARSOP and meteorological data, and are no longer disseminated.	Apr/1998 – Jun/2020 (VGT and PROBA-V)
	V2	Based on V1, but with following changes: (1) use of CGLOPS FAPAR 1km V2 as input, (2) update of CO_2 fertilization factor, (3) using FLUXNET-based biome specific LUE's, and (4) replacement of the autotrophic respiration factor. For details, see [CGLOPS1_ATBD_DMP1km-V2]	Apr/1998 – Jun/2020 (VGT and PROBA-V)
300m	V1.0	Same algorithm for DMP/GDMP as for 1km V2 but based on FAPAR 300m V1.0 derived from PROBA-V C1 Top-Of-Atmosphere daily composites. For details, see [CGLOPS1_ATBD_DMP300m-V1].	Jan/2014 – Jun/2020 (only PROBA-V)
	V1.1	No algorithm changes in the calculation of DMP/GDMP compared to 300m V1.0. The products are based on FAPAR V1.1 derived from Sentinel-3/OLCI TOC reflectance V1. NPP and GPP products were included from Jul/2023 onwards. For details, see [CGLOPS1_ATBD_DMP300m-V1.1].	From Jul/2020 onwards (only Sentinel-3/OLCI)

	V2	No algorithm changes in the calculation of DMP/ GDMP/NPP/GPP compared to 300m V1.0 and V1.1. For details, see [CGLOPS1_ATBD_DMP300m-V2]. The products are based on FAPAR 300m V2.0 derived from PROBA-V C2 TOC daily composites and Sentinel-3/OLCI TOC reflectance V2. See [CGLOPS1_ATBD_FAPAR300m-V2] for details on changes with FAPAR V1.0 and V1.1.	From Jan/2014 onwards (PROBA-V and Sentinel-3/OLCI)

4.1 RESEARCH QUESTIONS

From a user's perspective, one of the most relevant questions is the temporal stability over the product time series and the consistency between PROBA-V and Sentinel-3/OLCI parts of the DMP/GDMP/NPP/GPP 300m V2 time series. As stated above, the latter is evaluated by consistency analysis of a test dataset of PROBA-V based DMP/GDMP/NPP/GPP 300m V2 with the Sentinel-3/OLCI based DMP/GDMP/NPP/GPP 300m V2 over the year 2019. Temporal stability is analysed through evaluation of temporal smoothness, temporal pixel profiles and inter-annual precision, and by comparison with an external dataset.

Secondly, it is important to quantify and explain possible inconsistencies between the existing DMP/GDMP 300m V1.0 archive on PROBA-V C1, the DMP/GDMP 300m V1.1 archive on Sentinel-3/OLCI, and DMP/GDMP/NPP/GPP 300m V2 based on the combination of PROBA-V C2 and Collection 004 Sentinel-3/OLCI.

Users could require a longer time series. We therefore also investigate the consistency between the DMP/GDMP/ NPP/GPP 300m V2 dataset and the DMP/GDMP 1km V2 time series of SPOT4/VGT1, SPOT5/VGT2, and PROBA-V (1999 – 2020).

Furthermore, DMP/GDMP/NPP/GPP 300m V2 is compared to an external reference, namely the MOD17 V6.1 GPP product derived from MODIS/Terra. This allows evaluating the quality of DMP/GDMP/NPP/GPP 300m V2 with respect to the CLMS requirements in terms of acceptable differences with existing satellite-derived products.

The CLMS uncertainty requirement for DMP/GDMP/NPP/GPP 300m V2 is also assessed with respect to in-situ data from FLUXNET. FLUXNET provides GPP estimates based on in-situ eddy-covariance measurements.

In summary, the quality assessment of the DMP/GDMP/NPP/GPP 300m V2 product focuses on the following scientific questions:

1. What are the occurrences of quality flags in DMP/GDMP/NPP/GPP 300m V2 in terms of temporal and spatial distribution? Which quality flags are recommended for users and are applied in the subsequent analysis?

2. What are the temporal variation and spatial distribution of the product completeness of DMP/GDMP/NPP/GPP 300m V2? How do the results compare to those of DMP/GDMP 300m V1.0, DMP/GDMP 300m V1.1, DMP/GDMP 1km V2, and MOD17 V6.1 GPP?
3. What are the differences in DMP/GDMP/NPP/GPP 300m V2 between the different consolidation modes RT6, RT2, RT1, and RT0?
4. How do the DMP/GDMP/NPP/GPP 300m V2 PROBA-V and Sentinel-3/OLCI datasets compare visually to the DMP/GDMP 300m V1.0, DMP/GDMP 300m V1.1, and MOD17 V6.1 GPP datasets?
5. What is the statistical and spatial consistency between PROBA-V and Sentinel-3/OLCI for DMP/GDMP/NPP/GPP 300m v2?
6. How does DMP/GDMP/NPP/GPP 300m V2 compare with versions DMP/GDMP 300m V1.0 and DMP/GDMP 300m V1.1 spatially and statistically?
7. What is the statistical and spatial consistency with DMP/GDMP 1km V2?
8. What is the statistical and spatial consistency of DMP/GDMP/NPP/GPP 300m V2 in comparison to external MODIS data (MOD17 V6.1 GPP)?
9. What is the statistical and temporal consistency of the DMP/GDMP/NPP/GPP 300m V2 product compared to in-situ FLUXNET GPP data? Is the uncertainty between DMP/GDMP/NPP/GPP 300m V2 and FLUXNET within the requirement range (see Section 2.1)?
10. What is the temporal consistency of DMP/GDMP/NPP/GPP 300m V2 in comparison to MOD17 V6.1 GPP? How do the results of this analysis compare to those of DMP/GDMP 300m V1.0, DMP/GDMP 300m V1.1, and DMP/GDMP 1km V2? What is the stability of the DMP/GDMP/NPP/GPP 300m V2 product?

5 QUALITY ASSESSMENT METHODS

5.1 OVERALL PROCEDURE

5.1.1 Criteria

No standard procedure exists to assess the performance of Earth observation (EO)-derived DMP/GDMP/NPP/GPP products. Therefore, a procedure adapted from the guidelines, protocols, and metrics defined by the Land Product Validation (LPV) group of the Committee on Earth Observation Satellite (CEOS) for the validation of satellite-derived land products is used, as summarised in CGLOPS1_PQAD. The approach consists of 2 major components:

1. **Direct validation** compares satellite products to ground measurements of the corresponding biophysical variables. Direct validation is usually limited by a small number of reference sites.
2. **Indirect validation** includes inter-comparison between products as well as evaluation of their temporal and spatial consistency with similar datasets or using other suitable reference datasets. Indirect validation allows extending the evaluation over large areas or time periods but only relative discrepancies between existing datasets can be assessed.

Table 4 provides a description of the criteria under evaluation.

Table 4: Description of the criteria under evaluation

Criterion	Description
Product completeness	Product completeness is linked with the absence of gaps and the occurrences of quality flags in the product, both in space and in time. Gaps in the product inputs are caused by cloud or snow contamination, bad illumination conditions (e. g. in winter), poor atmospheric conditions, or technical problems during image acquisition, and – if propagated to the final product – are generally considered as a severe limitation. We focus on the temporal evolution and spatial distribution of product completeness, occurrences of quality flags, and on the frequency distribution of gap length.
Spatial consistency	Spatial consistency refers to the realism and repeatability of the spatial distribution of retrievals over the globe, including the absence of artefacts (e.g., missing data, stripes, unrealistic values, etc.), based on expert knowledge. The analysis is based on qualitative visual checks at full resolution and spatial distribution of validation metrics at global level (on a systematic subsample).
Statistical consistency	The evaluation of statistical consistency is based on (i) histograms of the overall bias between datasets, (ii) evaluation of the proportion of residuals with in-situ data below a goal (10%) and threshold (20%) uncertainty level[3], and (iii) the geometric mean regression between datasets and associated statistics.

[3] According to the requirements for DMP/NPP (see Section and [AD1])

Criterion	Description
Temporal consistency	The realism of temporal variations of the product is qualitatively assessed for point locations, well distributed over the globe. Temporal evolution of validation metrics is assessed. Frequency histograms of temporal smoothness are compared between datasets. Product stability is assessed based on the inter-annual precision.

5.1.2 Metrics

The validation metrics that are used for the assessment of statistical and temporal consistency are listed in Table 5. A selection of these validation metrics can be calculated per pixel over time (to evaluate spatial consistency) or per time step over grouped pixels (to evaluate temporal consistency). The dataset to be validated is *X* and the reference datasets are *Y*.

Table 5: Validation metrics used for quality assessment of DMP/GDMP/NPP/GPP

Metric	Description
R^2	The coefficient of determination (R^2) indicates agreement or covariation between two datasets with respect to a linear regression model. It summarizes the total data variation explained by this linear regression model. The result varies between 0 and 1 and higher R^2 values indicate higher covariation between the datasets. $$R^2 = \left(\frac{\sigma(X,Y)}{\sigma(X)\cdot\sigma(Y)}\right)^2,$$ with σ(X) and σ(Y) the standard deviation of X and Y, and σ(X, Y) the co-variation of X and Y. R^2 cannot be used for the comparison of the time series per pixel, due to temporal autocorrelation.

Metric	Description
GM regression	The geometric mean (GM) regression (GMR) model is used to identify the relationship between two datasets of remote sensing measurements, with both datasets subject to noise. The GMR model minimizes the sum of the products of the vertical and horizontal distances (errors on Y and X): $Y = a + b \cdot X$ By applying an eigen decomposition to the covariance metrics of X and Y, two eigenvectors v1 and v2 are obtained that describe the principal axes of the point cloud (Duveiller et al., 2016): $b = \frac{\lambda_1 - \sigma_X^2}{\sigma_{XY}}$ (GMR slope) $a = \bar{Y} - b \cdot \bar{X}$ (GMR intercept) with λ_1: the eigenvalue associated with the first eigenvector defining the principal axis σ_X: the standard deviation of X σ_{XY}: the covariance between X and Y $\bar{X}$: the mean value of X $\bar{Y}$: the mean value of Y
Bias	The bias is defined as the actual difference between two datasets: $bias = X - Y$
MBE or Accuracy (A)	The Mean Bias Error (MBE) or Accuracy (A) is defined as the actual difference or bias between two datasets, including positive and negative differences between observations: $MBE = \frac{1}{n}\sum_{i=1}^{n}(X_i - Y_i) = \bar{X} - \bar{Y}$ The mean bias provides an indication of accuracy, i.e. the degree of the "*closeness of the agreement between the result of a measurement and a true value of the measurand*" [JCGM, 2014]. Accuracy represents systematic errors. The lower the mean bias, the higher the accuracy.

Metric	Description
STD or Precision (P)	The standard deviation (STD) of the bias or Precision (P) is a measure for repeatability: $$STD = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(X_i - Y_i - A)^2}$$ The STD is the CEOS LPV best practice to report precision, i.e. the "*closeness of the agreement between the results of successive measurements of the same measurand carried out under the same conditions of measurement*" [JCGM, 2014]. Precision represents the dispersion of product retrievals around their expected value.
RMSD or Uncertainty (U)	The Root Mean Squared Deviation (RMSD) or Uncertainty (U) measures how far the difference between the two datasets deviates from zero: $$RMSD = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(X_i - Y_i)^2}$$ The RMSD is the CEOS LPV best practice to report uncertainty, i.e. a "*parameter, associated with the result of a measurement that characterizes the dispersion of the values that could reasonably be attributed to the measurand*" [JCGM, 2014]. Uncertainty includes systematic and random errors.
MD	The median bias or median deviation is defined as the median of the difference between two datasets: $$MD = median(X - Y)$$
MAD	The median absolute deviation is the median of the absolute difference between two datasets: $$MAD = median(\lvert X - Y \rvert)$$
Temporal smoothness	The temporal smoothness δ (Weiss et al., 2007) is evaluated by taking three consecutive observations and computing the absolute value of the difference between the centre $P(d_{n+1})$ and the corresponding linear interpolation between the two extremes $P(d_n)$ and $P(d_{n+2})$ as follows: $$\delta(d_n) = \left| P(d_{n+1}) - P(d_n) - \frac{P(d_n) - P(d_{n+2})}{d_n - d_{n+2}}(d_n - d_{n+1}) \right|$$
Time series noise	The time series noise can be estimated by averaging δ over the time series (Vermote et al., 2009): $$noise = \sqrt{\frac{\sum_{n=1}^{N-2}\delta(d_n)^2}{N-2}}.$$

Metric	Description
	Relative noise is computed by dividing the noise by the mean of the time series (Claverie et al., 2013): $$relative\ noise = 100 \times \frac{noise}{\bar{X}}$$
Inter-annual precision	The inter-annual precision is determined as the median absolute deviation of the 5th and 95th percentiles between consecutive years.
Stability	Stability is estimated as the slope of inter-annual precision over time.

The MBE, STD, and RMSD statistics are commonly referred to as APU (Accuracy, Precision, and Uncertainty) (Vermote and Kotchenova, 2008).

One of the tasks of this QAR is to determine if the DMP/GDMP/NPP/GPP 300m V2 data is conform with the uncertainty requirement (see §2.1). The uncertainty requirement is defined as a relative value and is here computed as the percentage of LANDVAL V2 pixels (see §5.3.5) or FLUXNET sites (see §5.2.3) for which the DMP/GDMP/NPP/GPP value lies within the goal or threshold distance from the GM regression line with respect to a certain reference dataset. The uncertainty for a certain site is thus defined as:

$$\boldsymbol{uncertainty} = \frac{\sqrt{|(\boldsymbol{a}+\boldsymbol{b}\cdot\boldsymbol{X})-\boldsymbol{Y}|^2}}{|\,\boldsymbol{a}+\boldsymbol{b}\cdot\boldsymbol{X}|}, \qquad \textbf{(Eq. 5)}$$

with $\boldsymbol{X}$ the DMP/GDMP/NPP/GPP 300m V2 value and $\boldsymbol{Y}$ the corresponding DMP/GDMP/NPP/GPP value of a reference dataset, and $\boldsymbol{a}$ and $\boldsymbol{b}$ are the intercept and slope of the GM regression between DMP/GDMP/NPP/GPP 300m V2 and the reference dataset. The uncertainty percentages are always presented for respectively all land cover types combined, for pixels in bare areas, and for all land cover types except for the bare area pixels. Bare area pixels have low GPP values and can hence have a large influence on the relative uncertainty value which is not representative for all pixels combined.

5.1.3 List of procedures

The overall validation procedures are summarized in Table 6. The procedures are designed to answer the scientific questions (Q1 to Q10) as described in Chapter 4. Depending on the scientific question, emphasis of the procedures is on statistical, spatial and/or temporal consistencies. Statistical consistency focuses on the distribution of values and the statistical relationship between two datasets. Spatial consistency investigates spatial artefacts, spatial distribution of validation metrics, and distribution of values per biome. Temporal consistency focuses on pixel profiles and temporal smoothness and stability.

Table 6: Procedures for the validation of GPP 300m V2

	Criterium	Method and/or Validation metric	Reference products	Spatial coverage	Temporal coverage
Q1	/	• Quantification (in %) of flag occurrences over land: temporal and spatial distribution	/	Global subsample	2019
Q2	Product completeness	• Temporal distribution of % of missing values or pixels qualified as 'invalid' over land • Spatial distribution of % of missing values or 'invalid' pixels over land	GPP 300m V1.0 GPP 300m V1.1 GPP 1km V2 MOD17 V6.1 GPP	Global subsample	2014 – 2024
		• Frequency distribution of the gap length		LANDVAL	2019/2023
Q3	Statistical consistency	• Histogram of overall bias • GM regression analysis	GPP 300m V2 RT0, RT1, RT2	LANDVAL Global subsample on selected tiles	2024
	Temporal consistency	• Temporal pixel profiles			
Q4	Spatial consistency	• Visual comparison	GPP 300m V1.0 GPP 300m V1.1 MOD17 V6.1 GPP	1°x1° tiles (FR) 10°x10° tiles (FR)	2019/2023
Q5	Statistical consistency	• Histogram of overall bias • GM regression analysis, overall and per biome	/	LANDVAL	2019
	Spatial consistency	• Histograms per biome • Spatial distribution of validation metrics		Global subsample	

Q6	Statistical consistency	• Histogram of overall bias • GM regression analysis, overall and per biome • Uncertainty requirement analysis	GPP 300m V1.0 GPP 300m V1.1	LANDVAL	2019/2023
	Spatial consistency	• Histograms per biome • Spatial distribution of validation metrics		Global subsample	
Q7	Statistical consistency	• Histogram of overall bias • GM regression analysis • Uncertainty requirement analysis	GPP 1km V2	LANDVAL	2019
	Spatial consistency	• Histograms per biome • Spatial distribution of validation metrics		Global subsample	
Q8	Statistical consistency	• Histogram of overall bias • GM regression analysis • Uncertainty requirement analysis	MOD17 V6.1 GPP	LANDVAL	2019
	Spatial consistency	• Histograms per biome • Spatial distribution of validation metrics		Global subsample	
Q9	Statistical consistency	• Histogram of overall bias • Uncertainty requirement analysis	FLUXNET	FLUXNET sites	2014 – 2024
	Temporal consistency	• Temporal pixel profiles			
Q10	Temporal consistency	• Temporal pixel profiles	GPP 300m V1.0 GPP 300m V1.1 GPP 1km V2 MOD17 V6.1 GPP	LANDVAL	2014 – 2024
		• Frequency distribution of the temporal smoothness			2019/2023
		• Spatio-temporal evolution of the validation metrics (Hovmöller plots)		Global subsample	2014 – 2024

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5.2 REFERENCE PRODUCTS

5.2.1 Overview

The DMP/GDMP product is an ecosystem variable developed for agronomic purposes (yield estimation), expressed as kgDM/ha/day. But more often NPP or GPP, in gC/m^2/day, are used in ecosystem modelling. Except for the naming and scaling, these variables can be considered as equally. Apart from the previous versions of the CLMS 300m DMP/GDMP product and the 1km DMP/GDMP product, two external datasets containing GPP data were gathered for this report to compare with the CLMS global GPP 300m V2 product (see Table 7). A brief description of each reference dataset is given in this chapter.

Table 7: Overview of the external reference datasets

Name	Type	Variable	Periodicity	Coverage	Resolution
MOD17 V6.1	EO model	GPP	8-day	Global grid	500m
FLUXNET	Flux measurements + respiration model	GPP	10-day	Point locations in Europe and America	n.a.

5.2.2 Satellite reference products

5.2.2.1 DMP/GDMP 300m V1.0

The DMP/GDMP 300m V1.0 product is derived from FAPAR 300m V1.0 [CGLOPS1_ATBD_FAPAR300m-V1] and exists for the period January 2014 – June 2020. More details on DMP/GDMP 300m V1.0 are available in [CGLOPS1_ATBD_DMP300m-V1]. It is to be noted that the product is provided in consolidation mode RT5 instead of RT6 (which is the case for all the other products in the comparison). This affects the smoothing and gap filling, and eventually (possibly) the accuracy of the product.

The PROBA-V based DMP/GDMP 300m V1.0 datasets used in the validation are global sub-sampled images, a selection of 1° x 1° and 10° x 10° tiles at full resolution, and the LANDVAL V2 sites over the year 2019 for statistical and spatial analyses, and over the full time series (January 2014 – June 2020) for the temporal analysis. The DMP 300m V1.0 product is converted to GPP 300m V1.0 by multiplying the DMP product by a factor of 0.1.

5.2.2.2 DMP/GDMP 300m V1.1

The DMP/GDMP 300m V1.1 product is derived from FAPAR 300m V1.1 [CGLOPS1_ATBD_FAPAR300m-V1.1] and the time series spans from July 2020 to the present. The dataset is described in more detail in [CGLOPS1_ATBD_DMP300m-V1.1].

The Sentinel-3/OLCI based DMP/GDMP 300m V1.1 datasets used in the validation are global sub-sampled images, a selection of 1° x 1° and 10° x 10° tiles at full resolution, and the LANDVAL V2 sites over the year 2023 for statistical and spatial analyses, and over the period July 2020 – December 2024 for the temporal analysis. The DMP 300m V1.1 product is converted to GPP 300m V1.1 by multiplying the DMP product by a factor of 0.1.

5.2.2.3 DMP/GDMP 1km V2

The DMP/GDMP 1km V2 product is derived from FAPAR 1km V2 [CGLOPS1_ATBD_FAPAR1km-V2]. The time series includes data from different sensors, i.e. SPOT4/VGT1 (May 1998 – January 2003), SPOT5/VGT2 (February 2003 – December 2013), and PROBA-V (January 2014 – June 2020). The dataset is described in more detail in [CGLOPS1_ATBD_DMP1km-V2].

The PROBA-V based DMP/GDMP 1km V2 datasets used in the validation are global sub-sampled images and the LANDVAL V2 sites over the year 2019 for statistical and spatial analyses. For the temporal consistency analysis, LANDVAL extractions (temporal profiles) and global subsamples (Hovmöller plots) for the period 2014 – 2024 are used. The DMP 1km V2 product is converted to GPP 1km V2 by multiplying the DMP product by a factor of 0.1.

5.2.2.4 MODIS MOD17 V6.1 GPP

The MODerate Resolution Imaging Spectroradiometer (MODIS) is mounted on two platforms: Earth Observing System (EOS) Terra and Aqua. The MODIS instruments on Terra and Aqua capture data in 36 bands with a spatial resolution of 250m (2 bands), 500m (5 bands), and 1km (29 bands). Both instruments image the same area on Earth approximately three hours apart: Terra has a morning equator overpass (descending node) at around 10:30 local time, while Aqua is timed to cross the equator (ascending node) at approximately 13:30 local time.

The MOD17A2HGF Version 6.1 dataset[4] is a GPP product at 500m resolution. It is a gap-filled 8-daily product based on 8-daily composites of FAPAR and Leaf Area Index (LAI) derived from the MOD15 LAI/FAPAR 500m product. The MOD17 algorithm uses the radiation use efficiency concept and includes a GPP and a Net Photosynthesis (PSN) layer and has a quality layer containing quality information for both (PSN QC).

MOD17 V6.1 is provided in sinusoidal projection. Hence the GPP and QC products were resampled to the 1km grid of the CLMS products in geographical projection system using a nearest neighbour resampling method. A Savitzky-Golay filter with window length of 3 consecutive dates and polynomial order of two (i.e. a linear interpolation) was performed to convert the 8-day GPP products to the same temporal resolution as the CLMS GPP 300m V2 dataset, i.e. 3 dekadal products per

[4] https://www.earthdata.nasa.gov/data/catalog/lpcloud-mod17a2hgf-061

month. The 8-day QC layer was converted to decadal cadence by bitwise combination of 8-day input QC before (or on) and after the target date. Only good quality pixels were retained for further analysis by filtering out back-up algorithm retrievals or filled values.

The MODIS MOD17 V6/1 GPP datasets used in this study are global sub-sampled images, a selection of 10° x 10° tiles at full resolution (i.e. 1km resampled) and the LANDVAL V2 sites over the years 2019 and 2023 for statistical and spatial analyses. For the temporal consistency analysis, LANDVAL V2 extractions (temporal profiles) and global subsamples (Hovmöller plots) for the period 2014 – 2024 are used.

5.2.2.5 Expected differences between datasets

The only difference between the various CLMS 300m and 1km DMP/GDMP/NPP/GPP datasets is that they are derived from different FAPAR versions based on either PROBA-V, Sentinel-3/OLCI, or a combination of both, or on SPOT/VGT and PROBA-V data in case of the 1km data. The algorithm differences between FAPAR 300m V1.0, FAPAR 300m V1.1, FAPAR 1km V2, and FAPAR 300m V2.0 are described in [CGLOPS1_ATBD_FAPAR300m-V2] and briefly summarised in Table 8. The GPP algorithm used for all these datasets is the same.

Table 8: Differences between various versions of FAPAR datasets at 300m and 1km resolution

FAPAR version	Input data	Spectral bands	LC map	Differences in algorithm
1km V2	SPOT-VGT C3 and PROBA-V C0 (Jan/2014 – Nov/2016) and C1 (Dec/2016 – Jun/2020) daily (S1) TOC reflectances	B0, B2, B3, MIR	GLOB COVER (Arino et al., 2007)	One neural network for EBF and one for non EBF biomes; Gap filling applied on all biomes
300m V1.0	PROBA-V C1 daily (S1) TOA reflectances	B0, B2, B3	GLOB COVER	One single neural network for all biomes
300m V1.1	Sentinel-3/OLCI TOC reflectances V1.0 (Jul/2020 – Jun/2022) and V1.1 (Jul/2022 – present)	Oa04 –12, Oa16 –18	CCI-LC[5]	One neural network for EBF and one for non EBF biomes
300m V2	PROBA-V C2 TOC reflectances & Sentinel-3/OLCI TOC reflectances V2.3	B0, B2, B3; Oa04 –12, Oa16 –18	CCI-LC	One neural network for EBF and one for non EBF biomes; Same neural networks parameters as V1.1 but few different compositing parameters; gap filling on EBF only.

[5] https://www.esa-landcover-cci.org/

The aspects with largest impact on the GPP values are the version of the input TOA/TOC reflectances, the land cover map used to classify EBF and non-EBF regions, and the specific FAPAR algorithm parameters. Known consequences of improved TOC input data include ~11% less pixels detected as cloud in PROBA-V C2 compared to C1 due to better cloud identification algorithm [PROBA-V C2 ACD] and better cloud and snow detection in Sentinel-3/OLCI TOC V2.3 compared to V1. It should also be noted that GPP 1km V2 is expected to be more complete than the other datasets, since FAPAR 1km V2 uses a climatology to fill gaps. Furthermore, FAPAR V2 corrects the EBF identification wrongly implemented in V1.1 [CGLOPS1_VR_FAPAR300m-V1.1].

Compared with MOD17 V6.1, considerable differences are expected since this dataset is created with different GPP modelling approach. MOD17 also uses a LUE model but with different input parameters and different meteorological data. The most significant difference, with the largest impact on the data, is the inclusion of a water/drought-stress related factor (vapor pressure deficit). As a result, the MOD17 V6.1 GPP values are overall expected to be lower compared to GPP values that have been computed without this water-stress limitation.

5.2.3 In-situ reference products: flux tower measurements

Flux towers are micrometeorology towers on which instruments are mounted that measure the gas and energy exchange between land ecosystems and the atmosphere using the Eddy Covariance (EC) method. The measured quantity is the Net Ecosystem Exchange (NEE) which is a measure for the total vertical fluxes of carbon between atmosphere and ecosystem. NEE can be separated in GPP and Ecosystem Respiration (Reco) with the “flux-partitioning algorithm” methodology from Reichstein et al. (Reichstein et al., 2005). This method relies on a short-term relationship between the respiration coefficient and temperature, which depends on the noisiness of the eddy covariance data and the range of temperatures encompassed during the short period.

There are multiple hundreds flux tower sites spread around the globe. A number of these towers are organised in larger networks at global (e.g. FLUXNET[6]), regional (e.g. Integrated Carbon Observation System, ICOS[7], for Europe and Ameriflux[8] for America), or national (e.g. CzechGlobe[9] for Czech Republic) scale. Flux data obtained directly from the measurements needs to be quality checked and post-processed before it can be intercompared with satellite data. The post-processing of flux data into variables such as GPP requires the necessary tools and expertise and hence should be performed by EC experts. Flux networks such as ICOS and Ameriflux safeguard the quality and

[6] https://fluxnet.org/
[7] https://www.icos-cp.eu/
[8] https://ameriflux.lbl.gov/
[9] https://www.czechglobe.cz/en/

processing of flux data for multiple sites and are hence the best available data sources for the CLMS GPP quality monitoring.

The locations of the ICOS and Ameriflux sites used for the in-situ validation are shown in Figure 2 and Figure 3 respectively. The sites are colour coded according to their biome type. Since the CLMS GPP V2 data has a resolution of 300m, the CLMS Global Land Cover classification (see §5.3.2) is used instead of the biome types from the flux towers themselves. For each site, 10-daily GPP for the period 2014 – 2024 is used in the following analysis.

To compare the flux tower measurements with the CLMS global GPP product, it is necessary to bridge the gap in spatial footprint of the two datasets. Each tower is representative for a specific area surrounding the tower. But currently, no detailed information is available to estimate each tower's spatial footprint. In heterogeneous landscapes, this footprint can be quite small (e.g. a small-scale crop parcel) where in homogeneous areas (e.g. a large forest) the measurements of the tower are representative for the surrounding landscape. Also, the wind direction and velocity play a major role. To compare the CLMS GPP products at 300m with the ICOS and Ameriflux datasets, it was decided to extract the 300m pixels in which the flux towers are located for the CLMS 300m GPP V2 dataset. It should be noted that this difference in spatial footprint adds uncertainty to the intercomparison of the satellite products with in-situ reference data.

An overview of all in-situ sites used in the analyses of this validation report is provided in Annex 2: ICOS and Ameriflux sites. In total, 194 Ameriflux sites and 96 ICOS sites are used in the analyses.

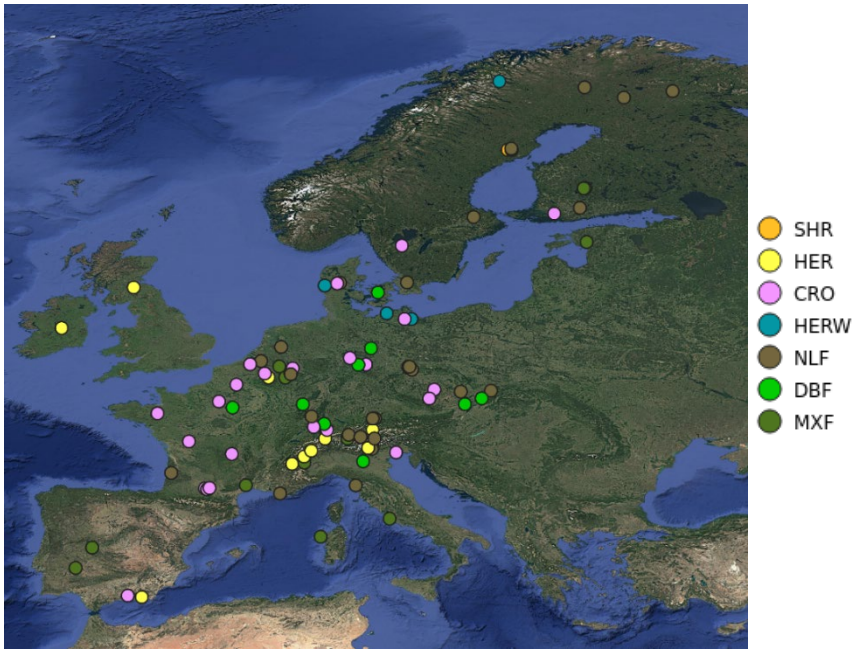


Figure 2: Global distribution of the ICOS sites. Colour coding according to biome (see §5.3.2).

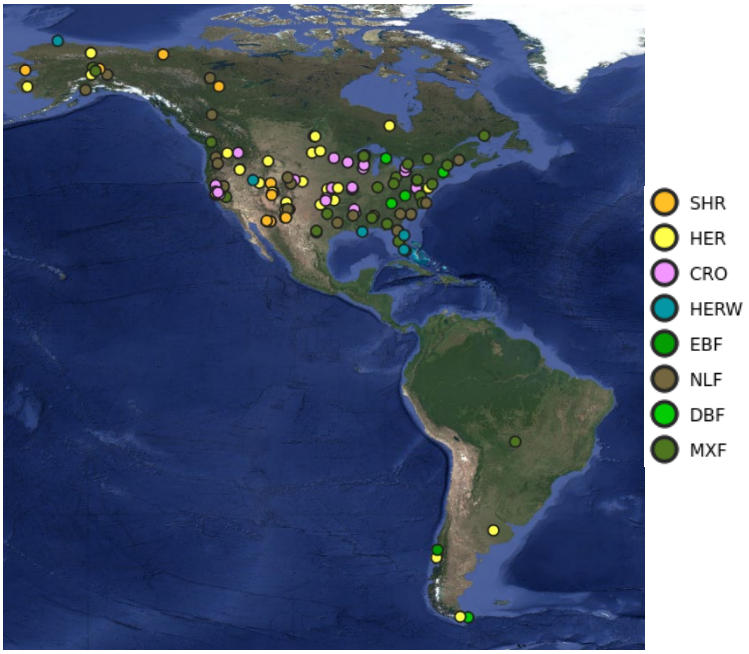


Figure 3: Global distribution of the Ameriflux sites. Colour coding according to biome (see §5.3.2).

5.3 REGIONAL/BIOME ASSESSMENT AND SPATIAL SUBSAMPLING

5.3.1 Introduction

The GPP 300m V2 datasets used in this QAR are global sub-sampled images (see §5.3.3), a selection of 10° x 10° tiles and 1° x 1° mini tiles at full resolution (FR) (see §5.3.4), and extractions over the LANDVAL V2 sites (see §5.3.5). For the spatial and statistical consistency analysis, the years 2019 and 2023 are selected to have overlap with the reference products (see §5.2). The temporal analysis is done on the full time series from 2014 up to the end of 2024.

5.3.2 Stratification per biome

An aggregated version of the CLMS Global Land Cover at 100m, epoch 2018 (Buchhorn et al., 2019) was used to distinguish between major land cover classes at the global scale (Figure 4). The classes were aggregated according to the scheme in Table 9. The map at 300m spatial resolution is generated from the CLMS-LC100 discrete classification map. For each of the 60 global UTM zones, a downscaling from 100m to 300m was performed by aggregation following the mode of the discrete classes. In case of equal occurrence of discrete classes, a set of expert rules based on cover fractions which can be retrieved from the CLMS-LC100 documentation is applied [CGLOPS1_ATBD_LC100mV3]. A new global tiling grid at 300m resolution in EPSG:4326 (WGS84) is created to which each of the UTM zones is transformed following the best-available-pixel approach.

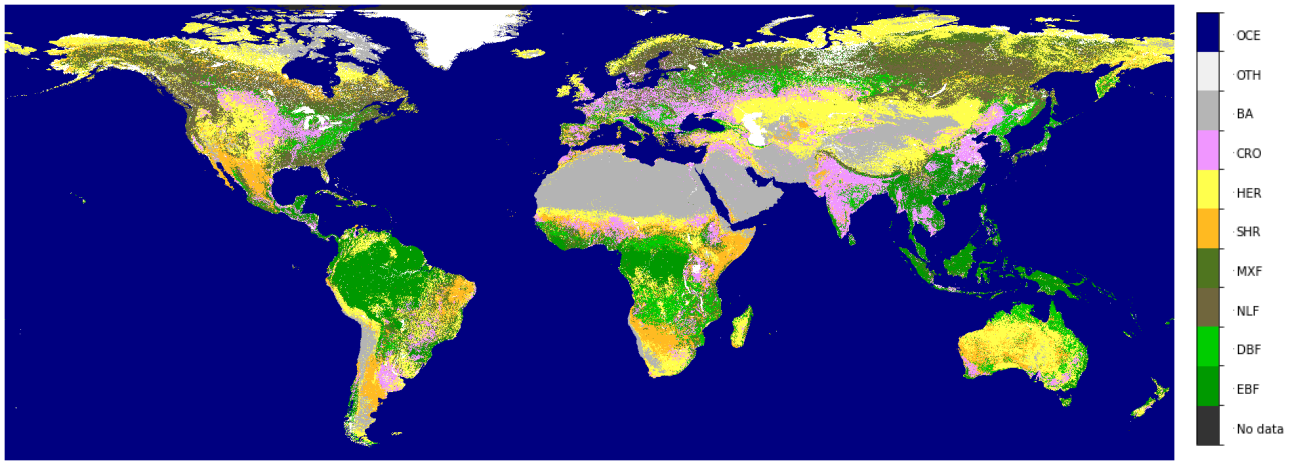


Figure 4: The CLMS LC100 epoch 2018 classification aggregated into eight classes

Table 9: Aggregation scheme for CLMS Land Cover 100m classes into eight major biomes and proportion of each biome at global scale

Abbreviation	Name	CLMS-LC100 classes	Proportion at global scale (%)
EBF	Evergreen broadleaf forest	112, 122	8.4
DBF	Deciduous broadleaf forest	114, 124	5.7
NLF	Needleleaf forest	111, 113, 121, 123	9.9
MXF	Mixed forest	115, 116, 125, 126	12.7
SHR	Shrubland	20	6.9
HER	Herbaceous	30	22.4
CRO	Crop	40	9.8
BA	Bare / sparse vegetation	60, 100	14.7
	Other (not considered in the analyses)	0, 50, 70, 80, 90	9.5

5.3.3 Global systematic subsample

Global images are systematically subsampled over the whole globe taking the central pixel in each (arbitrary) window of 51 by 51 pixels for 300m products, and 17 by 17 pixels for 1 km products. This subsample is representative for the global patterns of vegetation and considerably reduces the processing time.

To be able to perform pixelwise comparison at global scale between the GPP 300m V2 product, on the one hand, and the DMP/GDMP 1km V2 and MOD17 V6.1 GPP products, on the other hand, an

additional global subsample is extracted for the 300m data. This subsample is also created for every 51st pixel, but instead of using just one 300m x 300m pixel, the central 3 by 3 pixels are averaged to mimic a 1km pixel. The average is only computed when at least seven of the nine pixels have a valid GPP value, otherwise the value is discarded.

For the comparison analysis between RT modes in §6.3, a 'global subsample' is used with the same sampling strategy as described above, but only for a selection of tiles. The selected tiles represent all global biomes, and are located at various latitudes and longitudes. The tiles that are used are shown in Figure 5.

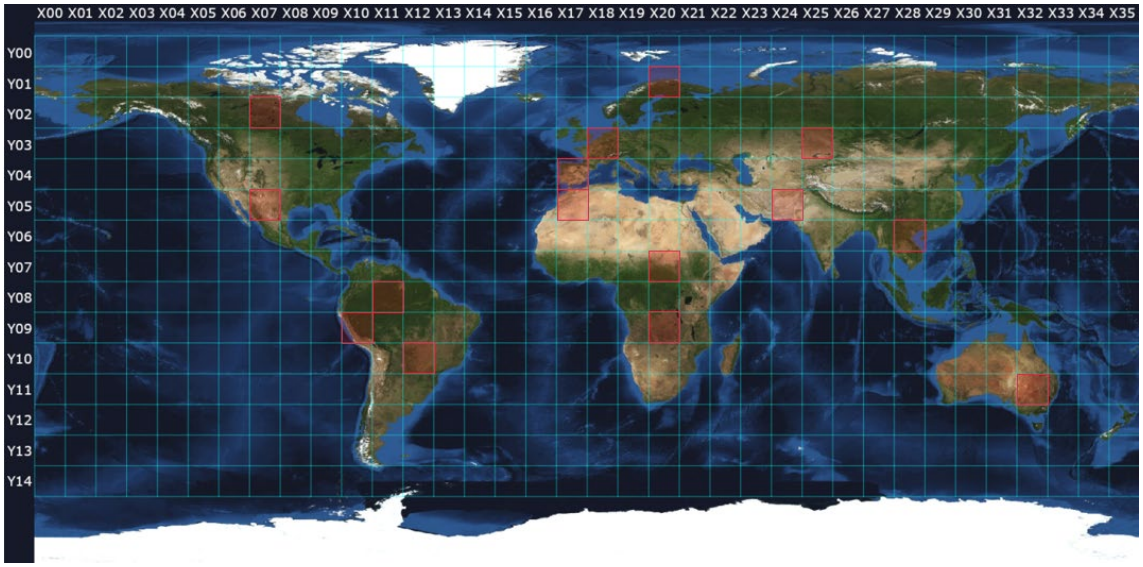


Figure 5: Global map with tiles used in consistency analysis of RT modes indicated in red

5.3.4 10° x 10° tiles and 1° x 1° mini-tiles at full resolution

To evaluate spatial patterns, a selection of the inter-comparison procedures can be applied on 10° x 10° tiles at full resolution (Figure 6). The CGLOPS Sentinel-3 processing tiling grid is used as reference.

To cover different climatic zones and vegetation conditions, the following tiles are considered:

- NAME: Northern America, tile X07Y02, coordinate ranges [65° - 55°N, 110° - 100° W]
- WEUR: Western Europe, tile X18Y03, coordinate ranges [55° - 45° N, 0° - 10° E]
- SAME: South America, tile X10Y08, coordinate ranges [5° N - 5° S, 80°W – 70° W]
- SAHE: Sahel, tile X17Y06, coordinate ranges [25° - 15° N, 10° - 0° W]
- WAFR: West Africa, tile X17Y07, coordinate ranges [15° - 5° N, 10° - 0° W]

For the Western Europe and West Africa tiles, also one 1° x 1° mini tile is depicted for in detail visual comparison between 300m products. The following mini-tiles are considered:

- BRU: Western Europe, region around Brussels, coordinate ranges [51° - 50° N, 4° - 5° E]
- IVO: West Africa, Ivory Coast, coordinate ranges [7° - 6°N, 6° - 5° W]

Quicklooks generated with Google satellite imagery are shown for each tile in Figure 7. These quicklooks will be used to compare spatial patterns in the visual comparison of the datasets.

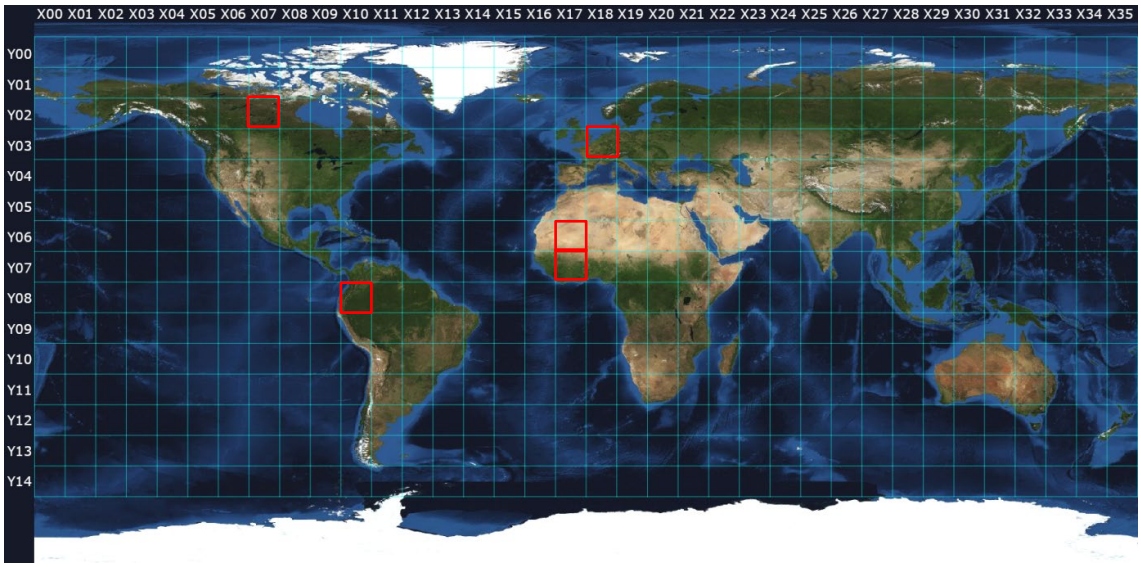


Figure 6: Grid tiles on world map[10]. Red squares indicate tiles under consideration.

[10]Background image source: http://server.arcgisonline.com/arcgis/rest/services/ESRI_Imagery_World_2D/MapServer

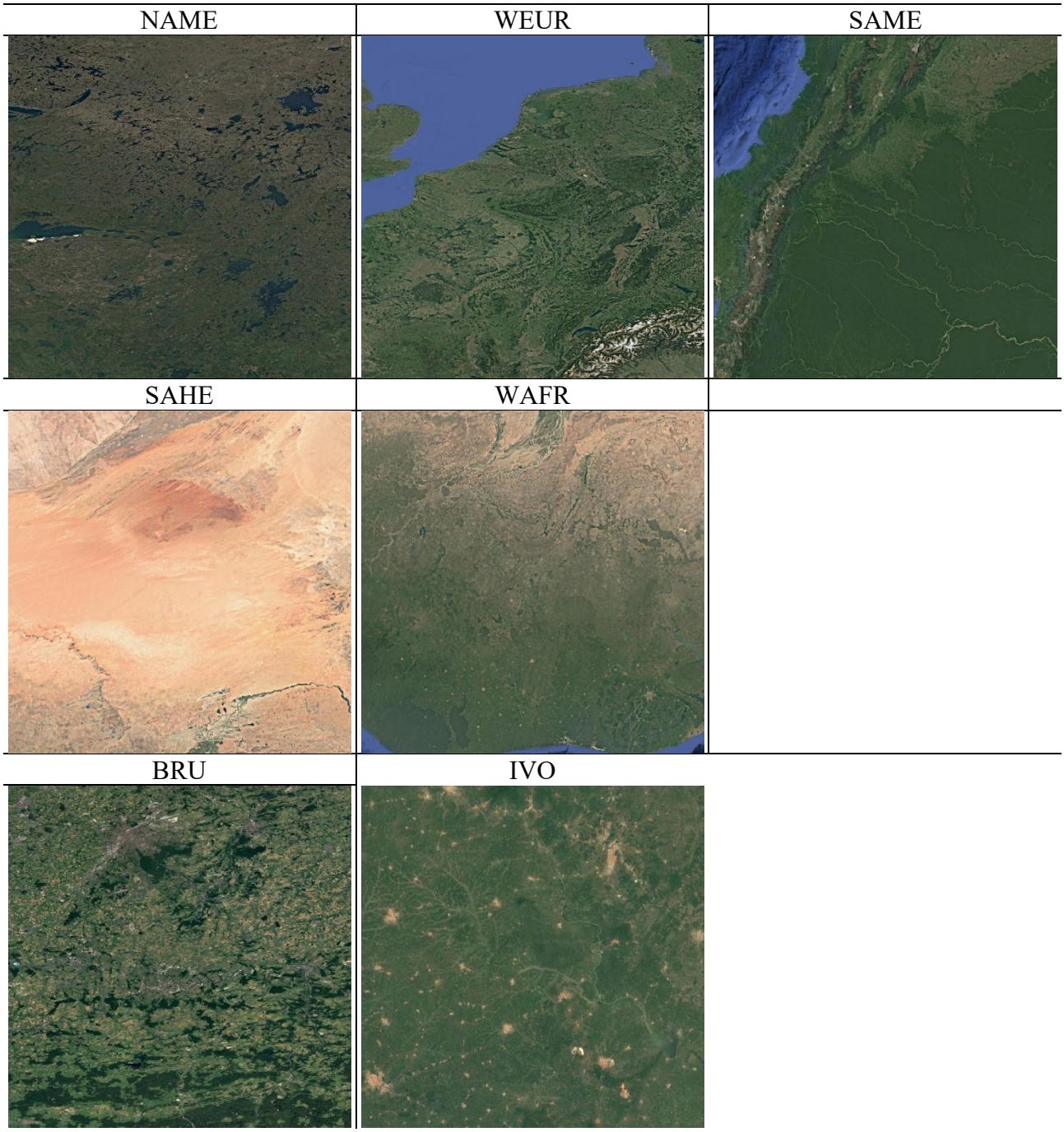


Figure 7: Google satellite quicklooks of 10°x10° tiles (top and middle rows) and 1°x10° mini-tiles (bottom row) considered in the validation analysis

5.3.5 LANDVAL V2 network

The LANDVAL V2 network of 2000 sites (Figure 8) is used to evaluate the statistical consistency among and temporal realism of the GPP time series (Martínez-Sánchez et al., 2024). The sites are

selected based on three criteria: (i) same land cover and ecoregion (> 90% homogeneity) over an area of 3km x 3km; (ii) distance between samples at least of 2 degrees on the hypotenuse in Plate Carrée projection; and (iii) terrain slope within the 3km x 3km area lower than 100m.

The LANDVAL V2 sites are used for the statistical analyses and for the generation of temporal plots. The 300m data extraction is based on 3x3 pixels covering the exact same footprint as the 1km pixel, calculating the average GPP only when at least seven (out of nine) values are valid. Stratification per biome is based on the aggregated version of the CLMS Global Land Cover at 100m (see §5.3.2).

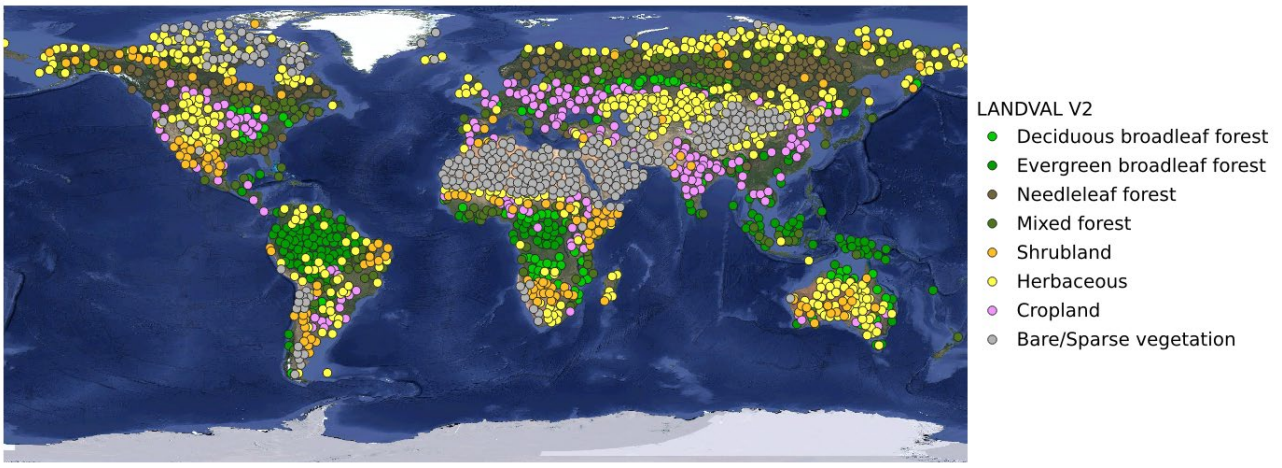


Figure 8: Global distribution of the selected LANDVAL V2 sites (Martínez-Sánchez et al., 2024)

6 RESULTS

6.1 Q1: QUALITY FLAGS – OCCURRENCES AND RECOMMENDATIONS

6.1.1 Introduction

The scientific questions to be answered are the following: what are the occurrences of quality flags in DMP/GDMP/NPP/GPP 300m V2 in terms of temporal and spatial distribution? Which quality flags are recommended for users and are applied in the subsequent analysis?

The analyses are performed over a one-year period (2019) for both PROBA-V and Sentinel-3/OLCI and are based on a global systematic subsample (see §5.3.3). Although the GPP 300m V2 dataset switches from PROBA-V to Sentinel-3/OLCI at the start of 2019, the year 2019 was processed for both sensors and this overlapping year is used for intercomparison.

6.1.2 Quality flags

The GPP 300m V2 product has an ancillary QFLAG layer which contains quality information for all land pixels (see Section 3). The specific focus of this question is on the information that is specifically defined for GPP, i.e. bits 1, 2, and 6 of the QFLAG layer (see Table 2). All other bits are copied from the FAPAR data set on which it is based.

QFLAG bit 1 indicates whether there is FAPAR 300m V2 data available or missing. Since GPP 300m V2 is not generated if FAPAR 300m V2 data are missing, this quality flag is never raised for pixels with valid GPP.

The same is true for QFLAG bit 2 which indicates whether meteorological data is available or missing. Similar as for FAPAR data, GPP 300m V2 is only computed for pixels for which meteorological data is available. This quality flag is hence never raised for valid GPP pixels.

Bit 6 of the QFLAG layer is set when the input FAPAR is lower than 0.15. Pixels with FAPAR lower than or equal to 0.05 are set to zero, while FAPAR values between 0.05 and 0.15 are linearly rescaled between these values (for details see [CGLOPS1_ATBD_DMP300m-V2]). This is done to mitigate unrealistic GPP values resulting from small (but non-negligible) FAPAR values over non vegetated areas (e.g. deserts) or periods without vegetation growth (e.g. during the winter period). The occurrences of QFLAG bit 6 are shown in Figure 9. Over desert areas, such as the Sahara, the low FAPAR correction bit is always set, both for PROBA-V and for Sentinel-3/OLCI. This is expected, as FAPAR values will always be very small (i.e. smaller than 0.05) in these regions. In winter months, the number of occurrences of bit 6 is lower for Sentinel-3/OLCI than for PROBA-V. This is related to the number of invalid observations (see §6.2.2): the Sentinel-3/OLCI product completeness is lower than that of the PROBA-V dataset, especially during winter, resulting in less pixels of the Sentinel-3/OLCI dataset with QFLAG bit6 raised.

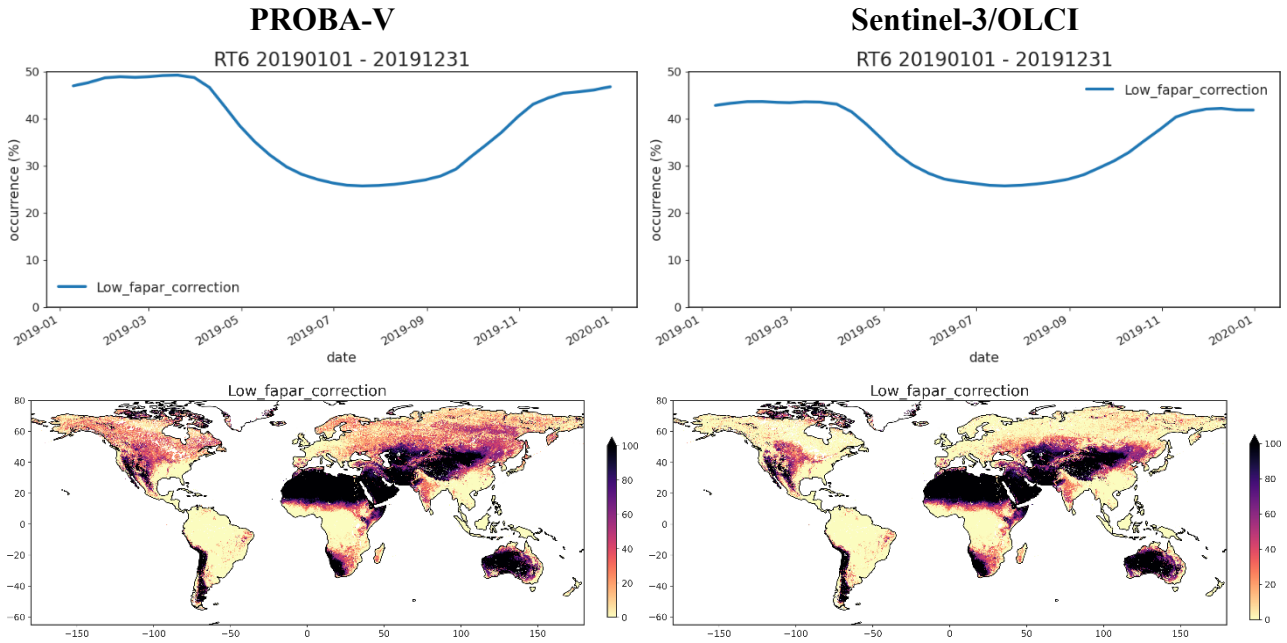


Figure 9: Temporal (top) and spatial (bottom) distribution of number of low FAPAR correction (QFLAG bit 6) occurrences over land pixels for GPP 300m V2 PROBA-V (left) and GPP 300m V2 Sentinel-3/OLCI (right) for the year 2019

6.1.3 Conclusion

What are the occurrences of quality flags in DMP/GDMP/NPP/GPP 300m V2 in terms of temporal and spatial distribution? Which quality flags are recommended for users and are applied in the subsequent analysis?

The only quality flags which are not directly propagated from the FAPAR 300m V2 product are bit 1, bit 2, and bit 6. These first two bits are never set for pixels with valid GPP 300m V2, since both FAPAR and meteorological data are needed to compute DMP/GDMP/NPP/GPP. The number of bit 6 occurrences changes during the year, and is lower for Sentinel-3/OLCI than for PROBA-V. This is related to lower completeness of Sentinel-3/OLCI (Section 6.2), especially during the Northern hemisphere winter. It is recommended to include all pixels of the DMP/GDMP/NPP/GPP 300m V2 product, regardless of the quality flag. The QFLAG layer does provide additional information on the FAPAR algorithm (e.g. EBF case or gap filled), and – in case a pixel is not filled – on the reason (e.g. due to missing FAPAR or missing meteorological data).

6.2 Q2: PRODUCT COMPLETENESS OF DMP/GDMP/NPP/GPP 300M V2

6.2.1 Introduction

The scientific questions to be answered are the following: what are the temporal variation and spatial distribution of the completeness of DMP/GDMP/NPP/GPP 300m V2? How do the results compare to those of DMP/GDMP 300m V1.0, DMP/GDMP 300m V1.1, DMP/GDMP 1km V2, and MOD17 V6.1 GPP?

The analyses are performed over two one-year periods (2019 and 2023), the latter for comparison with GPP 300m V1.1, and on the full period (2014 – 2024) for the temporal analysis. They are based on global systematic subsamples for the spatial and temporal distribution of gaps (see §5.3.3) and on extractions of valid GPP values for LANDVAL V2 sites for the evaluation of gap lengths (see §5.3.5).

The analyses include:

- Spatial distribution of gaps
- Temporal distribution of gaps
- Gap length frequency

6.2.2 Spatial distribution of gaps

Figure 10 shows the spatial patterns of gaps in GPP 300m V2 compared to the reference datasets for the year 2019, while Figure 11 shows the spatial gap patterns for the year 2023. As expected, all datasets show lower product completeness at high latitudes where there are no observations in winter. For GPP 1km V2, this is less pronounced because, in contrast to the other CLMS datasets, no cutoff at SZA > 75° is performed and a climatology was used to fill gaps.

There are less gaps in the tropics in GPP 300m V2 compared to GPP 300m V1.0. For the PROBA-V data, this can be explained by the difference in input data (see §5.2.2.5): PROBA-V C1 was used in GPP 300m V1.0, while PROBA-V C2 is used in GPP 300m V2. In PROBA-V C2, ~11% fewer pixels are detected as clouds and are hence valid observations (see PROBA-V C2 ACD). Additionally, there is a difference in RT mode between V1.0 and V2 of PROBA-V GPP: V1.0 uses RT5 while V2 adopts RT6. The longer consolidation period results in better outlier filtering and smoothing of the data. For the Sentinel-3/OLCI data, the difference in completeness is also a direct result of different TOC reflectance input versions with differences in cloud masking: TOC reflectance V1 was used in GPP 300m V1.1 while TOC reflectance V2.3 is used in GPP 300m V2 [CGLOPS1_VR_TOC-Reflectance300m-V2.3].

For GPP 300m V2, there are more gaps in Sentinel-3/OLCI based products than in PROBA-V based products. This is also observed in the FAPAR 300m V2 product [CGLOPS1_VR_FAPAR300m-V2] where the difference in completeness is attributed to the improved cloud detection algorithm in

PROBA-V C2 and the cloud-snow misclassification in OLCI-based TOC reflectances [CGLOPS1_VR_TOC-Reflectance300m-V2.3].

The MOD17 V6.1 dataset with quality flags applied as described above (see §5.2.2.4) has relatively higher amounts of missing observations, with low product completeness at mid to high latitudes and over areas in the Sahel and in Australia. There are also zones in the Sahara, the Arabian Peninsula, and in Asia where MOD17 V6.1 GPP is not determined at all.

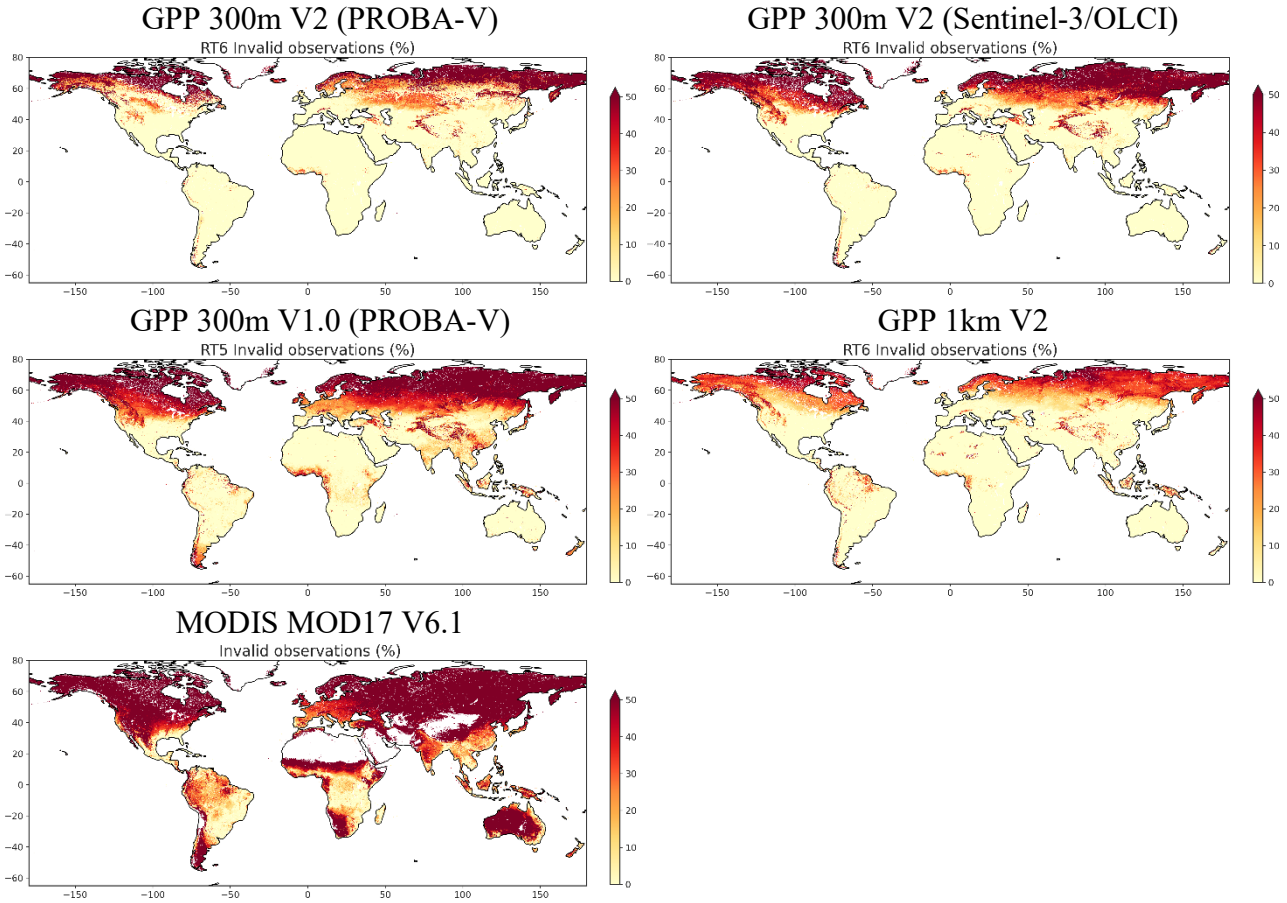


Figure 10: Spatial distribution of the number of missing values (%) of GPP 300m V2 PROBA-V, GPP 300m V2 Sentinel-3/OLCI, GPP 300m V1.0, GPP 1km V2, and MODIS MOD17 V6.1 GPP for the year 2019.

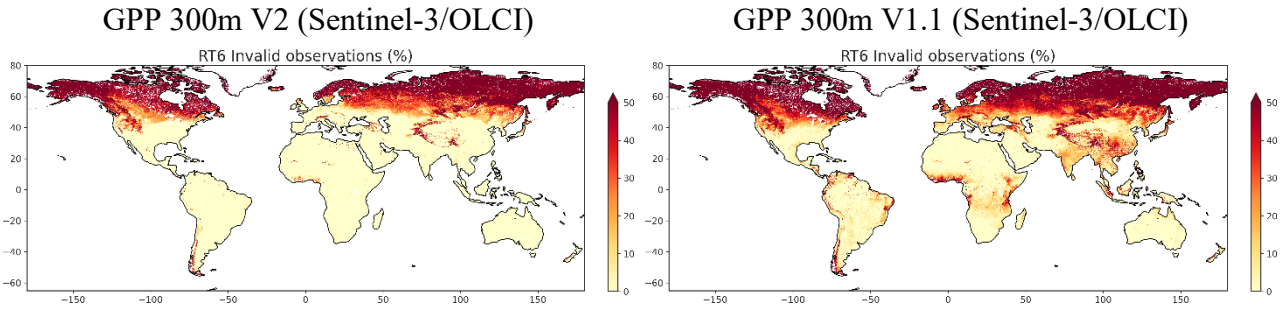


Figure 11: Spatial distribution of the number of missing values (%) of GPP 300m V2 Sentinel-3/OLCI and GPP 300m V1.1 for the year 2023.

6.2.3 Temporal distribution of gaps

The temporal distribution of the number of valid observations over land for the various datasets is shown in Figure 12. Similar conclusions can be drawn as for the spatial plots. All datasets have a higher completeness during the Northern hemisphere summer months compared to winter-time, because in summer the northern regions are not impacted by bad illumination conditions. The completeness of GPP 300m V2 is higher than for GPP 300m V1.0 and V1.1 (especially in winter) due to differences in input data and cloud masking. For GPP 1km V2, the completeness is high for a longer period in summer, due to gap filling with a climatology. The completeness of the MOD17 V6.1 dataset shows similar seasonal patterns but is overall lower compared to the other datasets.

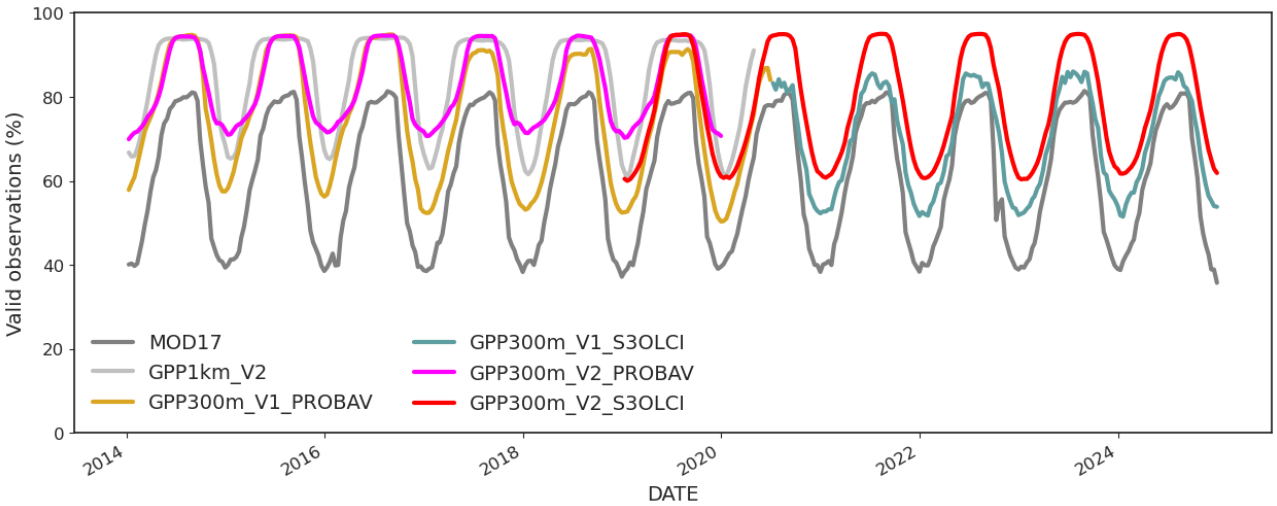


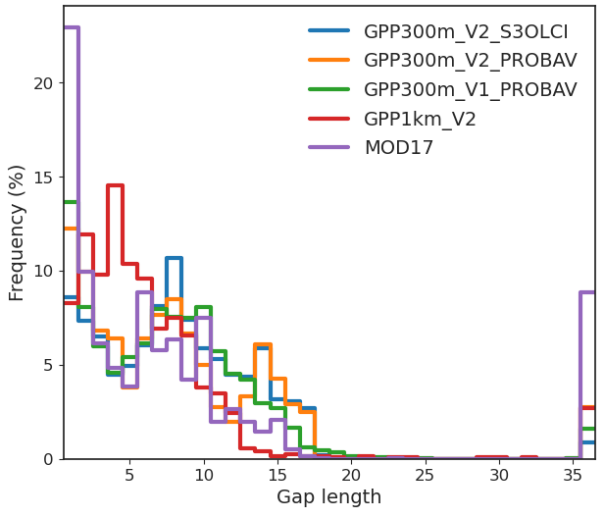
Figure 12: Temporal evolution of the number of valid observations over land (%) of GPP 300m V2 Sentinel-3/OLCI (red), GPP 300m V2 PROBA-V (pink), GPP 300m V1.0 (yellow), GPP 300m V1.1 (blue), GPP 1km V2 (light grey), and MODIS MOD17 V6.1 (grey)

6.2.4 Gap length frequency

A data gap is defined as a single missing data point or consecutive dates with missing data in the extractions over the LANDVAL V2 sites of the GPP 300m V2 dataset and reference datasets. The frequency of the data gap occurrences per length of the data gap is plotted in Figure 13 for the year 2019 and in Figure 14 for the year 2023.

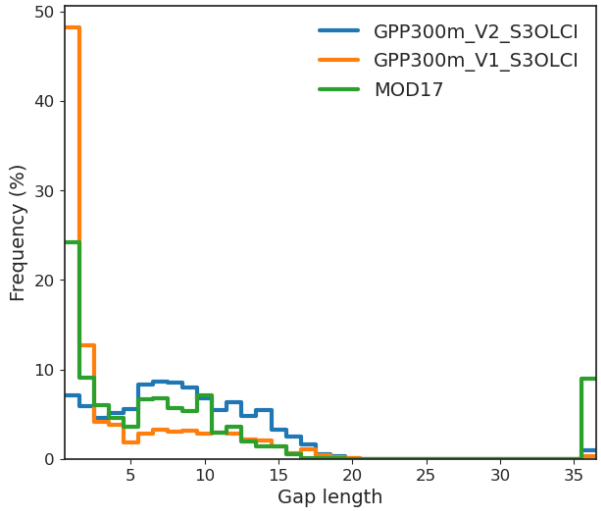
These figures show that for most datasets, gap lengths of 1 to 17 dekads are most common. Such long data gaps are due to the Northern hemisphere winter in which no observations are available for a long period. The MOD17 V6.1 and GPP 300m V1.1 datasets stand out against the other datasets, as these have a high number of gaps of length one. However, the number of gaps is also largest for these datasets.

The most recent GPP 300m V2 dataset contains less gaps than the previous 300m datasets, showing that a higher completeness is achieved for the new dataset. The increase in completeness with respect to previous versions is highest for the Sentinel-3/OLCI dataset (i.e. the completeness increases between V1.1 and V2 for Sentinel-3/OLCI is higher than between V1.0 and V2 for PROBA-V). The number of gaps is also low for the GPP 1km V2 dataset, which is due to the climatology that is used to fill gaps.



Dataset	Mean gap length (dekads)	Number of gaps
GPP 300m V2 Sentinel-3/OLCI	8.2	1678
GPP 300m V2 PROBA-V	8.0	1317
GPP 300m V1.0	7.6	2386
GPP 1km V2	6.4	1484
MODIS	8.2	3393

Figure 13: Histogram of gap length (left) and mean gap length and number of gaps (right) over LANDVAL V2 sites for GPP 300m V2 Sentinel-3/OLCI (blue), GPP 300m V2 PROBA-V (orange), GPP 300m V1.0 (green), GPP 1km V2 (red), and MODIS MOD17 V6.1 GPP (purple) for the year 2019



Dataset	Mean gap length (dekads)	Number of gaps
GPP 300m V2 Sentinel-3/OLCI	8.4	1616
GPP 300m V1.1	4.3	4512
MODIS	8.3	3346

Figure 14: Histogram of gap length (left) and mean gap length and number of gaps (right) over LANDVAL V2 sites for GPP 300m V2 Sentinel-3/OLCI (blue), GPP 300m V1.1 (orange), and MODIS MOD17 V6.1 GPP (green) for the year 2023

6.2.5 Conclusion

What are the temporal variation and spatial distribution of the completeness of DMP/GDMP/NPP/GPP 300m V2 in comparison to MOD17 V6.1? How do the results compare to those of DMP/GDMP 300m V1.0, DMP/GDMP 300m V1.1, and DMP/GDMP 1km V2?

The product completeness of DMP/GDMP/NPP/GPP 300m V2 is significantly higher than that of MOD17 V6.1. That is because certain regions are not processed in MOD17 V6.1 and related to the quality screening as applied in the current analysis. Apart from these regions, the product completeness of both datasets shows similar spatial patterns. Compared to previous 300m datasets, the DMP/GDMP/NPP/GPP 300m V2 also has higher completeness. This is related to the difference in TOC reflectance input data that has been used in each version and the corresponding cloud masking that has been applied. For all datasets, the product completeness is highest across desert sites and lower at higher latitudes during the Northern hemisphere winter period.

6.3 Q3: DIFFERENCES AMONG CONSOLIDATION MODES

6.3.1 Introduction

The scientific question to answer here is: what are the differences between RT6 (final consolidation mode) and the NRT modes (RT2, RT1, and RT0) in DMP/GDMP/NPP/GPP 300m V2?

The difference between the consecutive consolidation modes is directly related to the input FAPAR product. The analyses are only performed on GPP, as the results are the same for GDMP, NPP, and DMP, and are based on global systematic subsamples of a selection of tiles (see §5.3.3) for the statistical analysis and on extractions of valid GPP values for LANDVAL V2 sites (see §5.3.5) for the temporal analysis, both for the year 2024.

The analyses include:

- Statistical consistency analysis:
  - Histograms of overall bias
  - GM regression analysis, overall and per biome
- Temporal consistency analysis
  - Temporal plots over LANDVAL sites

6.3.2 Statistical consistency

Figure 15 shows the bias histograms and scatterplots between the final consolidation mode RT6 and the other modes RT0, RT1, and RT2. For all three modes, the bias is centred symmetrically around zero and the GM regression line is close to the 1:1 line. The spread is largest for the RT0 mode and decreases for RT1 and RT2 as expected, since longer consolidation periods increase the reliability of outlier detection and data cleaning. In the scatterplot for RT0 and, to a lesser extent in that of RT1, a large density of points is situated above the 1:1 line. Similar results were found for the FAPAR 300m V2 data (see Annex 3 in [CGLOPS1_VR_FAPAR300m-V2]). These are pixels for which the RT0 (or RT1) mode has a significantly higher GPP value than the RT6 mode. These are likely undetected clouds and other artefacts that are not removed by the outlier detection algorithm because the consolidation period is too short. With a longer consolidation period, the outlier detection algorithm is more performant.

A similar conclusion can be drawn from the statistics given in Table 10. The Pearson correlation coefficient (R) increases with consolidation mode (although it is high in all cases) while the other statistics (APU and others) decrease. The bias B is so small in all three cases that there is almost no difference between STD and RMSD.

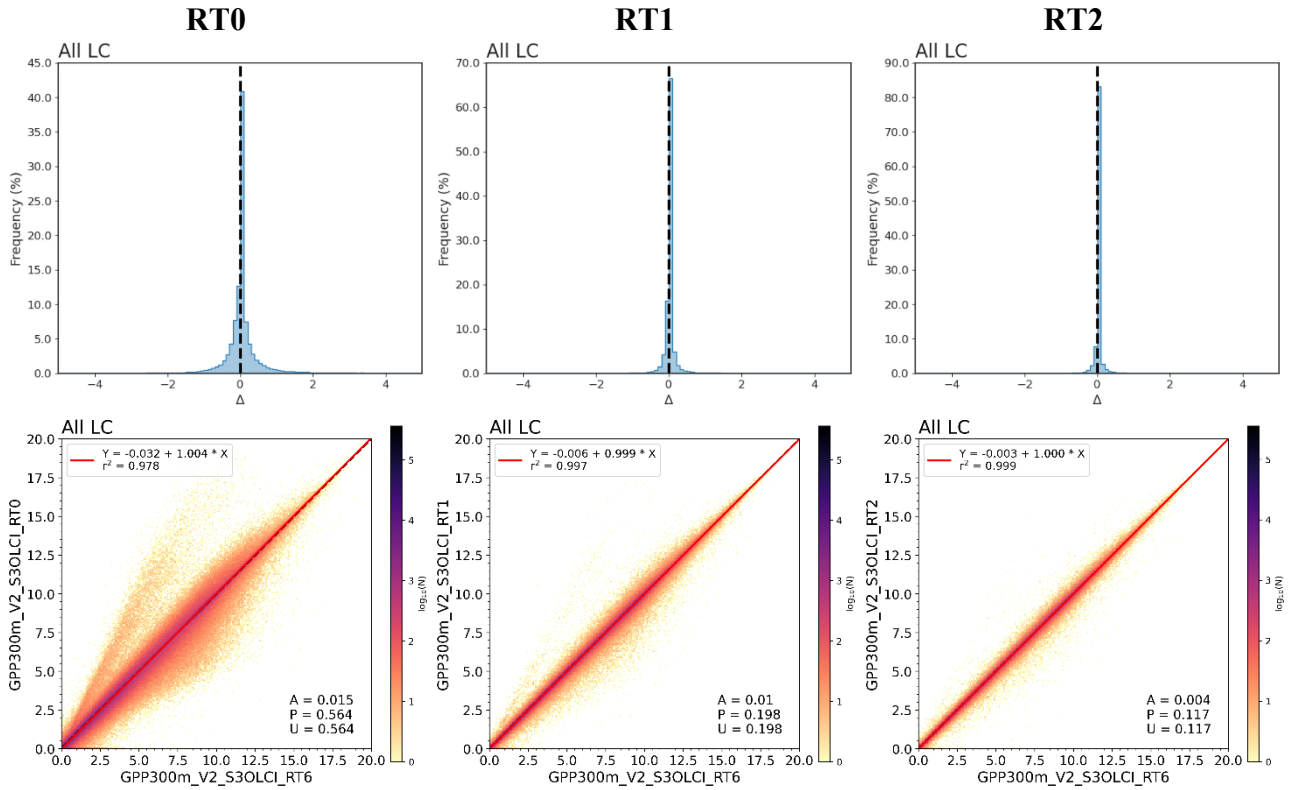


Figure 15: Bias histograms (top) and scatterplots (bottom) between RT6 and RT0 (left), RT1 (middle), and RT2 (right) for GPP 300m V2 (gC/m²/day) for the year 2024

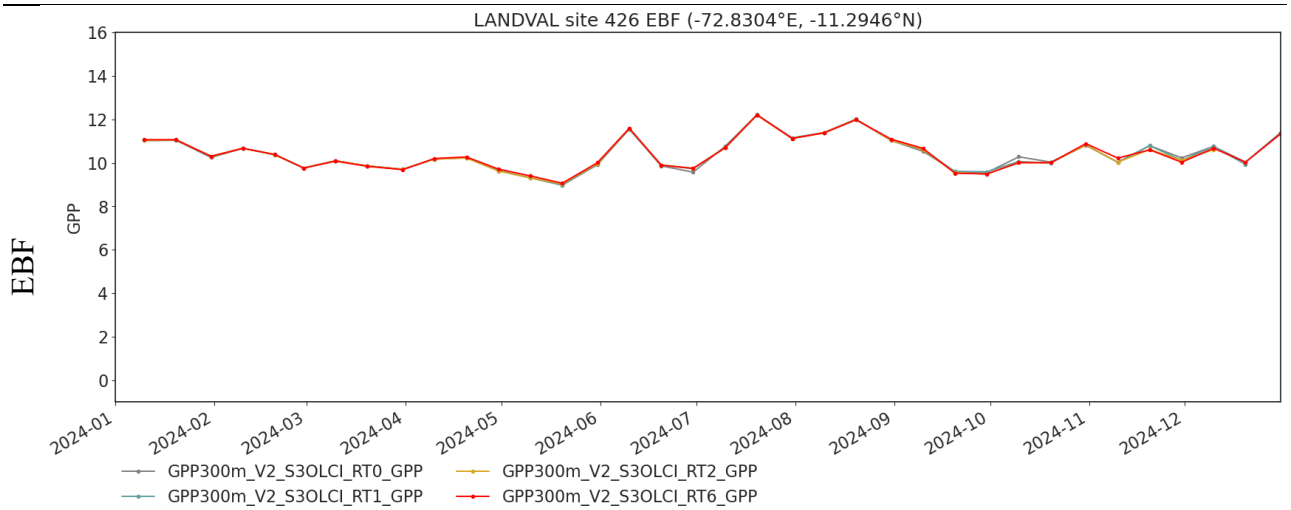
Table 10: Statistics of comparison between GPP 300m V2 Sentinel-3/OLCI RT6 and RT0, RT1, and RT2 over a global subsample for the year 2024. The values of the statistics are given in gC/m²/day.

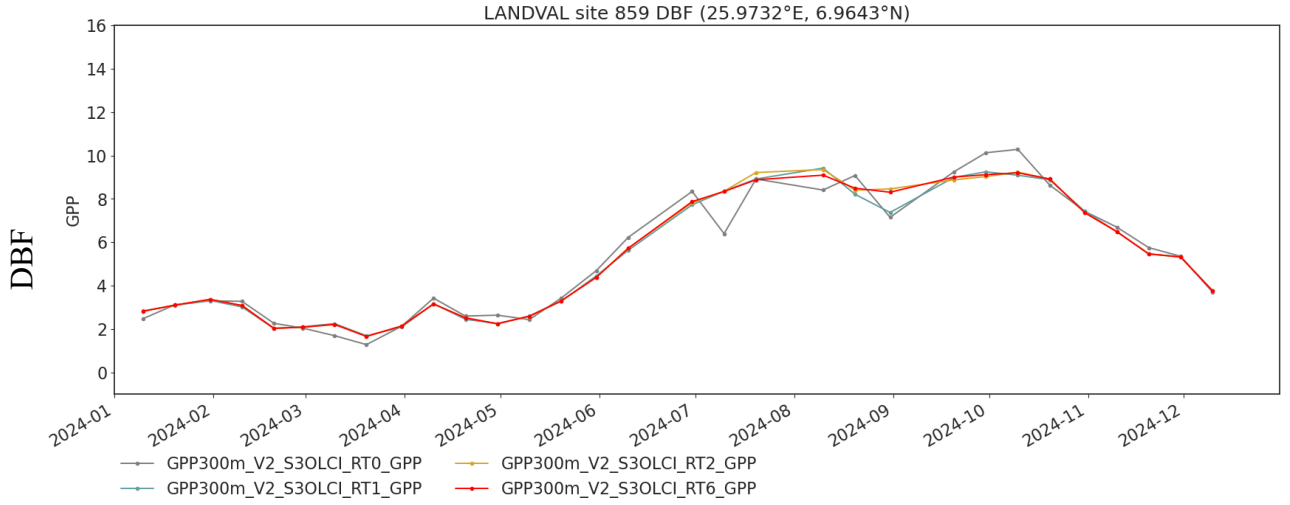
RT mode	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset / slope
RT6 vs RT0	1793938	0.99	0.015 (0.4%)	0.0 (0%)	0.6 (15%)	0.08 (2%)	0.6 (15%)	-0.032 / 1.004
RT6 vs RT1	1847669	> 0.99	0.010 (0.3%)	0.0 (0%)	0.2 (5%)	0.007 (0.19%)	0.2 (5%)	-0.006 / 0.999
RT6 vs RT2	1818743	> 0.99	0.004 (0.11%)	0.0 (0%)	0.12 (3%)	0.0 (0%)	0.12 (3%)	-0.003 / 1.000

6.3.3 Temporal pixel profiles

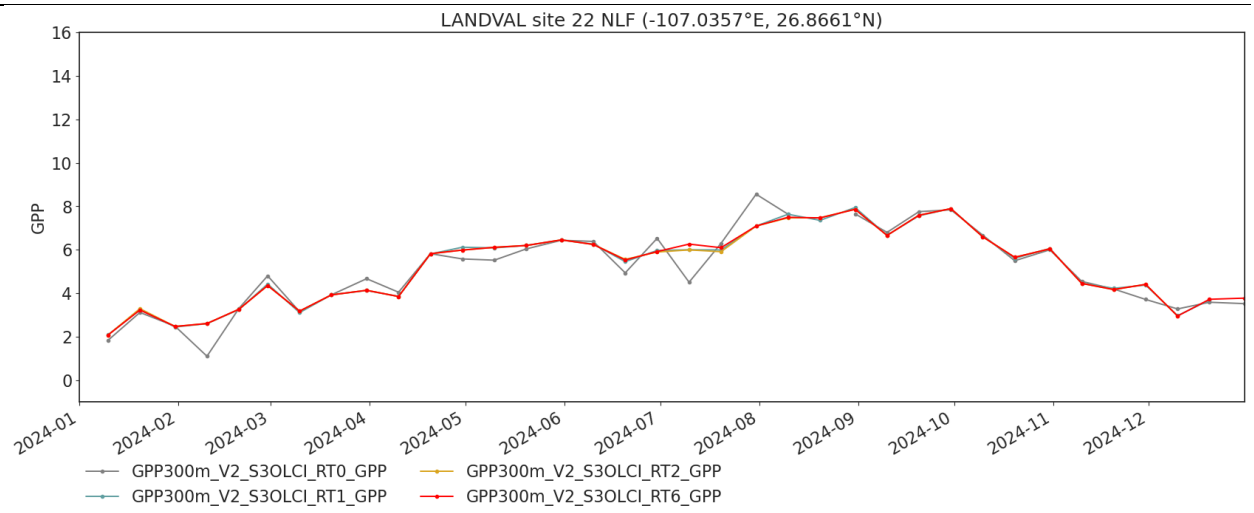
In Figure 16, temporal profiles of all four consolidation modes of GPP 300m V2 are shown for a selection of LANDVAL V2 sites. The figures illustrate that the temporal patterns of all consolidation modes are very similar, with some deviations for the RT0 mode. The GPP values in RT0 mode sometimes show high-frequency fluctuations (see for example LANDVAL sites 22 and 859) and is a

bit noisy, especially in regions affected by frequent cloud cover, while the temporal patterns become smoother with increasing consolidation mode. This is due to better outlier detection for larger windows (see above). Differences between RT1, RT2, and RT6 are generally very small. The large outlier in RT0 mode of LANDVAL site 1927 (CRO) is most likely an undetected cloud or another artefact, which is filtered out in the subsequent consolidation products.

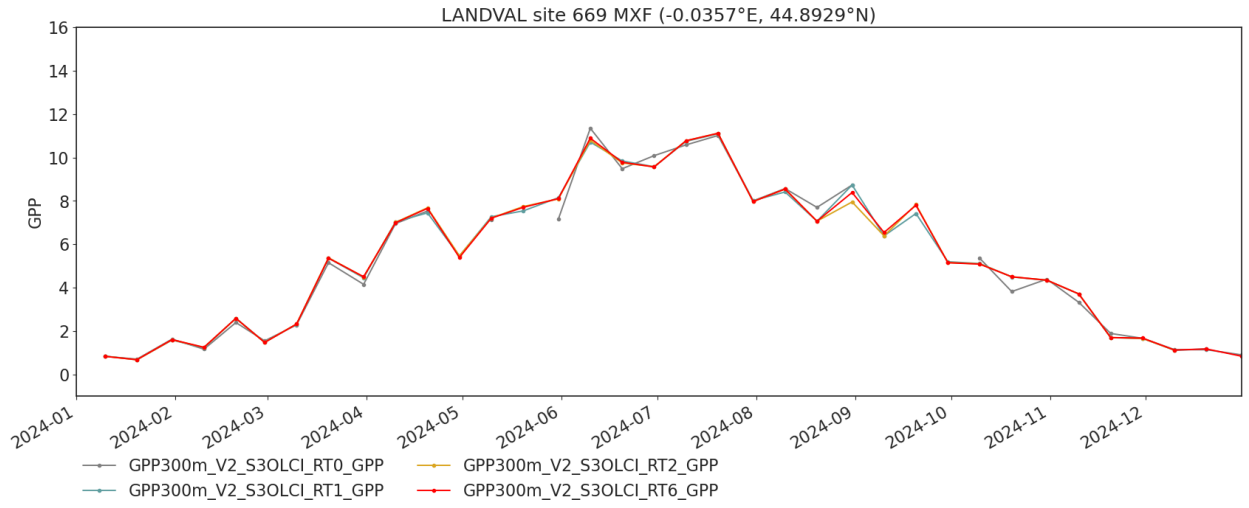




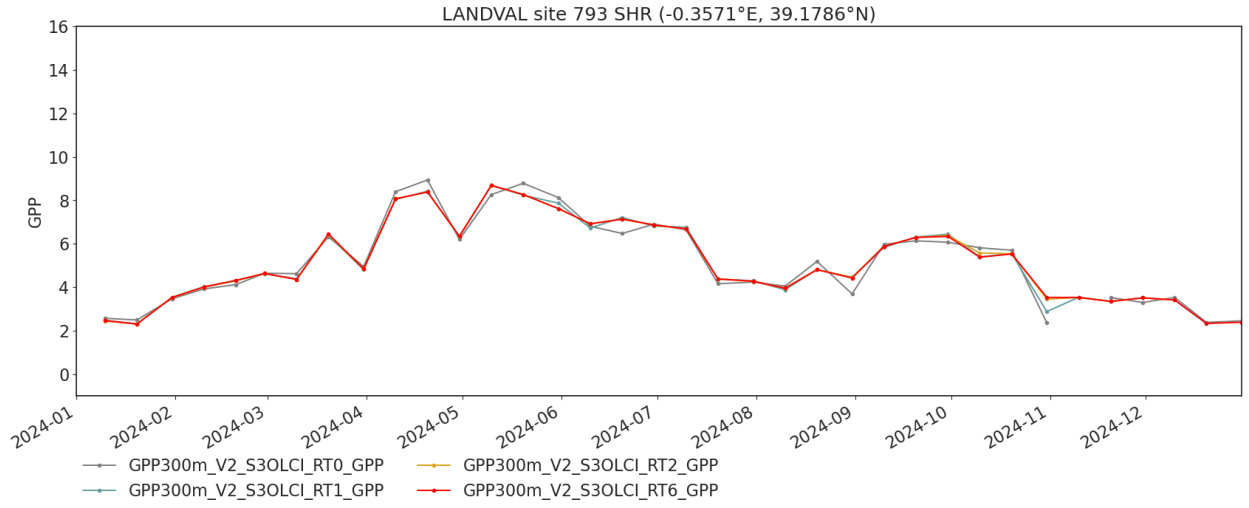
NLF


LANDVAL site 22 NLF (-107.0357°E, 26.8661°N)
GPP
GPP300m_V2_S3OLCI_RT0_GPP
GPP300m_V2_S3OLCI_RT1_GPP
GPP300m_V2_S3OLCI_RT2_GPP
GPP300m_V2_S3OLCI_RT6_GPP

MXF


LANDVAL site 669 MXF (-0.0357°E, 44.8929°N)
GPP
GPP300m_V2_S3OLCI_RT0_GPP
GPP300m_V2_S3OLCI_RT1_GPP
GPP300m_V2_S3OLCI_RT2_GPP
GPP300m_V2_S3OLCI_RT6_GPP

SHR


LANDVAL site 793 SHR (-0.3571°E, 39.1786°N)
GPP
GPP300m_V2_S3OLCI_RT0_GPP
GPP300m_V2_S3OLCI_RT1_GPP
GPP300m_V2_S3OLCI_RT2_GPP
GPP300m_V2_S3OLCI_RT6_GPP

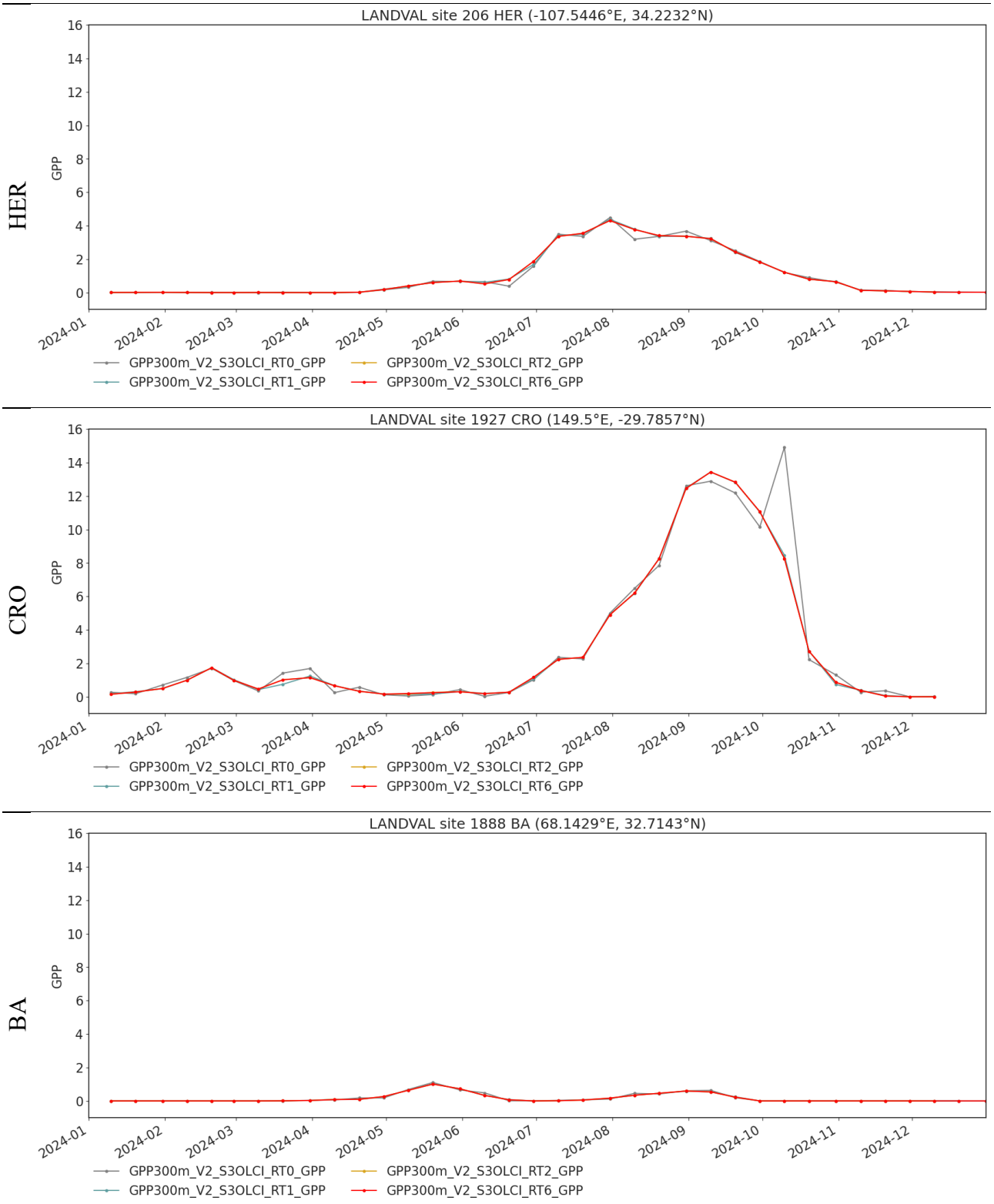


Figure 16: Temporal profiles over a selection of LANDVAL sites of GPP 300m V2 (gC/m²/day) RT6 (red), RT2 (yellow), RT1 (blue), and RT0 (grey).

6.3.4 Conclusion

What are the differences between the different consolidation modes of DMP/GDMP/NPP/GPP 300m V2?

The differences between the RT6 mode and other consolidation modes are minimal. The largest difference is found with the RT0 mode for which the short consolidation period is sometimes insufficient to remove all cloud contaminated observations. We therefore recommend users to use the largest RT mode available, and update NRT RT0 products with consolidated products once available.

6.4 Q4: VISUAL COMPARISON BETWEEN DMP/GDMP/NPP/GPP 300M V2 AND REFERENCE DATASETS

6.4.1 Introduction

The scientific question to be answered is the following: how do the DMP/GDMP/NPP/GPP 300m V2 PROBA-V and Sentinel-3/OLCI datasets compare visually to the DMP/GDMP 300m V1.0, DMP/GDMP 300m V1.1, and MOD17 V6.1 GPP datasets?

The analyses are performed over two one-year periods (2019 and 2023) and are based on a selection of 1° x 1° and 10° x 10° full resolution tiles (see §5.3.4). The results are shown for a few dekads; quicklooks for the whole period can be found in Annex 1: Digital annex.

6.4.2 Visual comparison at 1° x 1°

To visually compare the quality of GPP 300m V2 with previous versions V1.0 and V1.1, quicklooks of two 1°x1° mini tiles are visually inspected. Figure 17 and Figure 19 compare GPP 300m V2 with GPP 300m V1.0, and Sentinel-3/OLCI V2 with PROBA-V V2 for a region in Belgium and in Ivory Coast respectively. Figure 18 and Figure 20 inspect the same regions but for the comparison between GPP 300m V2 and GPP 300m V1.1.

When comparing V1.0 with V2, the latter clearly has a higher completeness and the GPP values of V2 are somewhat lower than in V1.0 but both versions have similar sharpness and show the same patterns. The difference in completeness and absolute values can be attributed to differences in the FAPAR versions, which are based on two different input TOC reflectance collections [CGLOPS1_VR_FAPAR300m-V2]. The two TOC reflectance collections include different cloud screening algorithms, with ~11% less pixels identified as cloud in PROBA-V C2, used in GPP 300m V2 [PROBA-V C2 ACD].

The quicklooks of V1.1 and V2 are also very similar with respect to sharpness, but more gaps are present in V1.1. The GPP values in V2 are higher than in V1.1, especially for the Ivory Coast region. Again, the only difference between the two GPP versions is the input FAPAR version and the TOC version on which they are based. Differences in cloud masking in the input TOC data can explain the better completeness in GPP 300m V2, while higher FAPAR 300m V2 values in comparison to V1.1 explain the difference in GPP values [CGLOPS1_VR_FAPAR300m-V2]. There is quite a big difference in the quicklooks of the Ivory Coast tile for dekad 20231110. When comparing to the Google satellite quicklooks in Figure 7, it can be concluded that GPP 300m V2 better captures the landscape patterns.

The differences between Sentinel-3/OLCI and PROBA-V GPP 300m V2 are very small. Similar patterns can be distinguished in the quicklooks, although the sharpness is somewhat higher for PROBA-V. This is likely related to differences in ground sampling distances between PROBA-V (<

100m at nadir) and Sentinel-3/OLCI (300m at nadir). The completeness and the GPP values differ a bit between PROBA-V and Sentinel-3/OLCI, especially for the Ivory Coast tile. GPP values are in general somewhat higher in Sentinel-3/OLCI, while more gaps are present in the PROBA-V dataset. The gaps are also not always located in the same locations for both sensors. Differences in completeness are directly related to differences in FAPAR product completeness, and can be attributed to different cloud masking algorithms, different input TOC data and differences in satellite/sensor.

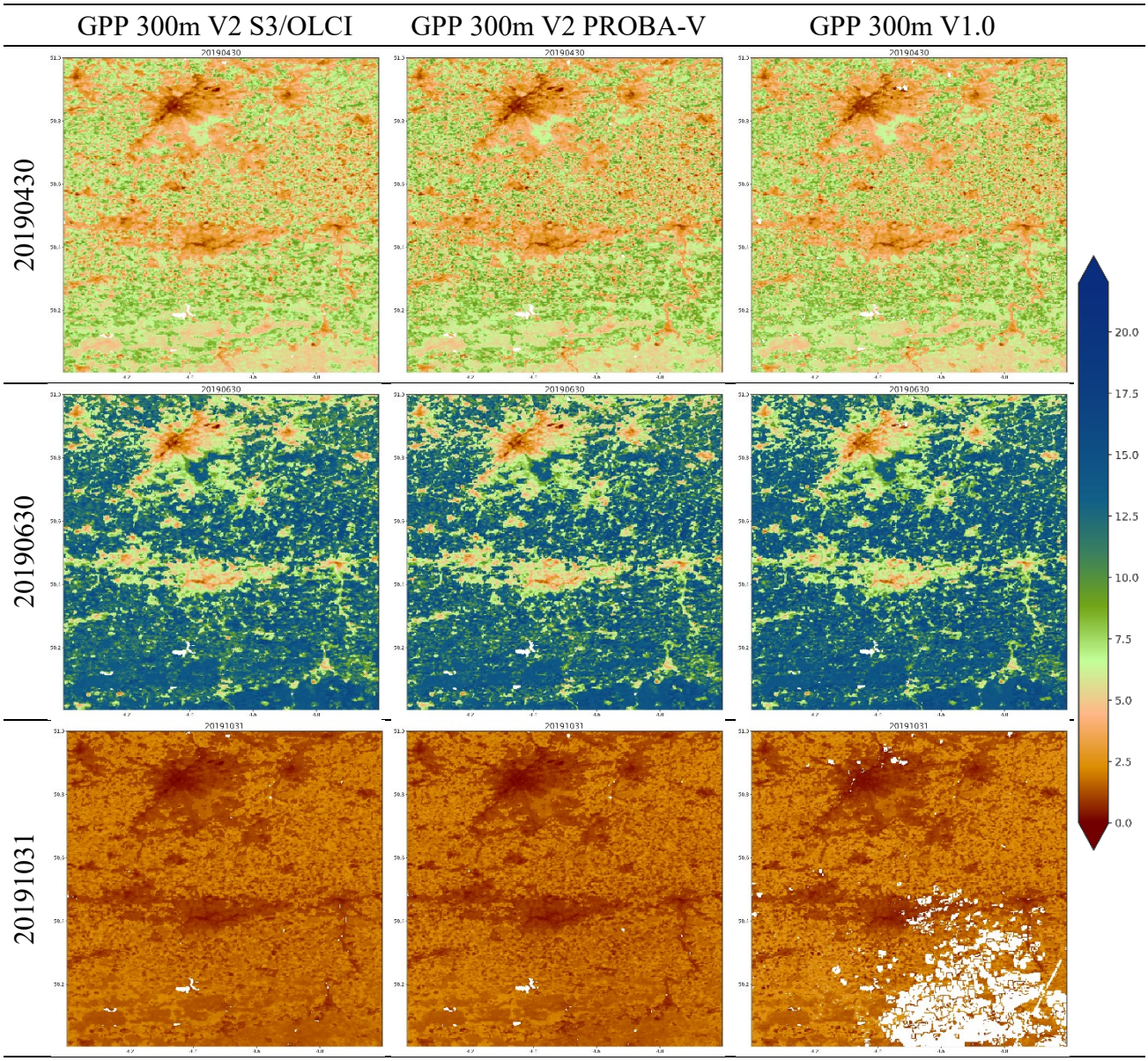


Figure 17: Visual comparison of GPP (gC/m²/day) for the Brussels 1° x 1° mini tile for the year 2019.

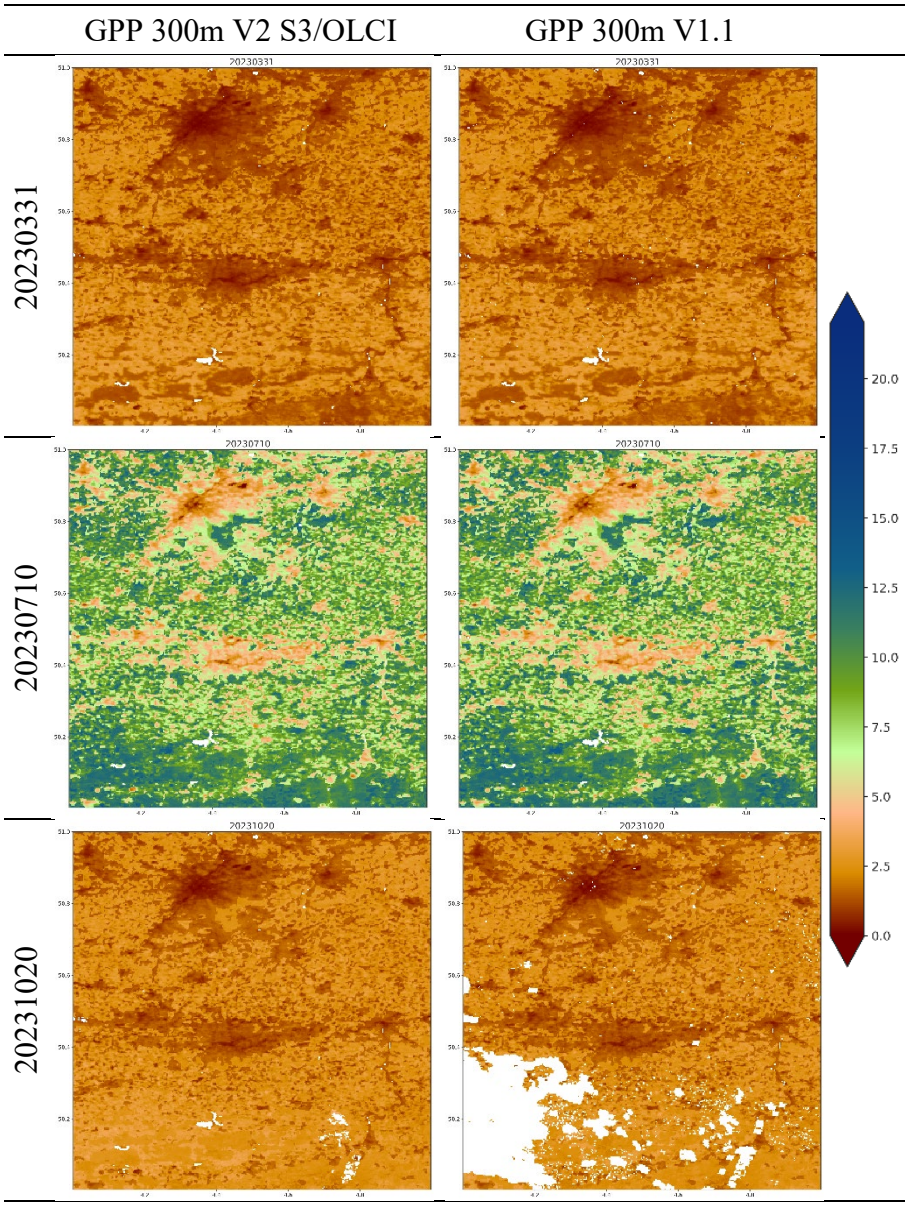


Figure 18: Visual comparison of GPP (gC/m^2/day) for the Brussels 1° x 1° mini tile for the year 2023.

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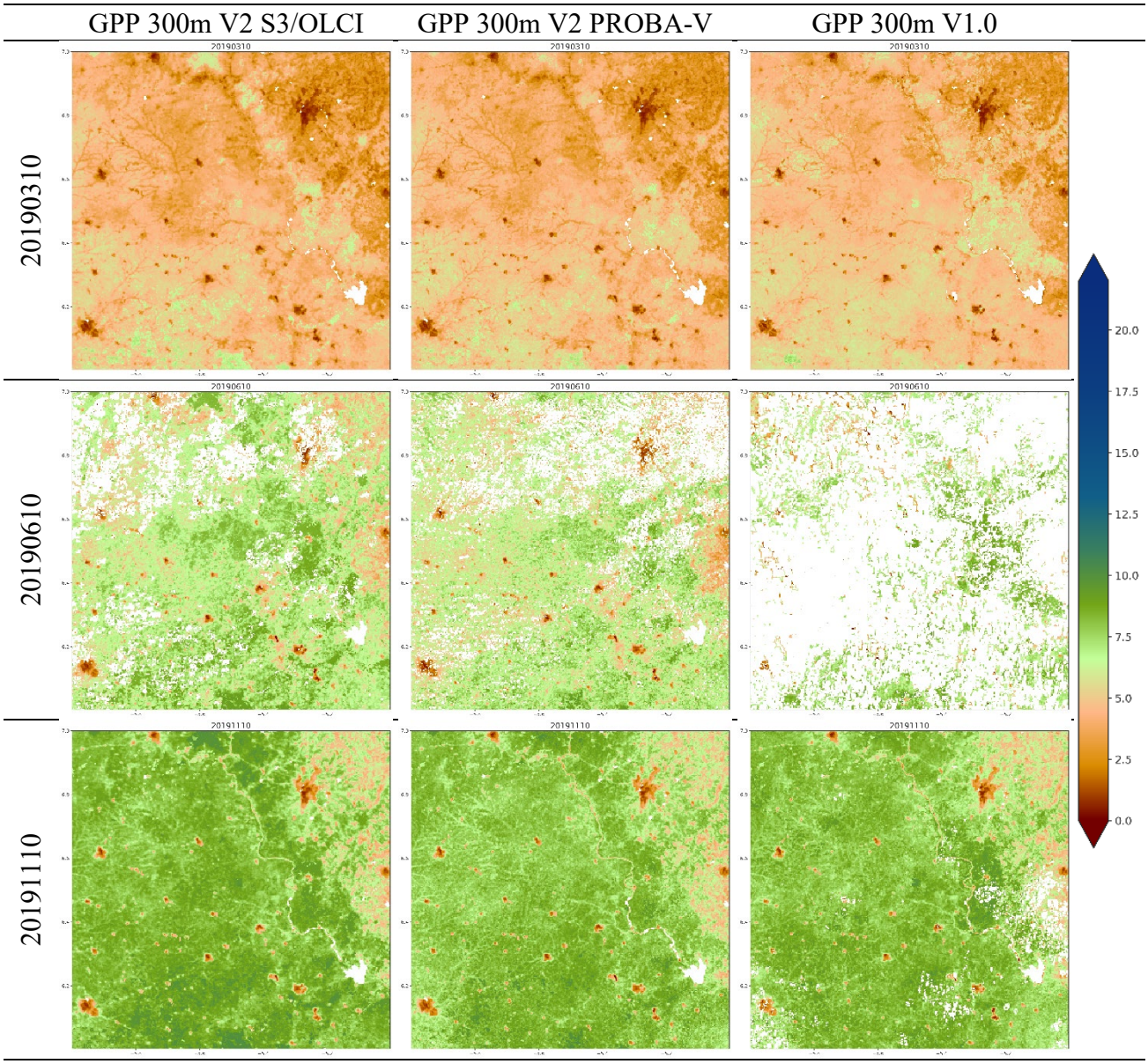

Figure 19: Visual comparison of GPP (gC/m^2/day) for the Ivory Coast 1° x 1° mini tile for the year 2019.

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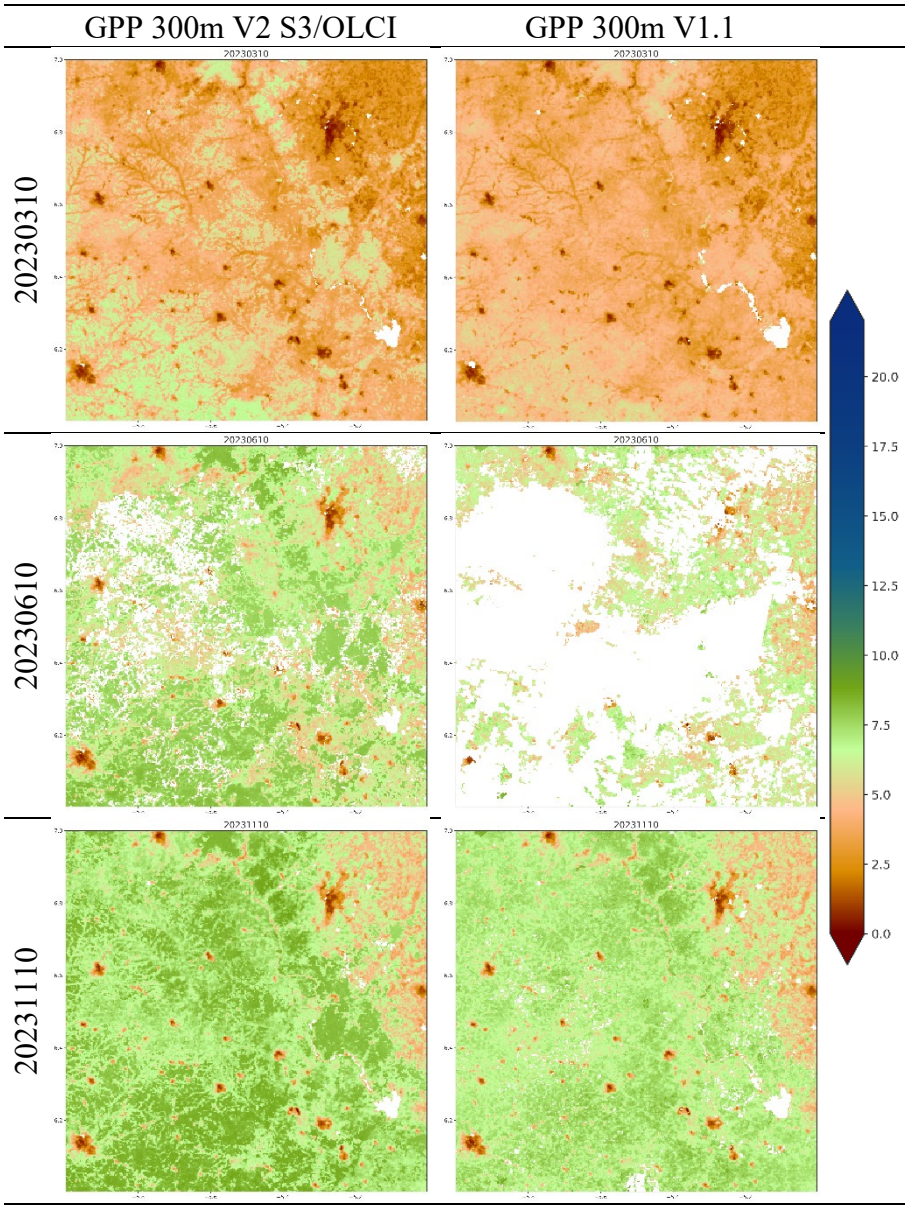

Figure 20: Visual comparison of GPP (gC/m^2/day) for the Ivory Coast 1° x 1° mini tile for the year 2023.

6.4.3 Visual comparison at 10° x 10°

Full resolution quicklooks of 10° x 10° are visually inspected to determine possible differences (i.e. spatial patterns, differences in completeness, …) between GPP 300m V2 and the external reference MOD17 V6.1 GPP. GPP 300m V1.1 quicklooks are shown as well to compare with the previous version of this CLMS product. Figure 21, Figure 22, Figure 23, Figure 24, and Figure 25 contain the visual comparison of valid GPP values for a tile in Northern America, Western Europe, South America, the Sahel, and West Africa, respectively.

There are no artefacts or major irregularities visible in GPP 300m V2. The spatial patterns of GPP are similar for the three datasets.

As mentioned before, the completeness increased from GPP 300m V1.1 to GPP 300m V2, with less gaps present in the latter, thanks to differences in input TOC versions (V1 is updated to V2.3 for GPP 300m V2). The improvement of snow classification is clearly visible over the Alps in March in Figure 22 and over the Andes Mountain range in Figure 23.

When comparing GPP 300m V2 to the MOD17 V6.1 dataset, the first major difference that can be noticed is the offset in GPP values between both datasets. Especially for the Western Europe and South America tile, much lower MOD17 V6.1 values are visible than for GPP 300m V2.The reason is that the MOD17 V6.1 dataset includes a water/drought-stress factor, which reduces GPP values and has a higher impact when FAPAR values are higher Related to the stringent flagging of gap filled data (see §5.2.2.4) MOD17 V6.1 shows a larger frequency of gaps, most clearly visible over the – frequently cloud covered – Tropics (South America, Figure 23, and West Africa, Figure 25). Since the MOD17 V6.1 GPP is not determined over the Sahara, also gaps are large in the Sahel tile (Figure 24).

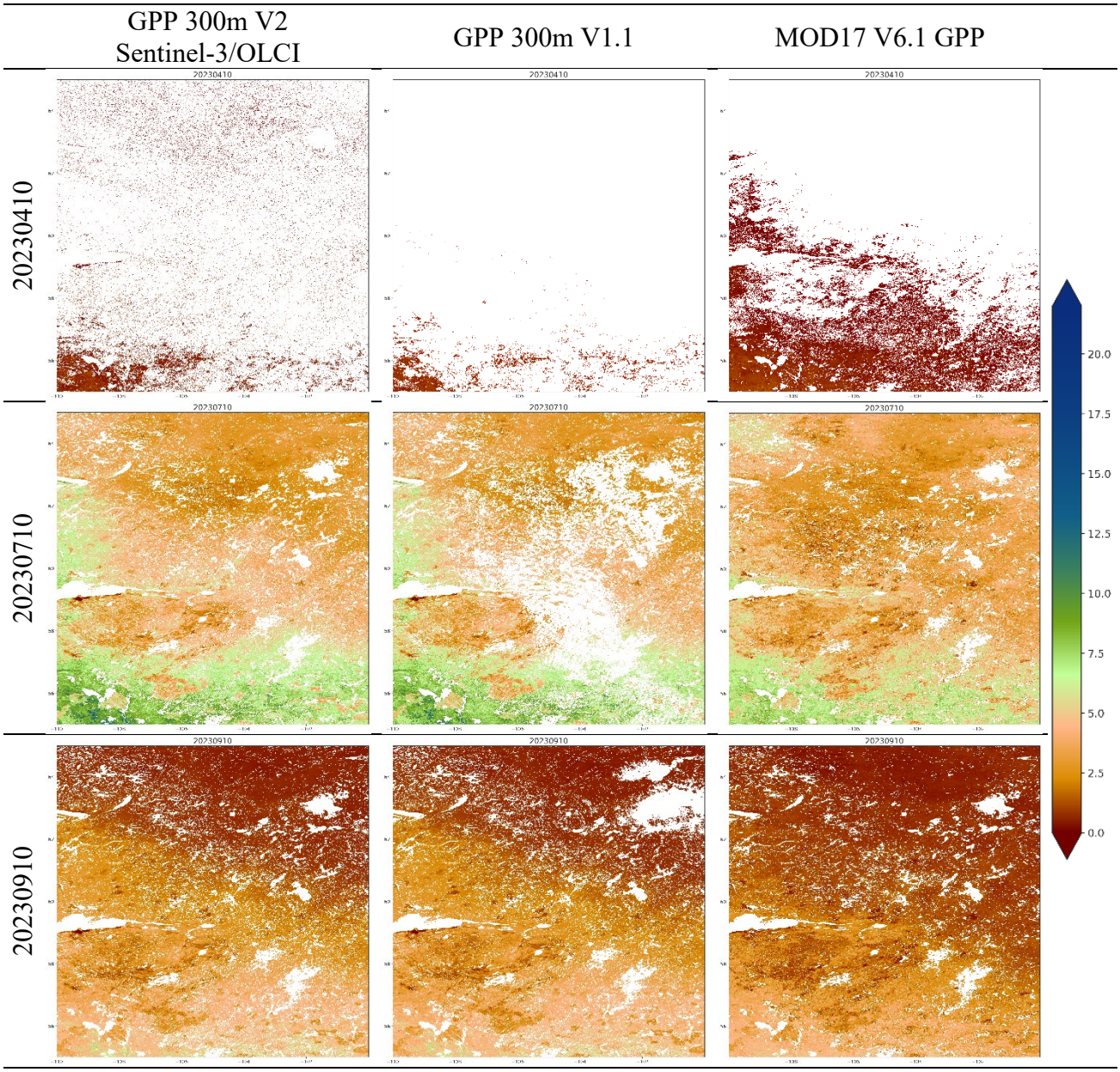

Figure 21: Visual comparison of GPP (gC/m^2/day) for the Northern America 10° x 10° tile for the year 2023.

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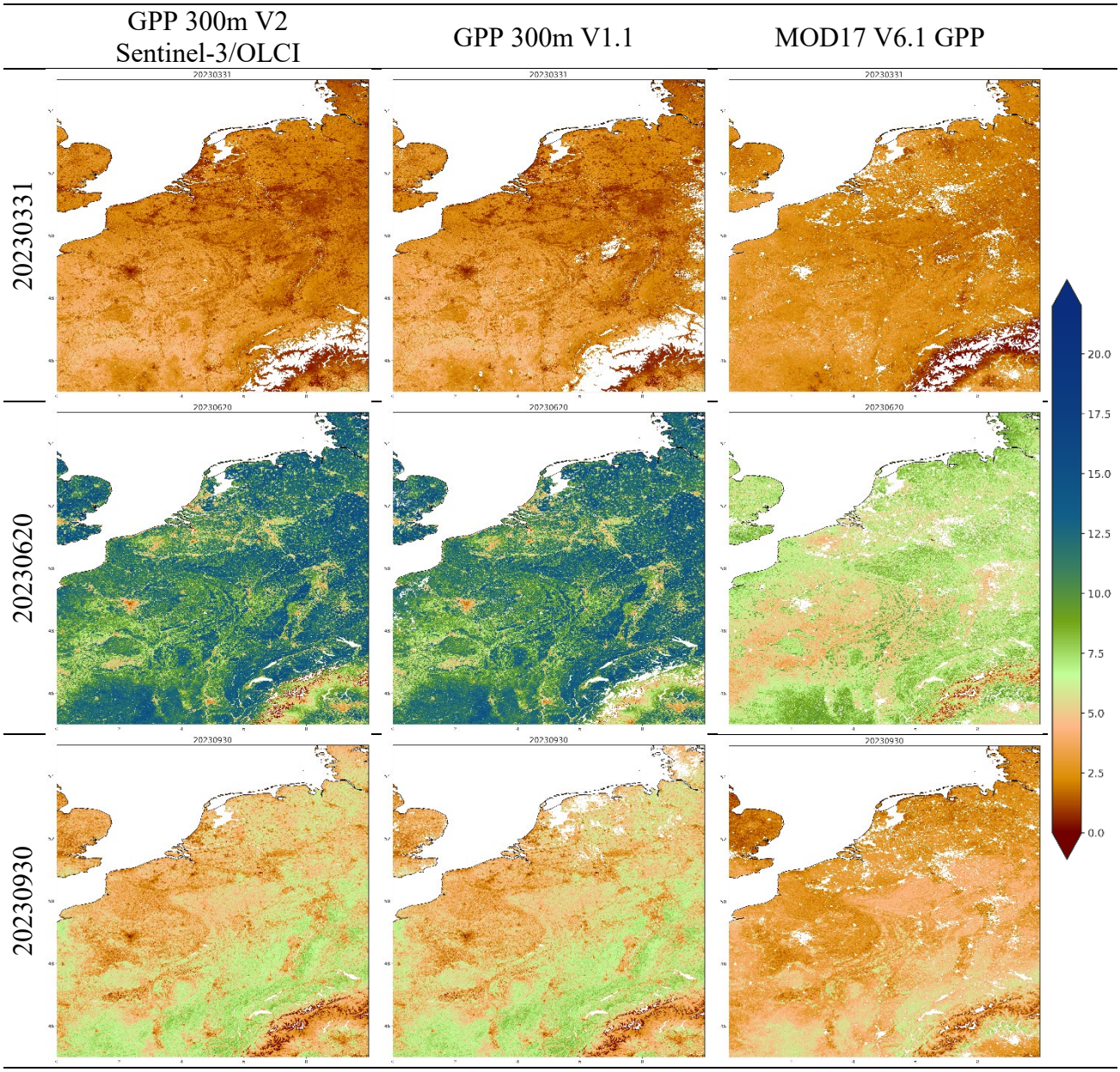

Figure 22: Visual comparison of GPP (gC/m^2/day) for the Western Europe 10° x 10° tile for the year 2023.

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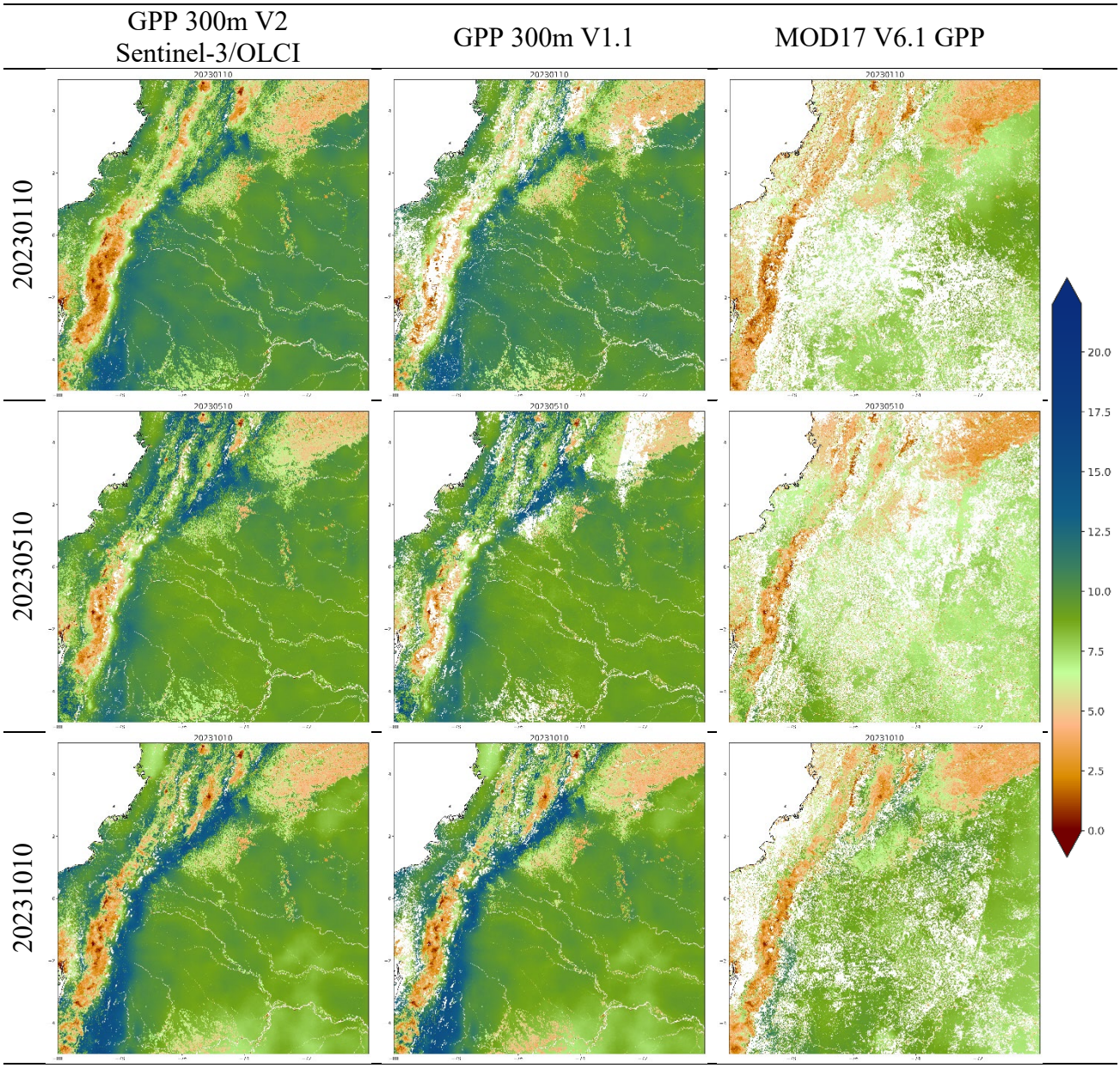

Figure 23: Visual comparison of GPP (gC/m^2/day) for the South America 10° x 10° tile for the year 2023.

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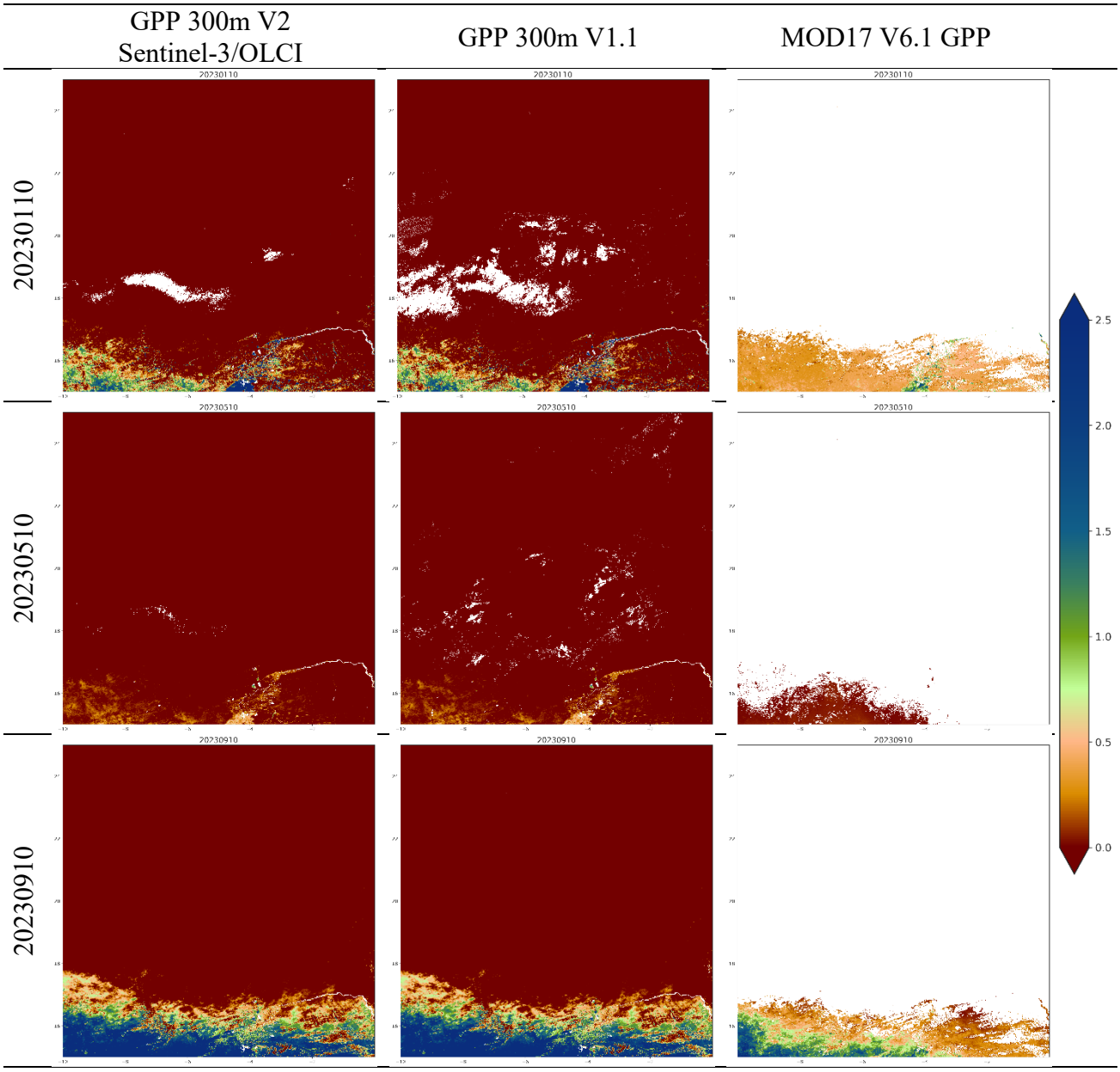

Figure 24: Visual comparison of GPP (gC/m²/day) for the Sahel 10° x 10° tile for the year 2023.

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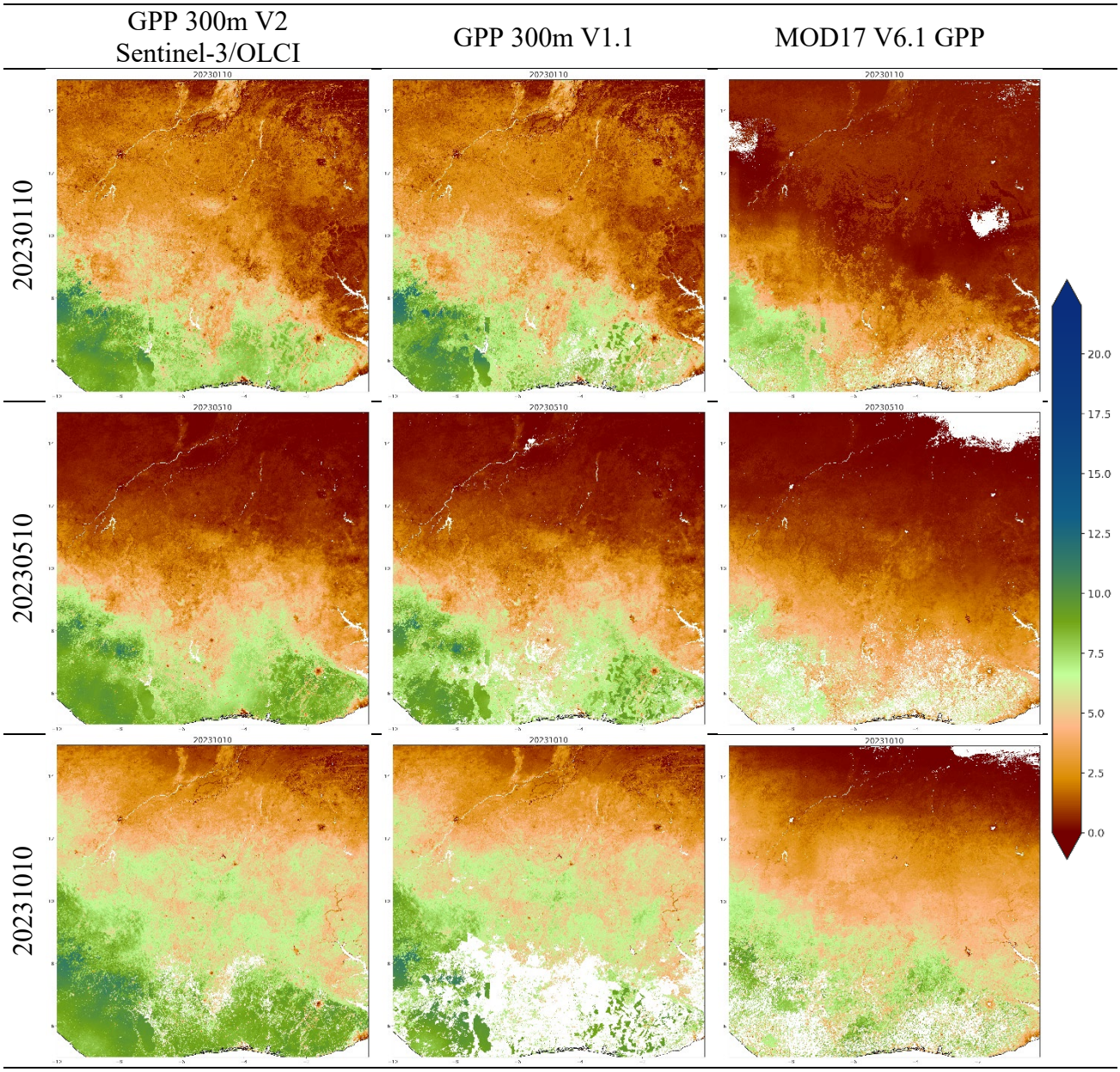


Figure 25: Visual comparison of GPP (gC/m^2/day) for the West Africa 10° x 10° tile for the year 2023.

6.4.4 Conclusion

How do the DMP/GDMP/NPP/GPP 300m V2 PROBA-V and Sentinel-3/OLCI datasets compare visually to the DMP/GDMP 300m V1.0, DMP/GDMP 300m V1.1 and MOD17 V6.1 GPP datasets?

The GPP 300m V2 quicklooks are similar in sharpness and in GPP values and patterns compared to the reference datasets. The completeness of GPP 300m V2 is higher than that of the previous version V1.0 and V1.1, with differences caused by differences in FAPAR version and hence also in input

TOC reflectances. Similar patterns are observed in MOD17 V6.1, but GPP values are on average lower than for GPP 300m V2. The reason is that the MOD17 V6.1 dataset includes a drought-stress related factor which as a lowering effect on GPP values, most effective during growing seasons.

6.5 Q5: STATISTICAL AND SPATIAL CONSISTENCY BETWEEN PROBA-V AND SENTINEL-3/OLCI FOR DMP/GDMP/NPP/GPP 300M V2

6.5.1 Introduction

The scientific question to be answered is the following: what is the statistical and spatial consistency between PROBA-V and Sentinel-3/OLCI based DMP/GDMP/NPP/GPP 300m V2?

This is an important question to answer, because it will demonstrate the consistency of the DMP/GDMP/NPP/GPP 300m V2 product over the whole time series, including the period 2014 – 2018 based on PROBA-V data and the period 2019 – 2024 (and future NRT observations) based on Sentinel-3/OLCI.

The analyses are performed on GPP 300m V2 over a one-year period, i.e. 2019 the year for which PROBA-V and Sentinel-3/OLCI observations overlap. For PROBA-V, a test dataset over the year 2019 is only available for the quality assessment and it is not included in the public product (see Section 4). In the disseminated product, the switch between PROBA-V and Sentinel-3 is made at the start of 2019. The datasets used in these analyses are based on a LANDVAL V2 extractions for the statistical consistency analysis (see §5.3.5) and on global systematic subsample created with 300m resolution pixels for the spatial consistency analysis (see §5.3.3).

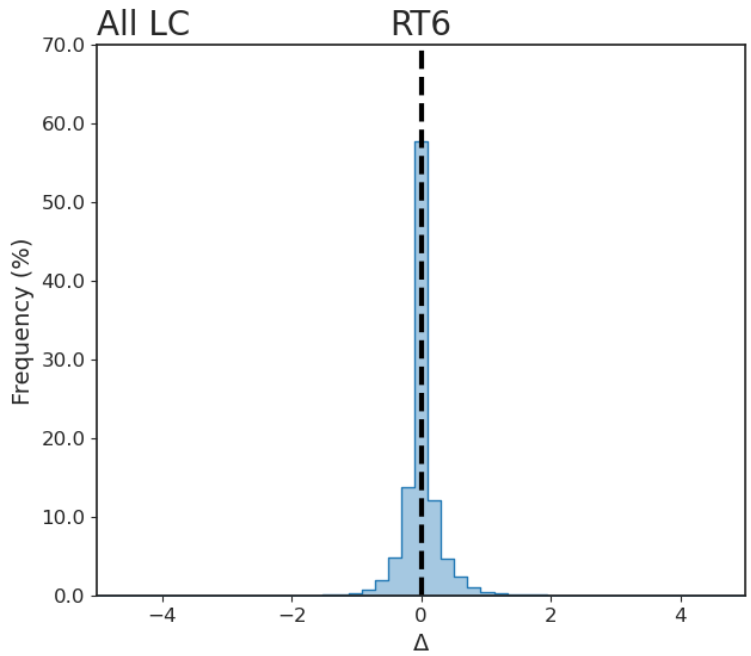
The analyses include:

- Statistical consistency analysis:
  - Histograms of overall bias
  - GM regression analysis, overall and per biome
- Spatial consistency analysis:
  - Histograms per biome
  - Spatial distribution of validation metrics

6.5.2 Statistical consistency

6.5.2.1 Overall bias

Figure 26 shows the frequency histogram of the overall bias between GPP 300m V2 Sentinel-3/OLCI and GPP 300m V2 PROBA-V. The bias histogram is symmetric with a peak at zero which indicates that the Sentinel-3/OLCI and PROBA-V values are almost identical.



Mean bias: 0.009 gC/m^2/day

Median bias: 0.000 gC/m^2/day

Figure 26: Frequency histogram of the overall bias between GPP 300m V2 Sentinel-3/OLCI and GPP 300m V2 PROBA-V for LANDVAL V2 sites.

6.5.2.2 GM regression

The consistency between GPP 300m V2 Sentinel-3/OLCI and GPP 300m V2 PROBA-V is also investigated based on a GM regression analysis as shown in Figure 27. This figure shows scatter density plots per biome and for all land cover types together (lower right plot). Statistics related to the GM regression over all land cover types are given in Table 11. The figures indicate that there is a linear relationship between both datasets with regression line close to the 1:1 line. The slope for all land cover types combined is equal to 0.99, meaning that GPP 300m V2 Sentinel-3/OLCI values are slightly higher than GPP 300m V2 PROBA-V. When looking at the biome-specific scatter plots, all biome types show good agreement between the PROBA-V and Sentinel-3/OLCI datasets, with the largest discrepancies for the EBF biome. This is a direct result of the differences in the FAPAR 300m V2 input data used to compute GPP, for which a slight discrepancy between PROBA-V and Sentinel-3/OLCI is also observed (see [CGLOPS1_VR_FAPAR300m-V2]).

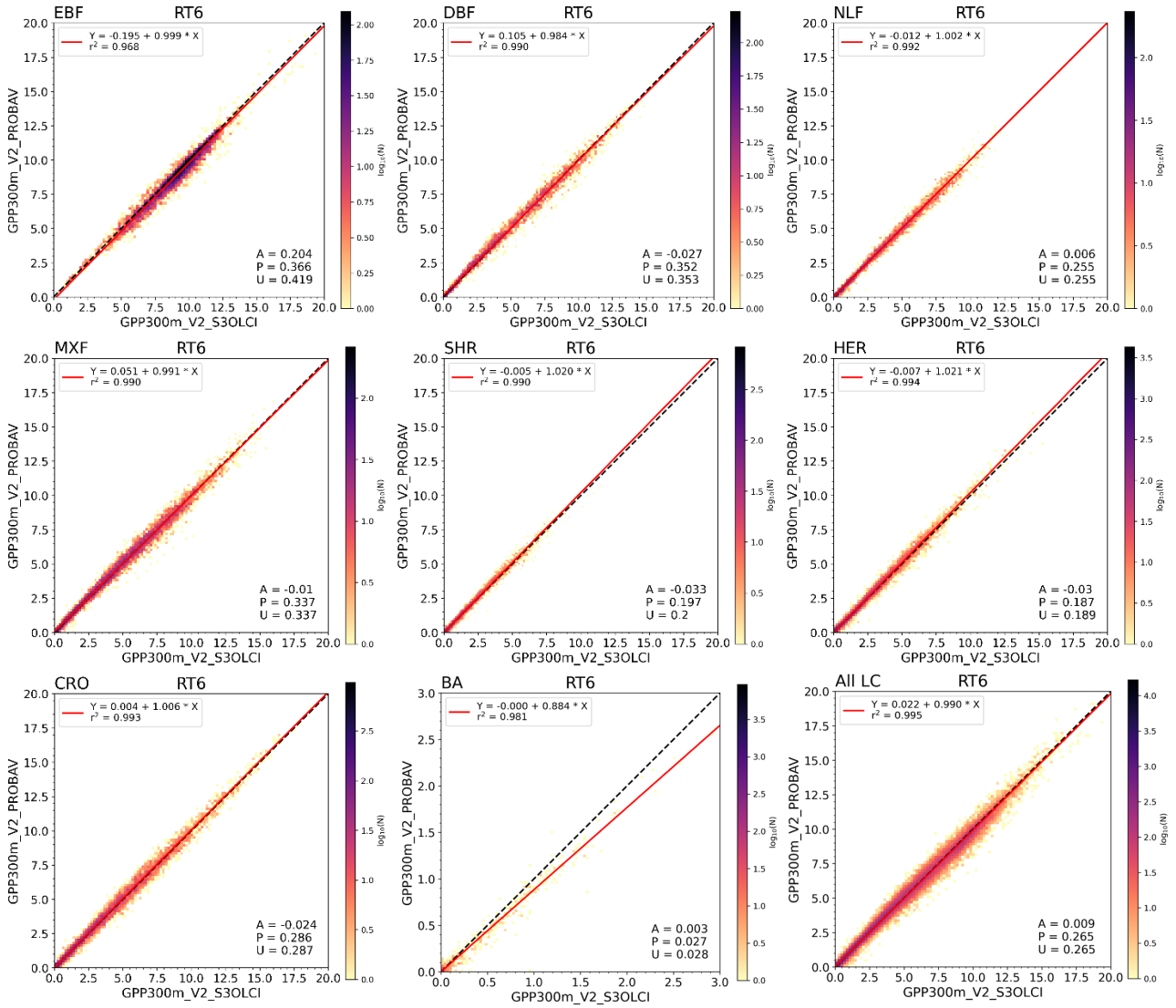


Figure 27: Scatter density plots between GPP 300m V2 Sentinel-3/OLCI (X) and GPP 300m V2 PROBA-V (Y) over LANDVAL V2 sites for the year 2019, per biome and for all land cover types.

Table 11: Statistics of comparison between GPP 300m V2 Sentinel-3/OLCI and GPP 300m V2 PROBA-V over LANDVAL V2 sites for the year 2019. The absolute values of the statistics are given in gC/m²/day.

N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset / slope
57449	0.99	0.009 (0.3%)	0.0 (0%)	0.3 (8%)	0.07 (2%)	0.3 (8%)	0.02 / 0.99

6.5.3 Spatial consistency

6.5.3.1 Histograms per biome

In Figure 28, the GPP frequency distributions of GPP 300m V2 Sentinel-3/OLCI and GPP 300m V2 PROBA-V are compared over different biomes. The frequency distributions are almost identical and only very small changes in distribution are observed. For biomes for which the distribution differ, the Sentinel-3/OLCI values are slightly higher than those of PROBA-V, similar as was found for the scatter plots.

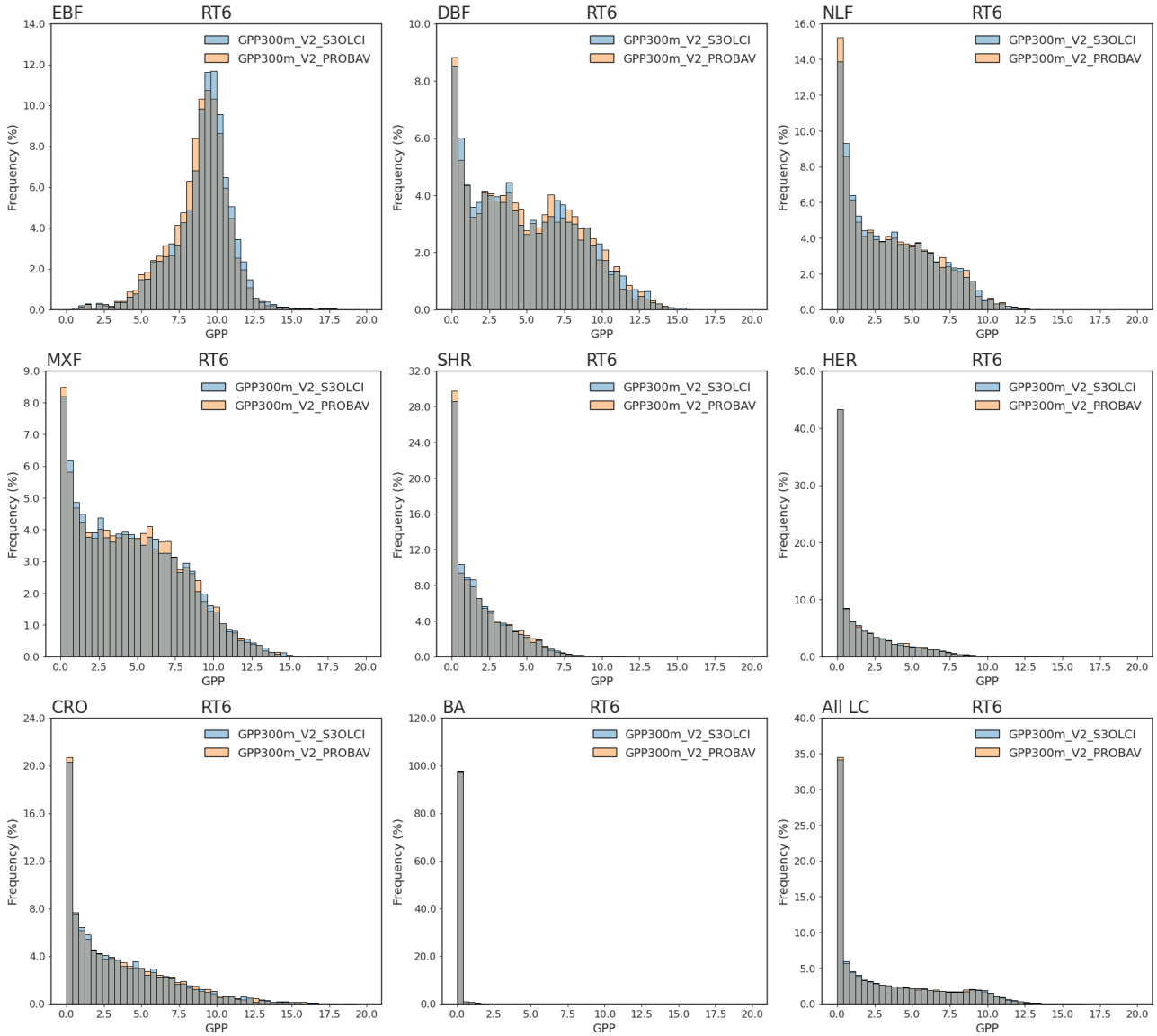


Figure 28: Frequency histograms per biome and for all land cover types comparing GPP 300m V2 Sentinel-3/OLCI (blue) and GPP 300m V2 PROBA-V (orange). Grey colour indicates where both histograms overlap.

6.5.3.2 Spatial distribution of validation metrics

The spatial distributions of accuracy, precision, and uncertainty between GPP 300m V2 Sentinel-3/OLCI and GPP 300m V2 PROBA-V are shown in Figure 29. The spatial patterns are strongly linked to vegetation densities, with higher bias and uncertainty in areas with dense vegetation (e.g. Amazon, central Africa, South-East Asia, etc.). These are regions with EBF biome type for which slightly higher Sentinel-3/OLCI values are observed with respect to PROBA-V (see [CGLOPS1_VR_FAPAR300m-V2]). The difference between both datasets is very small over deserts and other bare areas, e.g. the Sahara, the Arabian Peninsula, and Australia.

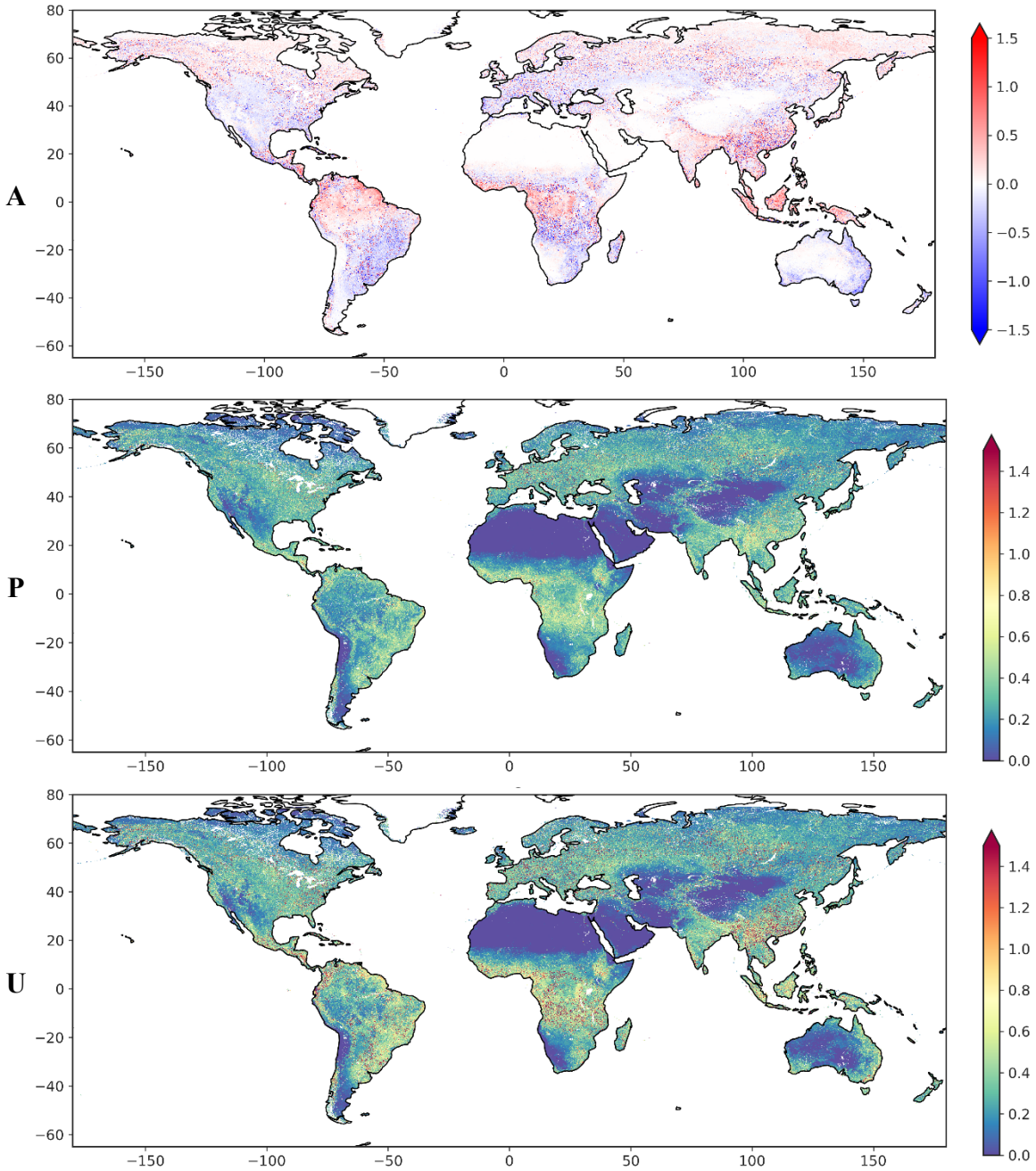


Figure 29: Spatial patterns of pixel-wise APU between GPP 300m V2 Sentinel-3/OLCI and GPP 300m V2 PROBA-V for all pairwise valid GPP values for the year 2019.

6.5.4 Conclusion

What is the statistical and spatial consistency between PROBA-V and Sentinel-3/OLCI based DMP/GDMP/NPP/GPP 300m V2?

There is a very small positive bias (mean bias = 0.009 gC/m^2/day, median bias = 0.000 gC/m^2/day) between Sentinel-3/OLCI and PROBA-V for GPP 300m V2, with Sentinel-3/OLCI GPP values being slightly higher than PROBA-V values.

Overall, there is good consistency between the two datasets, considering that they are derived from different sensors with different spectral characteristics and different ground resolution. This indicates that the DMP/GDMP/NPP/GPP 300m V2 product is consistent over the whole period 2014 - NRT. This is investigated further in §6.10.

6.6 Q6: STATISTICAL AND SPATIAL CONSISTENCY BETWEEN DMP/GDMP/NPP/GPP 300M V2, DMP/GDMP 300M V1.1, AND DMP/GDMP 300M V1.0

6.6.1 Introduction

The scientific question to be answered is the following: what are the differences between DMP/GDMP/NPP/GPP 300m V2, DMP/GDMP 300m V1.0, and DMP/GDMP 300m V1.1 in terms of statistical and spatial consistency?

The analyses are performed over two one-year periods (2019 for the comparison with DMP/GDMP 300m V1.0, and 2023 for the comparison with DMP/GDMP 300m V1.1) and are based on LANDVAL V2 extractions for the statistical consistency (see §5.3.5) and on pairwise comparison of valid GPP values for a global systematic subsample created with 300m resolution pixels (see §5.3.3) for the spatial consistency analysis. Since GPP is not available for the DMP/GDMP 300m V1.1 and DMP/GDMP 1km V2 products, for intercomparison it was calculated based on the GDMP using the methodology as described in Chapter 3.

The analyses include:

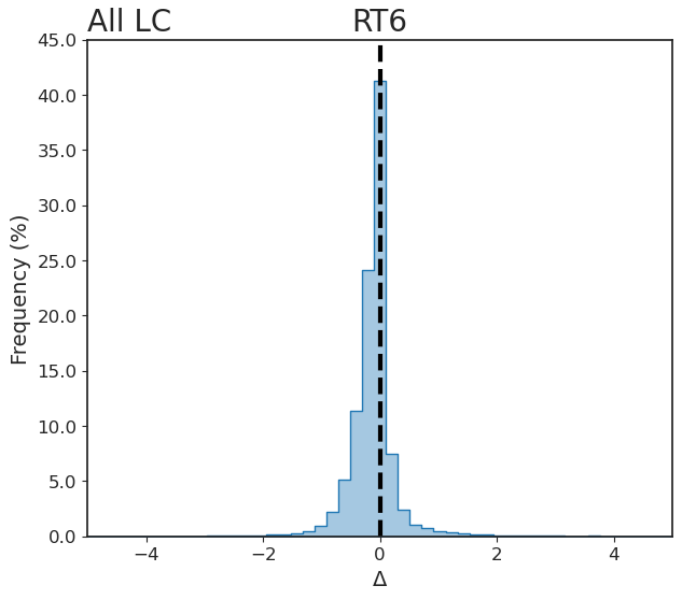
- Statistical consistency analysis:
  - Histograms of overall bias
  - GM regression analysis, overall and per biome
- Spatial consistency analysis:
  - Histograms per biome
  - Spatial distribution of validation metrics

6.6.2 Comparison of GPP 300m V2 with GPP 300m V1.0

6.6.2.1 Statistical consistency

6.6.2.1.1 Overall bias

The frequency histogram of the overall bias between GPP 300m V2 PROBA-V and GPP 300m V1.0 is shown in Figure 30. The bias histogram is slightly skewed towards negative values with a peak at -0.08 gC/m^2/day. This indicates that the GPP values in V2 are a bit lower than those in V1.0. The same trend is observed in the FAPAR 300m V2 versus FAPAR 300m V1.0 analysis (see [CGLOPS1_VR_FAPAR300m-V2]).



Mean bias: -0.12 gC/m^2/day

Median bias: -0.08 gC/m^2/day

Figure 30: Frequency histogram of the overall bias between GPP 300m V2 PROBA-V and GPP 300m V1.0 for the year 2019 for LANDVAL V2 sites

6.6.2.1.2 GM regression

Scatter density plots demonstrating the statistical consistency between GPP 300m V2 PROBA-V and GPP 300m V1.0 per biome type and for all land cover types are shown in Figure 31 and Figure 32. For all biome types, except BA sites that have low GPP values, the GM regression lines almost coincide with the 1:1 line. The largest scatter is present in the EBF plot, again related to the FAPAR data on which GPP is based. Differences between FAPAR 300m V1.0 and FAPAR 300m V2 are e.g. the land cover map used to classify EBF vs non-EBF pixels, the processing of the data (e.g. only one neural network is used in V1.0, while separate neural networks are used for EBF and non-EBF pixels in V2), and the post-processing steps, along with different input collections of PROBA-V reflection data (see §5.2.2.5).

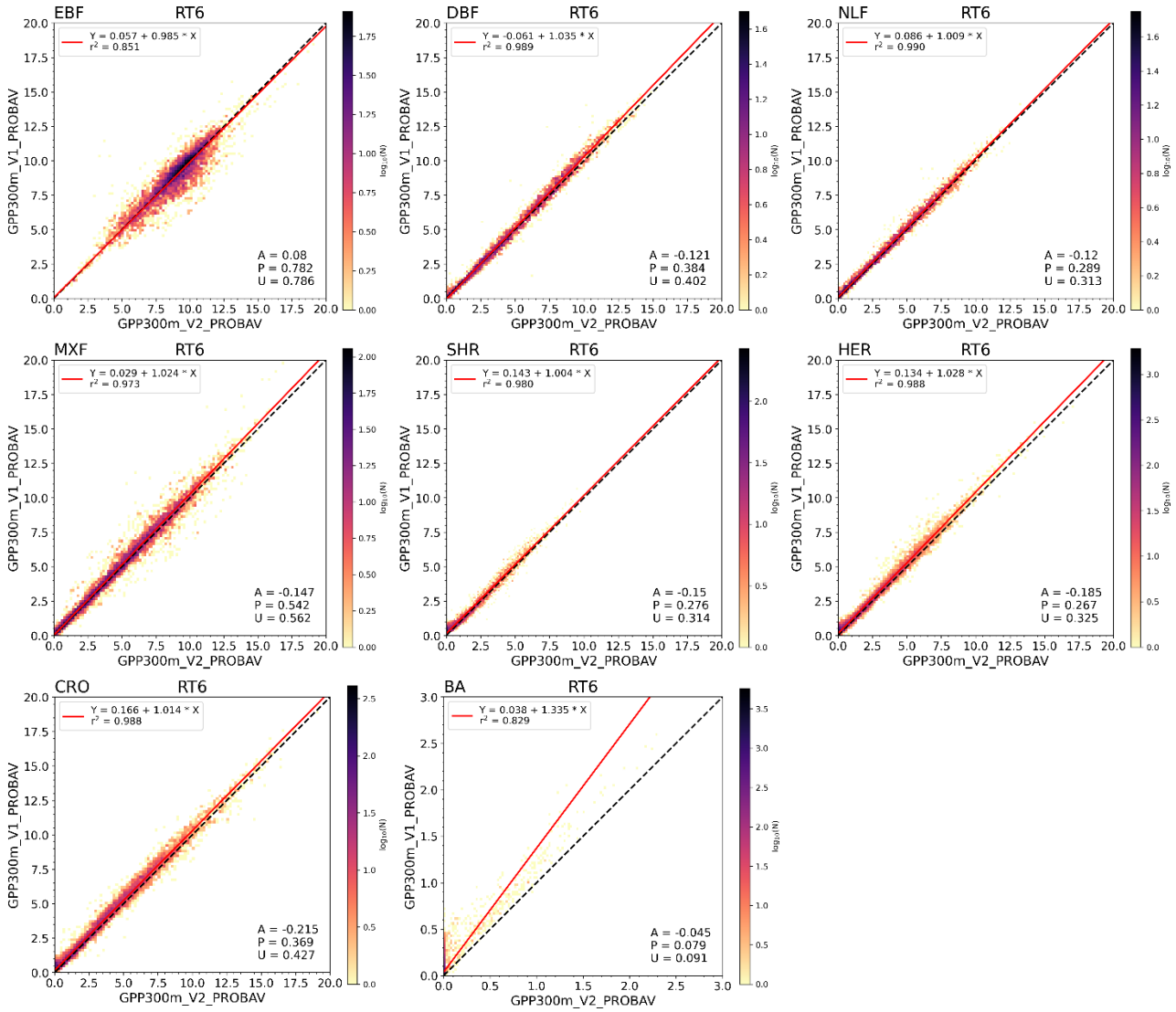


Figure 31: Scatter density plots between PROBA-V GPP 300m V2 (X) and GPP 300m V1.0 (Y) per biome type over LANDVAL V2 sites for the year 2019.

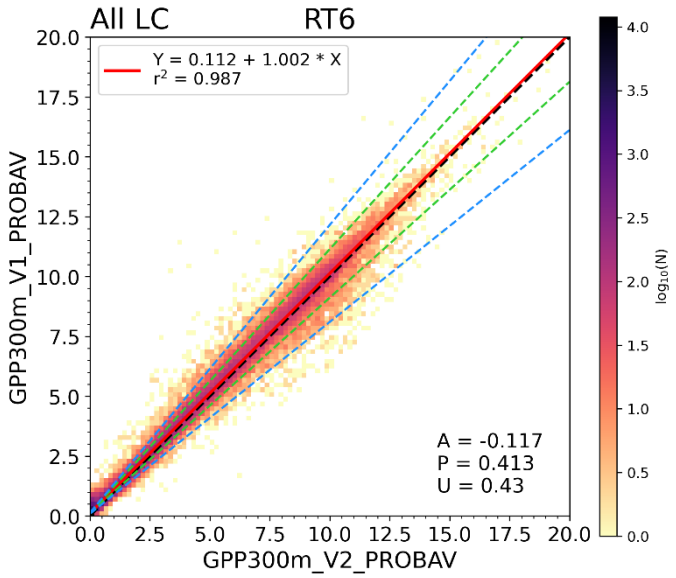


Figure 32: Scatter density plot between PROBA-V GPP 300m V2 (X) and GPP 300m V1.0 (Y) over LANDVAL V2 sites for the year 2019. Green and blue dashed lines indicate the goal and threshold uncertainty requirement.

Table 12 contains statistics related to the GM regression over all land cover types. The last column in the table are the percentages of LANDVAL V2 sites for which the uncertainty of GPP V2 compared to GPP V1.0 is within the goal and threshold limits. These are the LANDVAL V2 sites for which the GPP lies within the green and blue dashed lines of Figure 32, respectively. About half resp. two thirds of the sample shows comparison statistics within the goal and threshold range. It should be noted that the good correspondence for low GPP values (GMR equation offset close to zero) is not reflected in this statistic because of its relative character. This is most notorious for BA sites, where the very low bias is not reflected in high occurrence of observations within uncertainty requirements.

Table 12: Statistics of comparison between GPP 300m V2 PROBA-V and GPP 300m V1.0 over all LANDVAL V2 sites, all BA sites, and all sites except BA sites for the year 2019. The last column contains the percentage of pixels with uncertainty within the goal (10%) and threshold (20%) range. The absolute values of the statistics are given in gC/m²/day.

	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset / slope	% Goal / Threshold
All LC	53534	0.99	-0.12 (-3%)	-0.08 (-2%)	0.4 (12%)	0.14 (4%)	0.4 (13%)	0.11 / 1.0	52 / 67
BA	9516	0.91	-0.05 (-67%)	-0.010 (-15%)	0.08 (117%)	0.010 (15%)	0.09 (135%)	0.04 / 1.3	4 / 10
No BA	44018	0.99	-0.13 (-3%)	-0.11 (-3%)	0.5 (11%)	0.19 (5%)	0.5 (11%)	0.14 / 0.99	62 / 79

6.6.2.2 Spatial consistency

6.6.2.2.1 Histograms per biome

The GPP frequency distributions of GPP 300m V2 PROBA-V and GPP 300m V1.0 are compared per biome type in Figure 33. The differences between distributions are minimal and only small changes are observed.

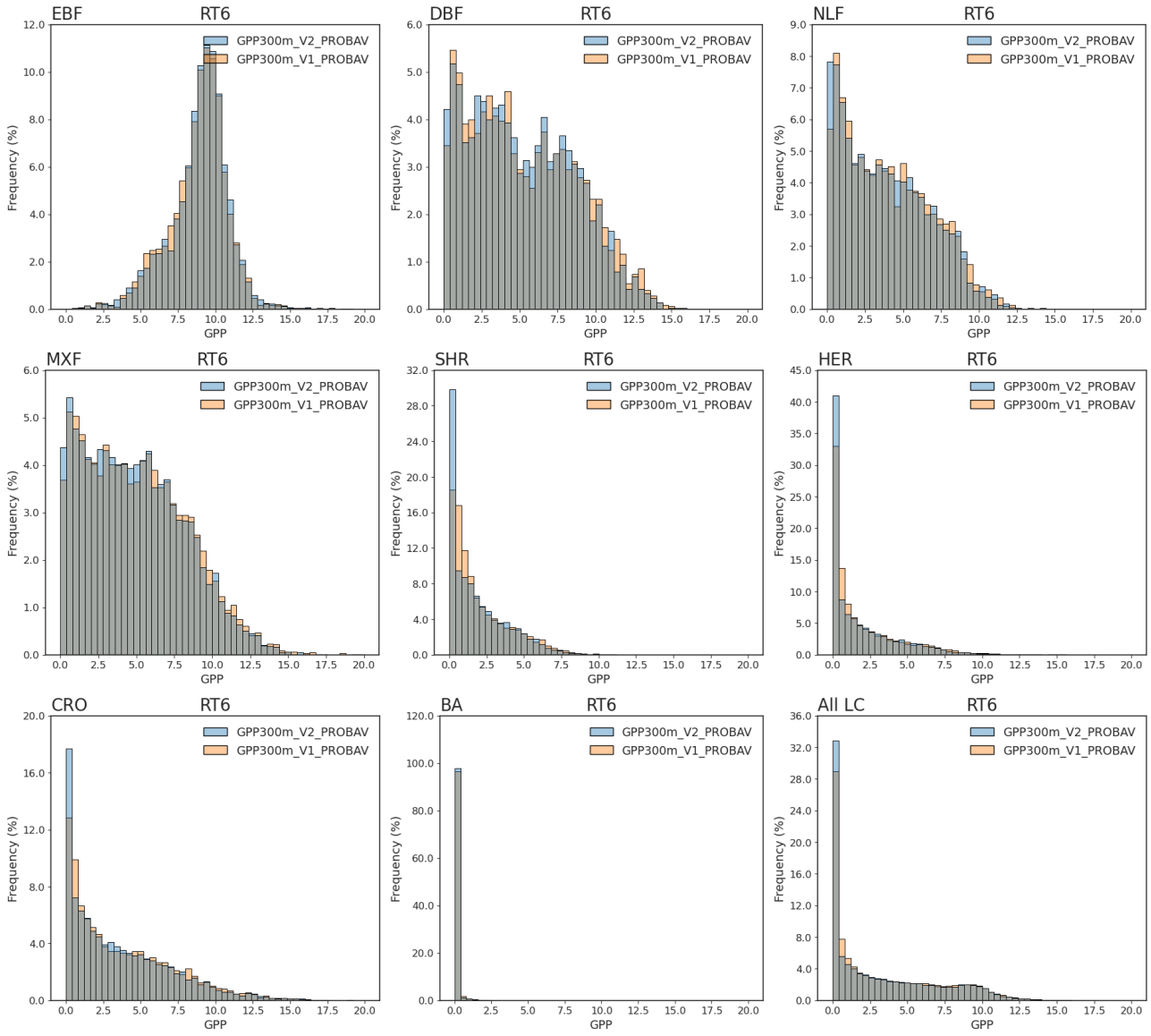
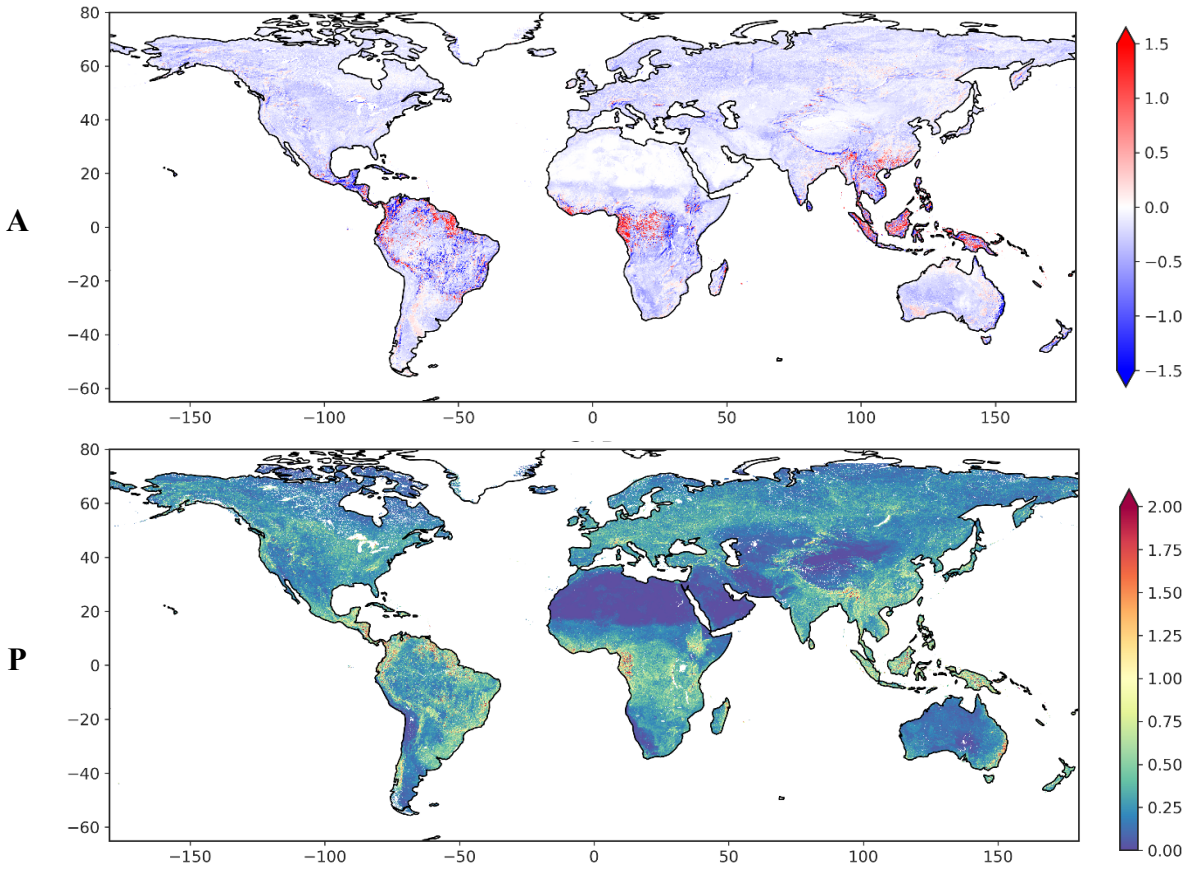


Figure 33: Frequency histograms per biome and for all land cover types comparing GPP 300m V2 PROBA-V (blue) and GPP 300m V1.0 (orange) for the year 2019. Grey colour indicates where both histograms overlap.

6.6.2.2.2 Spatial distribution of validation metrics

The spatial distribution of differences between GPP 300m V2 PROBA-V and GPP 300m V1.0, shown in Figure 34, is strongly linked with vegetation densities. The APU statistics are much higher in areas with dense vegetation such as the Amazon, central Africa, and South-East Asia, while they are close to zero over the Sahara, the Arabian Peninsula, and Australia. Largest differences can again be found in areas with EBF biome, and can hence be attributed to differences in EBF processing in the two FAPAR versions on which the GPP versions are based (see above).



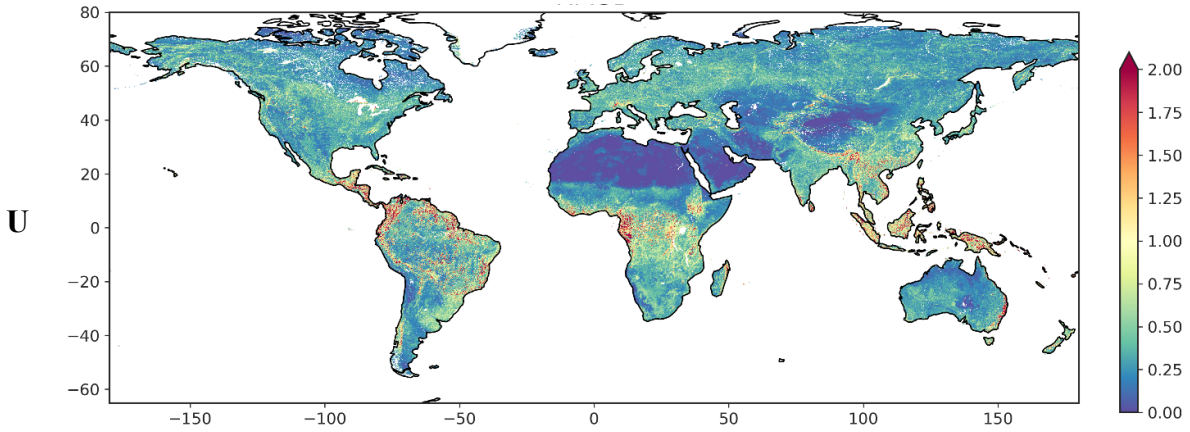


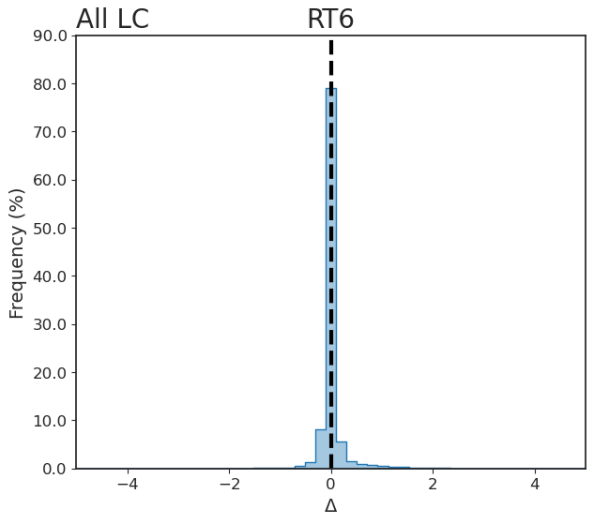
Figure 34: Spatial pattern of pixel wise APU between GPP 300m V2 PROBA-V and GPP 300m V1.0 for all pairwise valid GPP values for the year 2019.

6.6.3 Comparison of GPP 300m V2 with GPP 300m V1.1

6.6.3.1 Statistical consistency

6.6.3.1.1 Overall bias

The bias histogram for comparison between GPP 300m V2 Sentinel-3/OLCI and GPP 300m V1.1 is shown in Figure 35. It peaks around zero and is symmetrical. Hence, the values of both versions are very similar. The agreement is better than for GPP 300m V2 PROBA-V and GPP 300m V1.0 (see § 6.6.2.1.1), which is expected since there is almost no change between V1.1 and V2.



Mean bias: 0.018 gC/m^2/day

Median bias: 0.000 gC/m^2/day

Figure 35: Frequency histogram of the overall bias between GPP 300m V2 Sentinel-3/OLCI and GPP 300m V1.1 for the year 2023 for LANDVAL V2 sites

6.6.3.1.2 GM regression

Scatter plots with GM regressions for statistical comparison between GPP 300m V2 Sentinel-3/OLCI and GPP 300m V1.1 are shown in Figure 36 per biome type and in Figure 37 for all land cover types combined. The correlation between these two datasets is very good, with regression lines close to the 1:1 line. The GPP values in V2 are generally somewhat higher than in V1.1, which is mostly associated with EBF pixels. The same conclusion was found for comparison between FAPAR 300m V2 and FAPAR 300m V1.1 (see [CGLOPS1_VR_FAPAR300m-V2]).

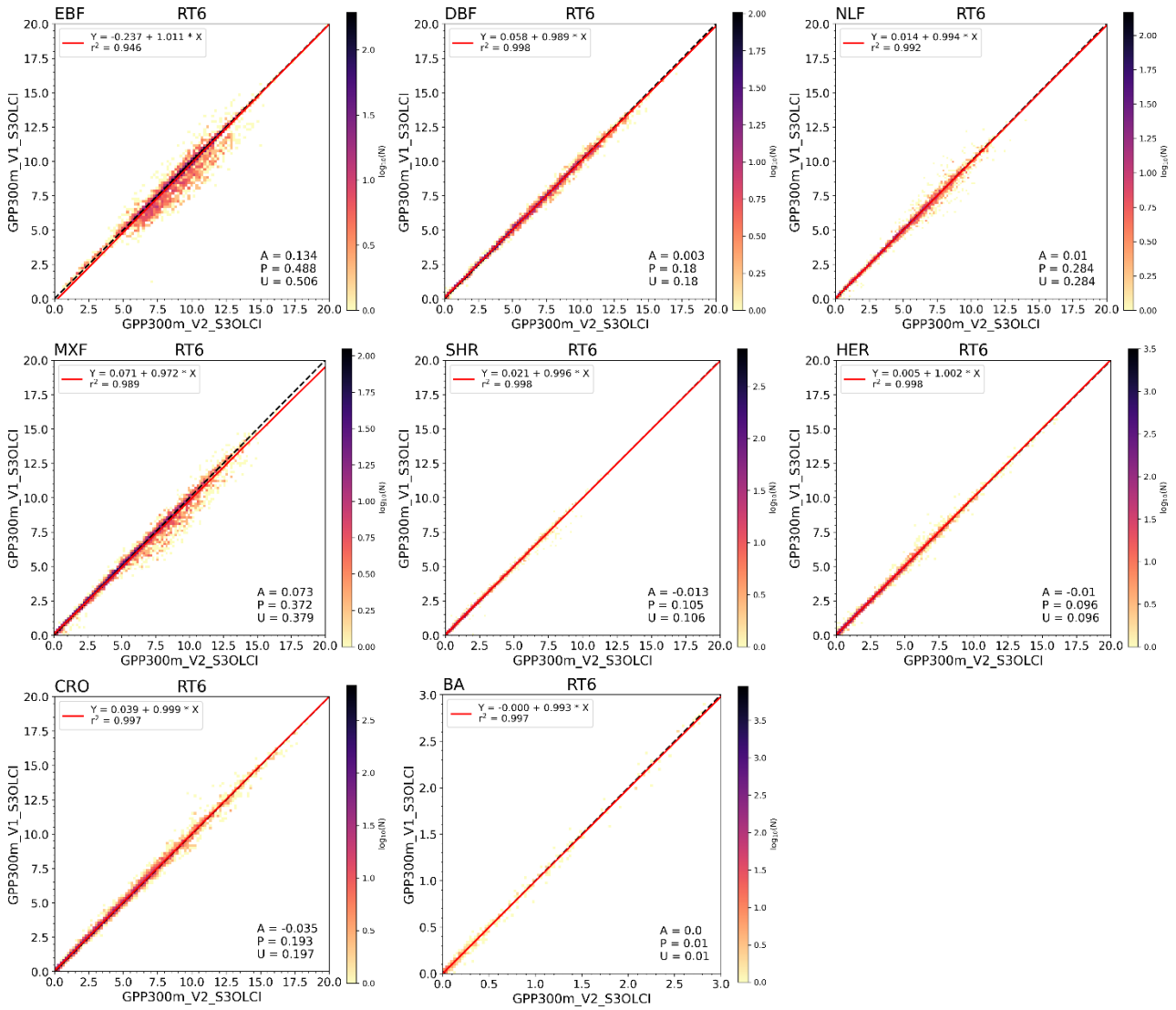


Figure 36: Scatter density plots between Sentinel-3/OLCI GPP 300m V2 (X) and GPP 300m V1.1 (Y) per biome type over LANDVAL V2 sites for the year 2023.

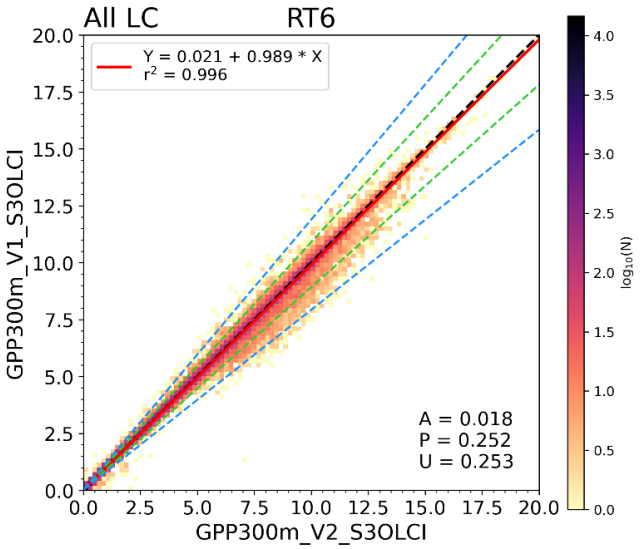


Figure 37: Scatter density plot between Sentinel-3/OLCI GPP 300m V2 (X) and GPP 300m V1.1 (Y) over LANDVAL V2 sites for the year 2023. Green and blue dashed lines indicate the goal and threshold uncertainty requirement.

Table 13 contains statistics related to the GM regression over all land cover types. The last column in the table are the percentages of LANDVAL V2 sites for which the uncertainty of GPP V2 compared to GPP V1.1 is within the goal and threshold limits. These are the LANDVAL V2 sites for which the GPP lies within the green and blue dashed lines of Figure 37, respectively. 68% resp. 73% of the sample shows comparison statistics within the goal and threshold range. It should be noted that the good correspondence for low GPP values (GMR equation offset close to zero) is not reflected in this statistic because of its relative character. This is most notorious for BA sites, where the very low bias is not reflected in high occurrence of observations within uncertainty requirements.

Table 13: Statistics of comparison between GPP 300m V2 Sentinel-3/OLCI and GPP 300m V1.1 over all LANDVAL V2 sites, all BA sites, and all sites except BA for the year 2023. The last column contains the percentage of pixels with uncertainty within the goal (10%) and threshold (20%) range. The absolute values of the statistics are given in gC/m²/day.

	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset / slope	% Goal / Threshold
All LC	52386	0.99	0.018 (0.5%)	0.0 (0%)	0.3 (7%)	0.02 (0.7%)	0.3 (7%)	0.02 / 0.99	68 / 73
BA	9513	0.99	0.0003 (1.0%)	0.0 (0%)	0.010 (38%)	0.0 (0%)	0.010 (38%)	0.0 / 0.99	3 / 4
No BA	42873	0.99	0.02 (0.5%)	-0.003 (-0.08%)	0.3 (7%)	0.04 (0.9%)	0.3 (7%)	0.3 / 0.99	81 / 88

6.6.3.2 Spatial consistency

6.6.3.2.1 Histograms per biome

The comparison of GPP frequency distributions between GPP 300m V2 Sentinel-3/OLCI and GPP 300m V1.1 is shown in Figure 38. The histograms mostly overlap and are very similar to the comparison between GPP 300m V2 PROBA-V and GPP 300m V1.0 (see Figure 33).

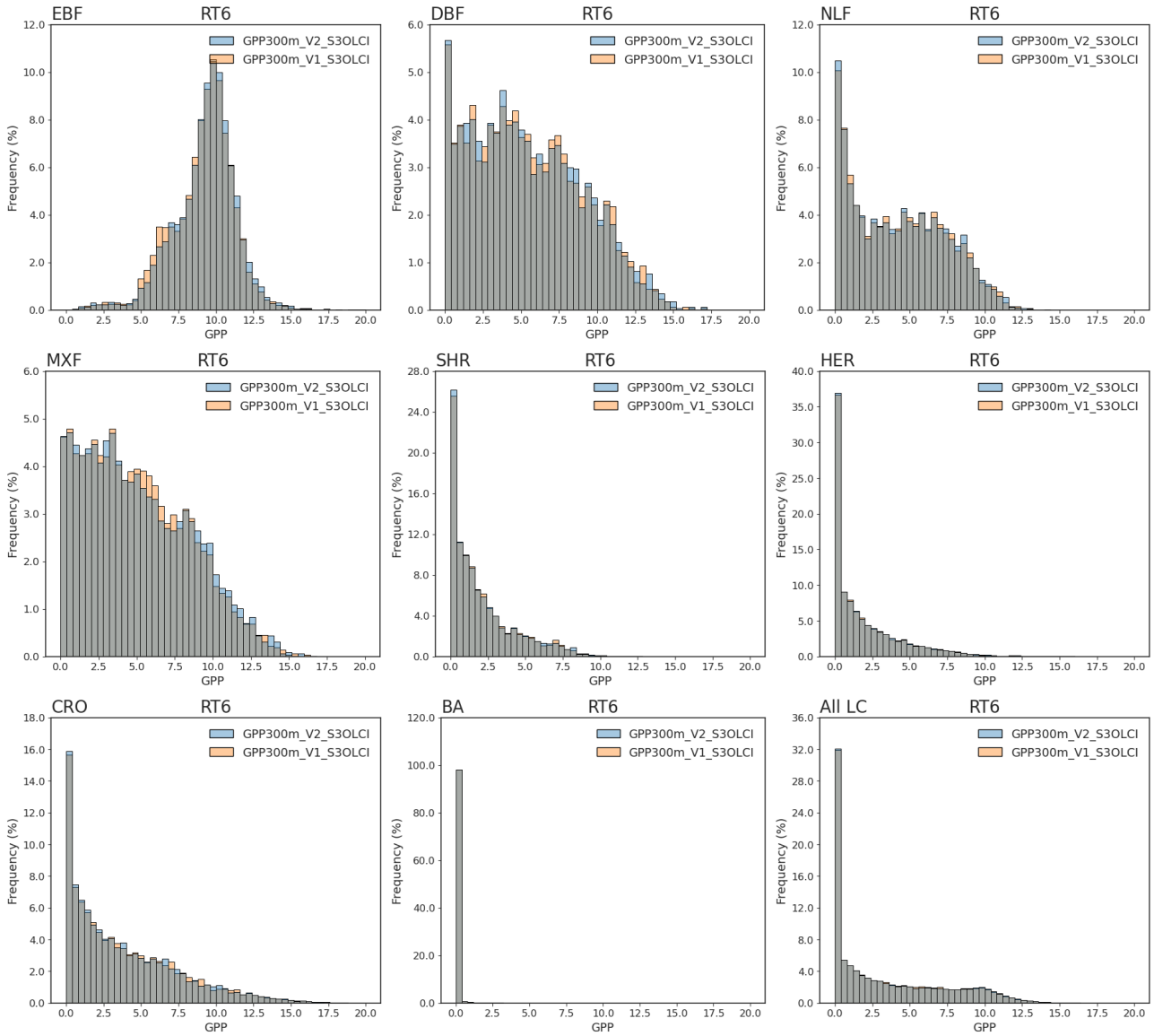
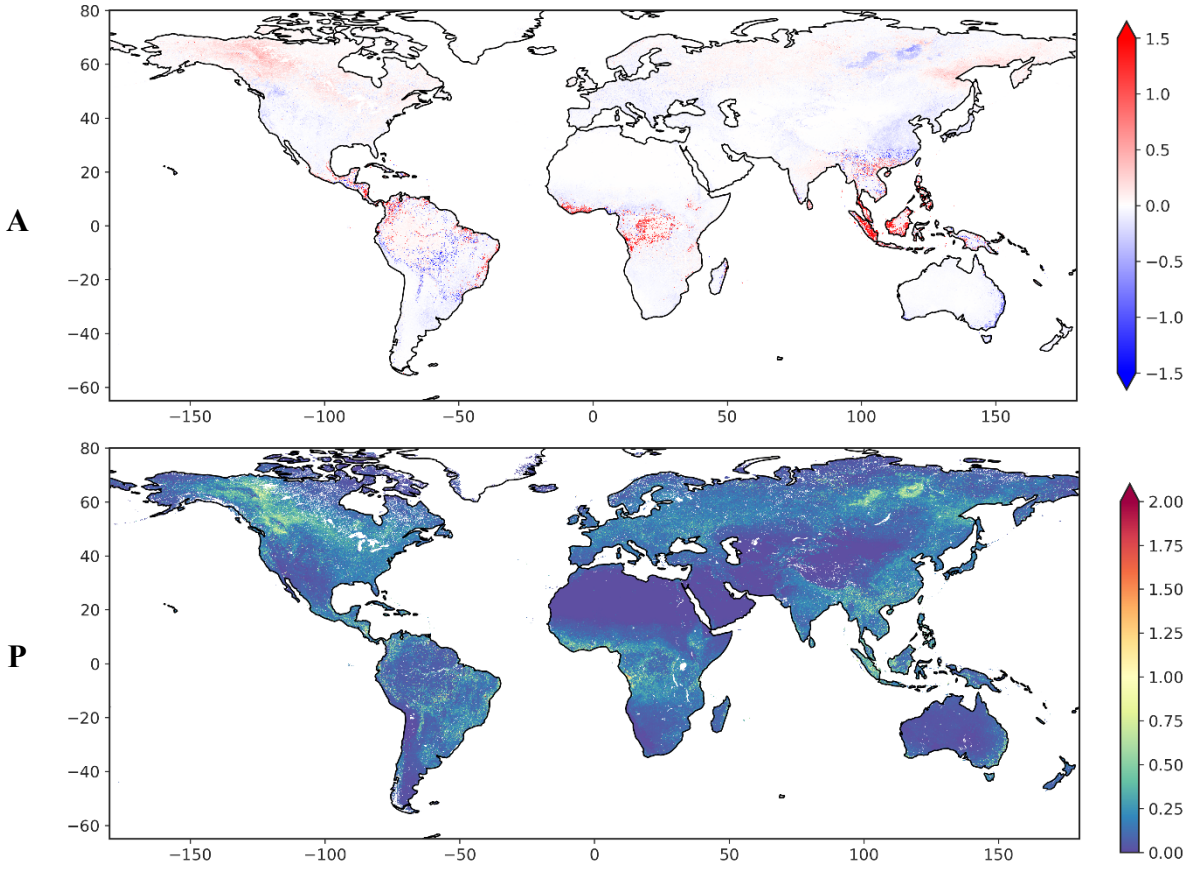


Figure 38: Frequency histograms per biome and for all land cover types comparing GPP 300m V2 Sentinel-3/OLCI (blue) and GPP 300m V1.1 (orange) for the year 2023. Grey colour indicates where both histograms overlap.

6.6.3.2.2 Spatial distribution of validation metrics

The spatial distributions of the APU statistics between GPP 300m V2 Sentinel-3/OLCI and GPP 300m V1.1 are shown in Figure 39. The difference between both datasets is very low globally, except for some tropical regions such as Indonesia and Central Africa, both regions with EBF biome type. Compared to GPP 300m V1.0, the APU statistics of GPP 300m V1.1 versus GPP 300m V2 are much lower. That is because for PROBA-V, apart from a new collection of input data and a different cloud screening method, also large changes in the FAPAR algorithm and processing chain are made in V2. On the other hand, the differences in the FAPAR algorithm for Sentinel-3/OLCI V1.1 and V2 are smaller, although also different input TOC versions are used.



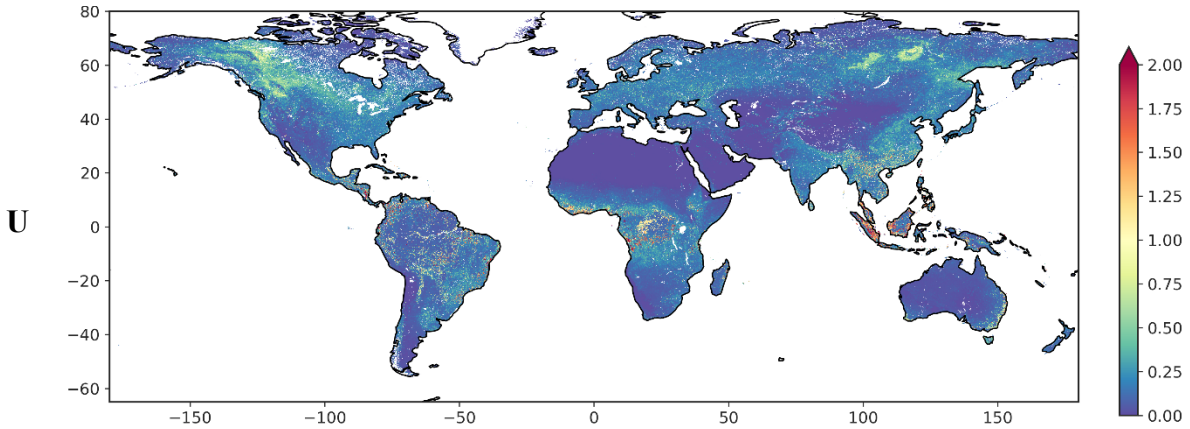


Figure 39: Spatial pattern of pixel wise APU between GPP 300m V2 Sentinel-3/OLCI and GPP 300m V1.1 for all pairwise valid GPP values for the year 2023

6.6.4 Conclusion

What is the difference between GPP 300m V2 and GPP 300m V1.0 in terms of statistical and spatial consistency?

There is a small negative bias between GPP 300m V2 and GPP 300m V1.0 (PROBA-V) with mean bias = -0.12 gC/m^2/day and median bias = -0.08 gC/m^2/day. The bias histogram is skewed towards negative values, indicating that in general GPP values in V2 are smaller than GPP values in V1.0. The differences are largest for the EBF biome type which is a result of different processing of EBF pixels in FAPAR 300m versions V1.0 and V2 (see also [CGLOPS1_VR_FAPAR300m-V2]).

What is the difference between GPP 300m V2 and GPP 300m V1.1 in terms of statistical and spatial consistency?

The mean bias = 0.018 gC/m^2/day and median bias = 0.0 gC/m^2/day between GPP 300m V2 and GPP 300m V1.1 indicate that the consistency between these two datasets is much better than for comparison with GPP 300m V1.0. This is expected, as the FAPAR processing and post-processing steps of V1.1 and V2 are almost identical, and the only major difference is the change in Sentinel-3 TOC input data from V1 to V2.3. The largest differences are found for the EBF biome, which is again caused by the different EBF identification between V1.1 and V2 and the separate processing of this biome type.

6.7 Q7: Statistical and spatial consistency between DMP/GDMP/NPP/GPP 300m V2 and DMP/GDMP 1km V2

6.7.1 Introduction

The scientific question to be answered is the following: what is the statistical and spatial consistency with DMP/GDMP 1km V2?

The answer to this question is of particular interest to users that require a longer time series, i.e. including the time series of SPOT/VGT1, SPOT/VGT2 and PROBA-V at 1km resolution, e.g. for looking at long term vegetation productivity dynamics. Since GPP is not available for the DMP/GDMP 1km V2 product, for intercomparison it was calculated based on the GDMP using the methodology as described in Chapter 3.

The analyses are performed over a one-year period (2019) and are based on LANDVAL V2 extractions for the statistical consistency analysis (see §5.3.5) and on a global systematic subsample created with 1km resolution pixels (see §5.3.3) for the spatial consistency analysis.

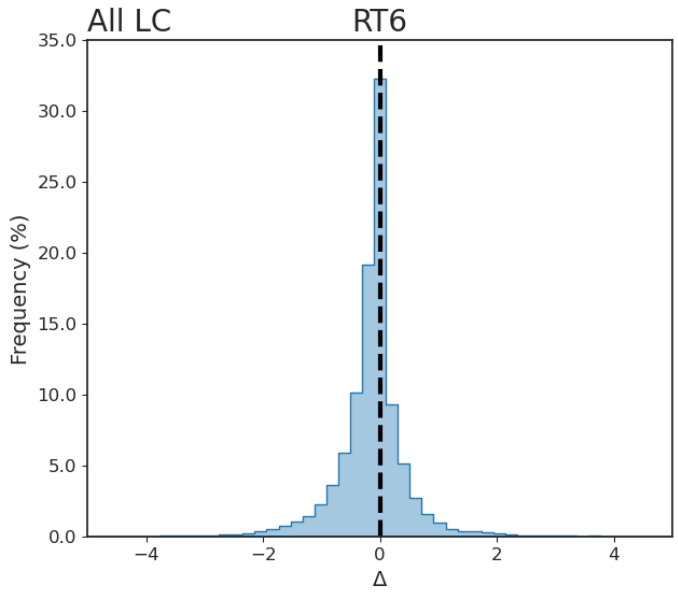
The analyses include:

- Statistical consistency analysis:
  - Histograms of overall bias
  - GM regression analysis
- Spatial consistency analysis:
  - Histograms of retrieval per biome
  - Spatial distribution of validation metrics

6.7.2 Statistical consistency

6.7.2.1 Overall bias

The frequency histogram of the overall bias between LANDVAL V2 sites of Sentinel-3/OLCI GPP 300m V2 and GPP 1km V2 is shown in Figure 40. The histogram is slightly skewed towards negative values with a median bias of -0.07 gC/m^2/day, meaning that the GPP 1km V2 values are a bit higher than the GPP 300m V2 Sentinel-3/OLCI values.



Mean bias: -0.13 gC/m^2/day

Median bias: -0.07 gC/m^2/day

Figure 40: Frequency histogram of the overall bias between GPP 300m V2 Sentinel-3/OLCI and GPP 1km V2 for LANDVAL V2 sites for the year 2019

6.7.2.2 GM regression

A scatterplot between GPP 300m V2 and GPP 1km V2 for LANDVAL V2 sites is shown in Figure 41. The GM regression line is very close to the 1:1 line, but there is an offset, and some scatter is present as well. The statistics related to this regression are given in Table 14. The last column in the table are the percentages of LANDVAL V2 sites for which the goal and threshold uncertainty requirement values are met. A bit more than half of the sample shows comparison statistics within the threshold range. It should be noted that the good correspondence for low GPP values (GMR equation offset close to zero) is not reflected in this statistic because of its relative character. This is most notorious for BA sites, where the very low bias is not reflected in high occurrence of observations within uncertainty requirements. The percentages are significantly lower than for the comparison with the 300m data in the previous sections. Possible causes are the difference in spatial footprint, where the 300m data is averaged into 1km pixels, the differences between the two instruments (PROBA-V for the 1km dataset versus Sentinel-3/OLCI for the 300m dataset in this comparison), and differences in the algorithm of the FAPAR inputs (see §5.2.2.5).

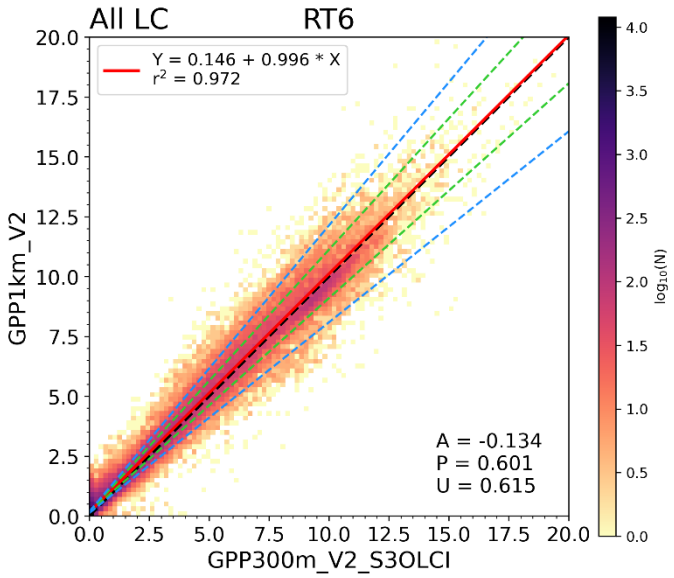


Figure 41: Scatter density plot between Sentinel-3/OLCI GPP 300m V2 (X) and GPP 1km V2 (Y) over LANDVAL V2 sites for the year 2019. Green and blue dashed lines indicate the goal and threshold uncertainty requirement.

Table 14: Statistics of comparison between GPP 300m V2 Sentinel-3/OLCI and GPP 1km V2 over all LANDVAL V2 sites, all BA sites, and all sites except BA for the year 2019. The last column contains the percentage of pixels with uncertainty within the goal (10%) and threshold (20%) range. The absolute values of the statistics are given in gC/m²/day.

	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset / slope	% Goal / Threshold
All LC	56736	0.99	-0.13 (-4%)	-0.07 (-2%)	0.6 (18%)	0.2 (6%)	0.6 (19%)	0.15 / 0.99	38 / 58
BA	9542	0.70	-0.10 (-79%)	-0.05 (-41%)	0.14 (117%)	0.05 (44%)	0.17 (141%)	0.09 / 1.4	6 / 12
No BA	47194	0.98	-0.14 (-4%)	-0.10 (-2%)	0.7 (17%)	0.3 (7%)	0.7 (17%)	0.17 / 0.99	45 / 67

6.7.3 Spatial consistency

6.7.3.1 Histograms per biome

The frequency histograms per biome and for all landcover types between GPP 300m V2 and GPP 1km V2 are presented in Figure 42. They show good consistency, albeit with higher GPP 1km V2 values for all biome types except for EBF, for which the FAPAR algorithm is different than for non-EBF pixels.

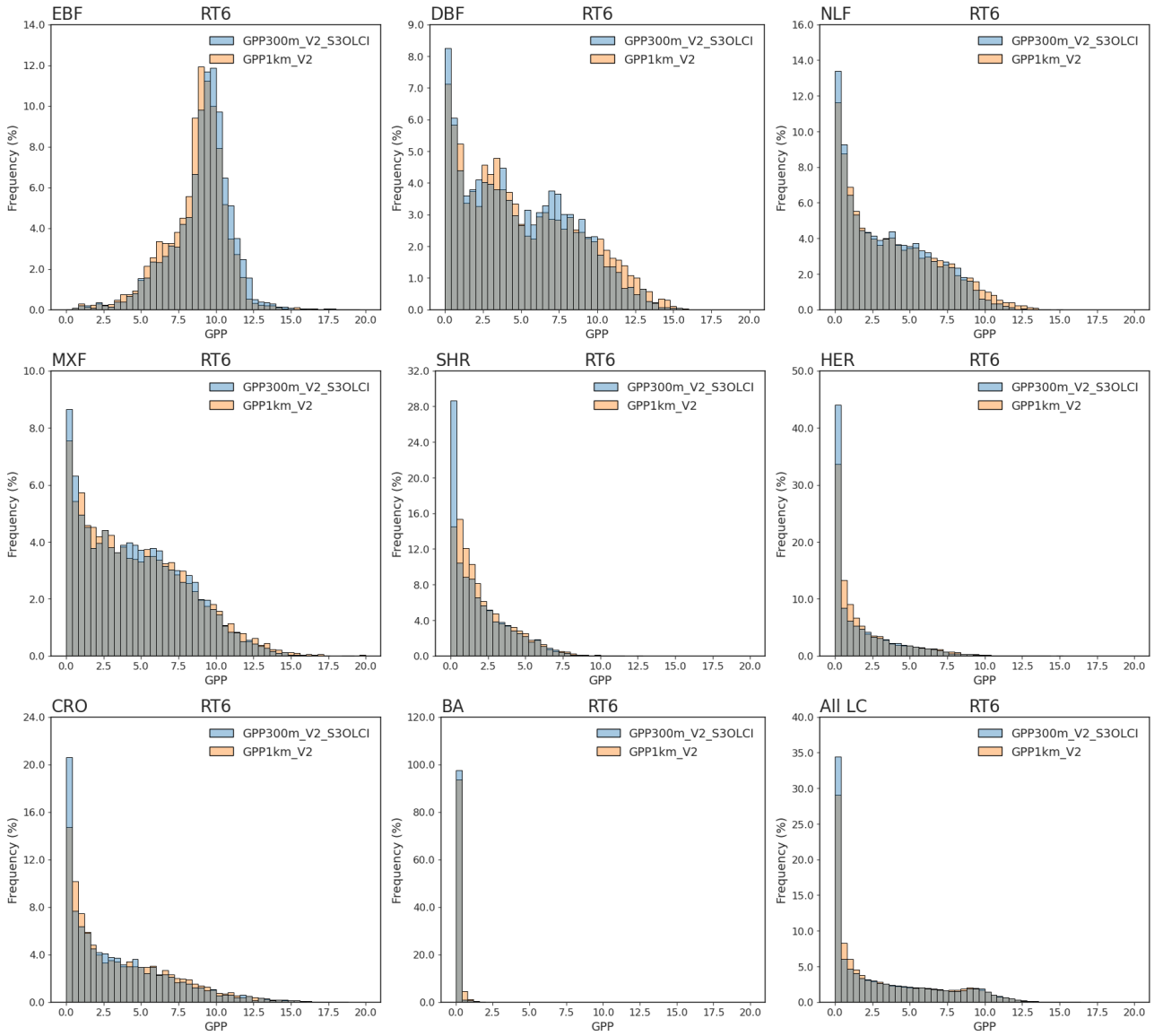


Figure 42: Frequency histograms per biome and for all land cover types comparing GPP 300m V2 Sentinel-3/OLCI (blue) and GPP 1km V2 (orange) for the year 2019. Grey colour indicates where both histograms overlap.

6.7.3.2 Spatial distribution of validation metrics

The spatial patterns of APU statistics between GPP 300m V2 and GPP 1km V2 are shown in Figure 43. A large positive bias is visible over tropical areas, in Equatorial Africa and in the region around the Amazon forest, while the bias is negative in the (dry) Southern Andes, in East Africa, Australia, and in the region south of the Congo Basin. The spatial patterns of uncertainty and precision agree well with vegetation amount (i.e. lower uncertainty and precision values for regions with lower vegetation density), except for EBF regions where the uncertainty and precision statistics are

relatively low. This is due to the fact that both GPP 300m V2 and GPP 1km V2 are gap filled for pixels in the EBF biome, while there is a different approach in non-EBF pixels (where GPP 1km V2 is gap filled, and GPP 300m V2 is not).

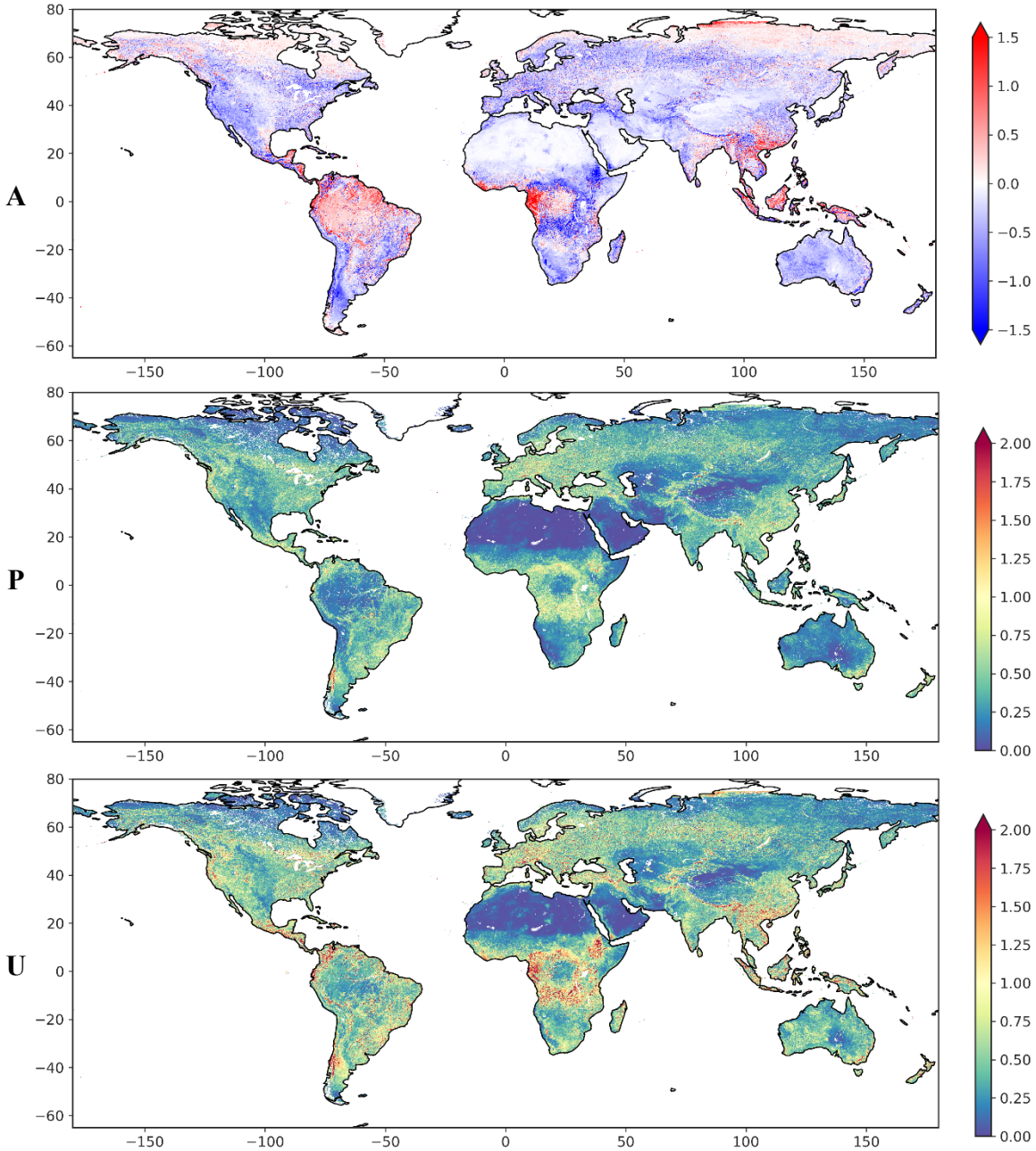


Figure 43: Spatial pattern of pixel APU between GPP 300m V2 Sentinel-3/OLCI and GPP 1km V2 for all pairwise valid GPP for the year 2019

6.7.4 Conclusions

What is the statistical and spatial consistency with DMP 1km V2?

GPP 300m V2 shows overall large statistical and spatial consistency with GPP 1km V2, although the GPP 1km V2 values are slightly higher than the GPP 300m V2 values. The EBF biome shows different results than the other biomes, with lower uncertainty and precision statistics, thanks to the fact that both products are in this case gap filled. For non-EBF biomes, there is more unsystematic bias since in these cases GPP 1km V2 is gap filled and GPP 300m V2 is not. It can be concluded that the GPP 300m V2 dataset resampled to 1km resolution can be used as a continuation of the DMP/GDMP 1km V2 dataset.

6.8 Q8: Statistical and spatial consistency between DMP/GDMP/NPP/GPP 300m V2 and MOD17 V6.1

6.8.1 Introduction

The scientific question to answer in this section is: what is the spatial and statistical consistency of DMP/GDMP/NPP/GPP 300m V2 in comparison to MOD17 V6.1 GPP?

The analyses are performed over a one-year period (2019) and are based on LANDVAL V2 extractions for the statistical consistency analysis (see §5.3.5) and on a global systematic subsample created with 1km resolution pixels (see §5.3.3) for the spatial consistency analysis.

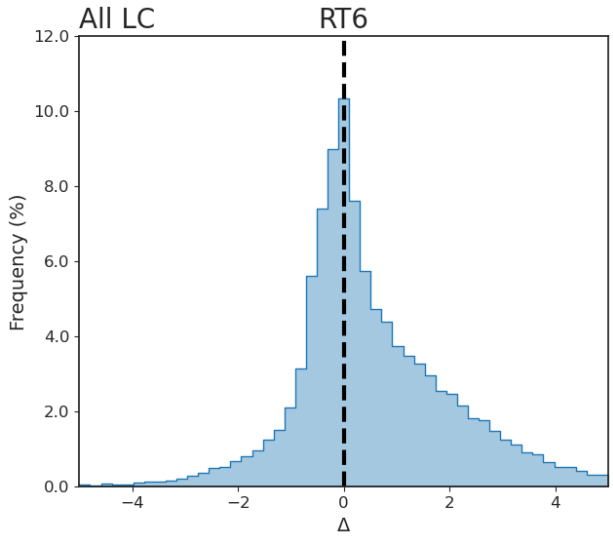
The analyses include:

- Statistical consistency analysis:
  - Histograms of overall bias
  - GM regression analysis, overall and per biome
- Spatial consistency analysis:
  - Histograms of retrievals per biome
  - Spatial distribution of validation metrics

6.8.2 Statistical consistency

6.8.2.1 Overall bias

The bias frequency histogram in Figure 44 indicates that, in general, GPP 300m V2 is somewhat higher than the MOD17 V6.1 GPP as the distribution is skewed towards positive values. The mean and median bias however are very low (0.6 $gC/m^2/day$ and 0.2 $gC/m^2/day$, respectively). The positive tail in the bias histogram is due to a drought-stress related factor which is taken into account in MOD17 V6.1 but not in GPP 300m V2, leading to lower GPP values for the former.



Mean bias: 0.6 gC/m^2/day

Median bias: 0.2 gC/m^2/day

Figure 44: Frequency histogram of the overall bias between GPP 300m V2 Sentinel-3/OLCI and MOD17 V6.1 GPP for LANDVAL V2 sites for the year 2019

6.8.2.2 GM regression

Figure 45 shows the scatterplot between GPP 300m V2 and MOD17 V6.1 GPP. The regression line clearly deviates from the 1:1 line although the intercept is close to zero. This figure confirms the findings from the bias histogram, namely that there is a systematic bias between GPP 300m V2 and MOD17 V6.1, with the former having generally higher values than the latter.

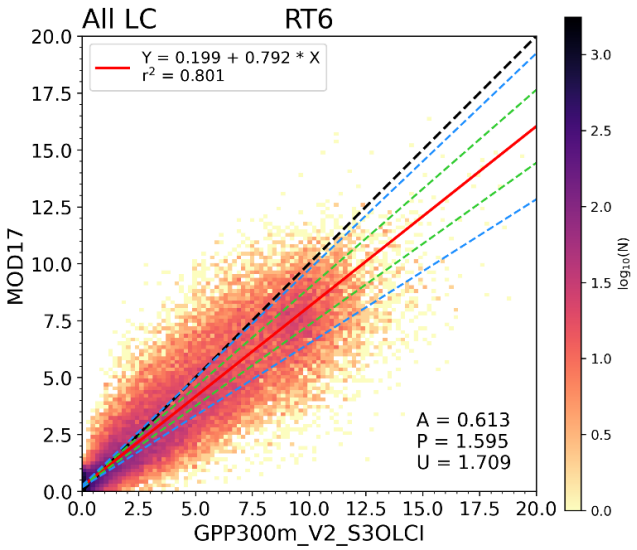


Figure 45: Scatter density plot between GPP 300m V2 Sentinel-3/OLCI (X) and MOD17 V6.1 GPP (Y) over LANDVAL V2 sites for the year 2019. Green and blue dashed lines indicate the goal and threshold uncertainty requirement.

The statistics related to the GM regression are given in Table 15. The last column in the table are the percentages of LANDVAL V2 sites for which the uncertainty of GPP 300m V2 compared to MOD17 V6.1 is within the goal and threshold limits. These are the LANDVAL V2 sites for which the GPP lies within the green and blue dashed lines of Figure 45, respectively. A bit more than one third of the sample shows comparison statistics within the threshold range, related to the high scatter observed in Figure 45. It should be noted that the good correspondence for low GPP values (GMR equation offset close to zero) is not reflected in this statistic because of its relative character.

Table 15: Statistics of comparison between GPP 300m V2 Sentinel-3/OLCI and MOD17 V6.1 GPP over all LANDVAL V2 sites, all BA sites, and all sites except BA for the year 2019. The last column contains the percentage of pixels with uncertainty within the goal (10%) and threshold (20%) range. The absolute values of the statistics are given in gC/m²/day.

	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset / slope	% Goal / Threshold
All LC	43068	0.9	0.6 (19%)	0.2 (7%)	1.6 (48%)	0.7 (22%)	1.7 (52%)	0.2 / 0.8	19 / 36
BA	919	0.55	-0.3 (-56%)	-0.3 (-59%)	0.4 (79%)	0.3 (70%)	0.4 (97%)	0.3 / 0.8	10 / 19
No BA	42149	0.9	0.6 (19%)	0.3 (8%)	1.6 (48%)	0.8 (22%)	1.7 (51%)	0.2 / 0.8	19 / 37

6.8.3 Spatial consistency

6.8.3.1 Histograms per biome

Histograms per biome (Figure 46) show again that GPP 300m V2 values are higher than MOD17 V6.1. The differences are most pronounced for the EBF biome and other forest biomes, and for the CRO biome. They can be related to differences in instrument/sensor, in the FAPAR algorithm that is used, and in the way that GPP is computed. Most importantly, for MOD17 V6.1 GPP a drought-related stress factor is incorporated in the GPP algorithm, while this is not done for the CLMS GPP computation. This drought-related stress factor results in lower GPP values, as can be seen in Figure 46. The impact is higher on forest biomes and croplands than for bare areas and grasslands

with little vegetation.

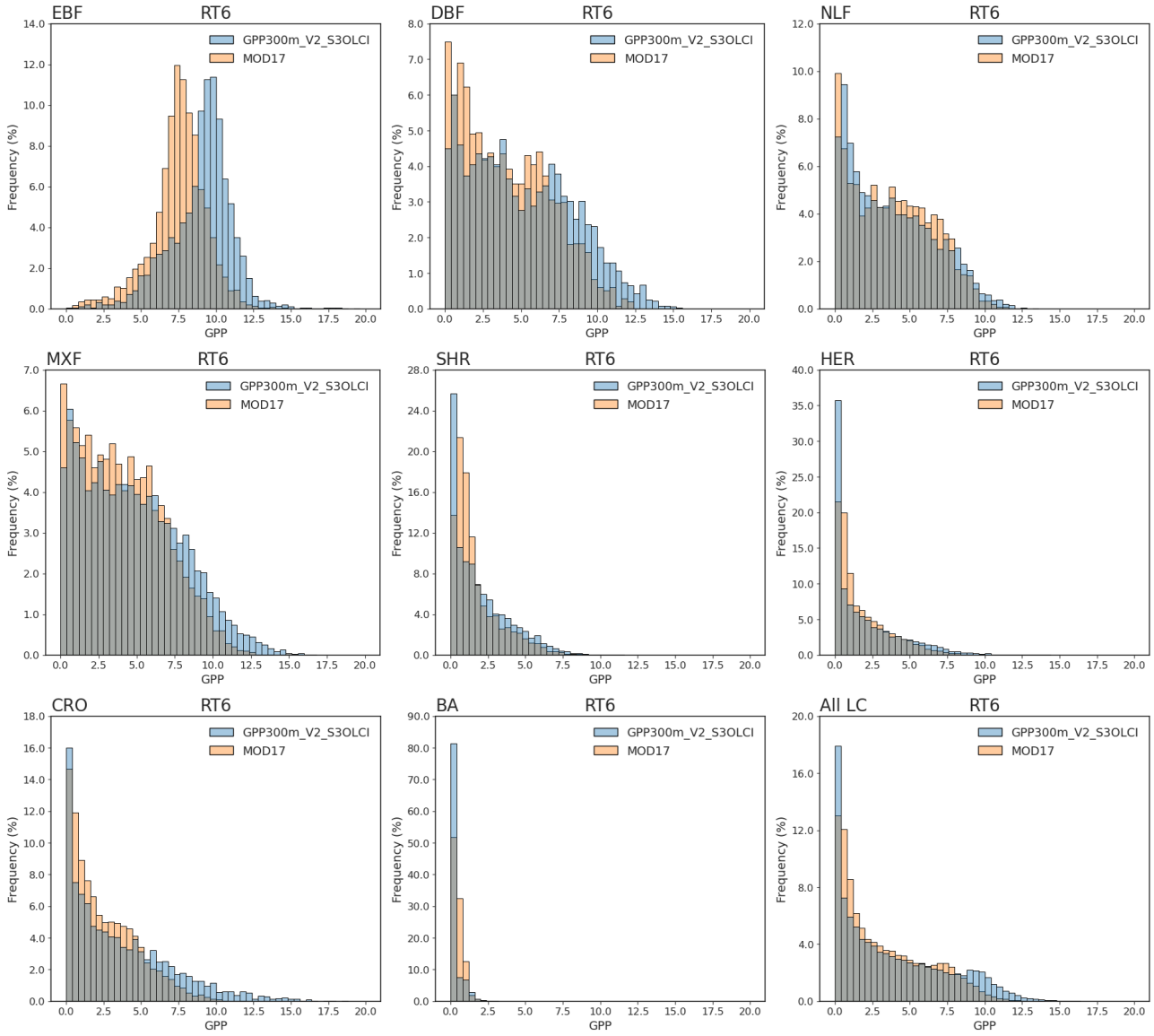


Figure 46: Frequency histograms per biome and for all land cover types comparing GPP 300m V2 Sentinel-3/OLCI (blue) and MOD17 V6.1 GPP (orange) for the year 2019. Grey colour indicates where both histograms overlap.

6.8.3.2 Spatial distribution of validation metrics

Spatial patterns of accuracy, precision, and uncertainty between GPP 300m V2 and MOD17 V6.1 (see Figure 47) show that differences between both datasets have a well-defined global distribution. The bias is mostly positive, indicating that GPP 300m V2 values are higher than MOD17 V6.1 values. Positive biases are mostly apparent over areas with forest cover (notably EBF, DBF) or cropland

areas (CRO) (see Figure 4, §5.3.2), where MOD17 V6.1 is overall lower compared to the GPP 300m V2 (see above). More sparsely vegetated regions (SHR, HER, BA) show lower bias.

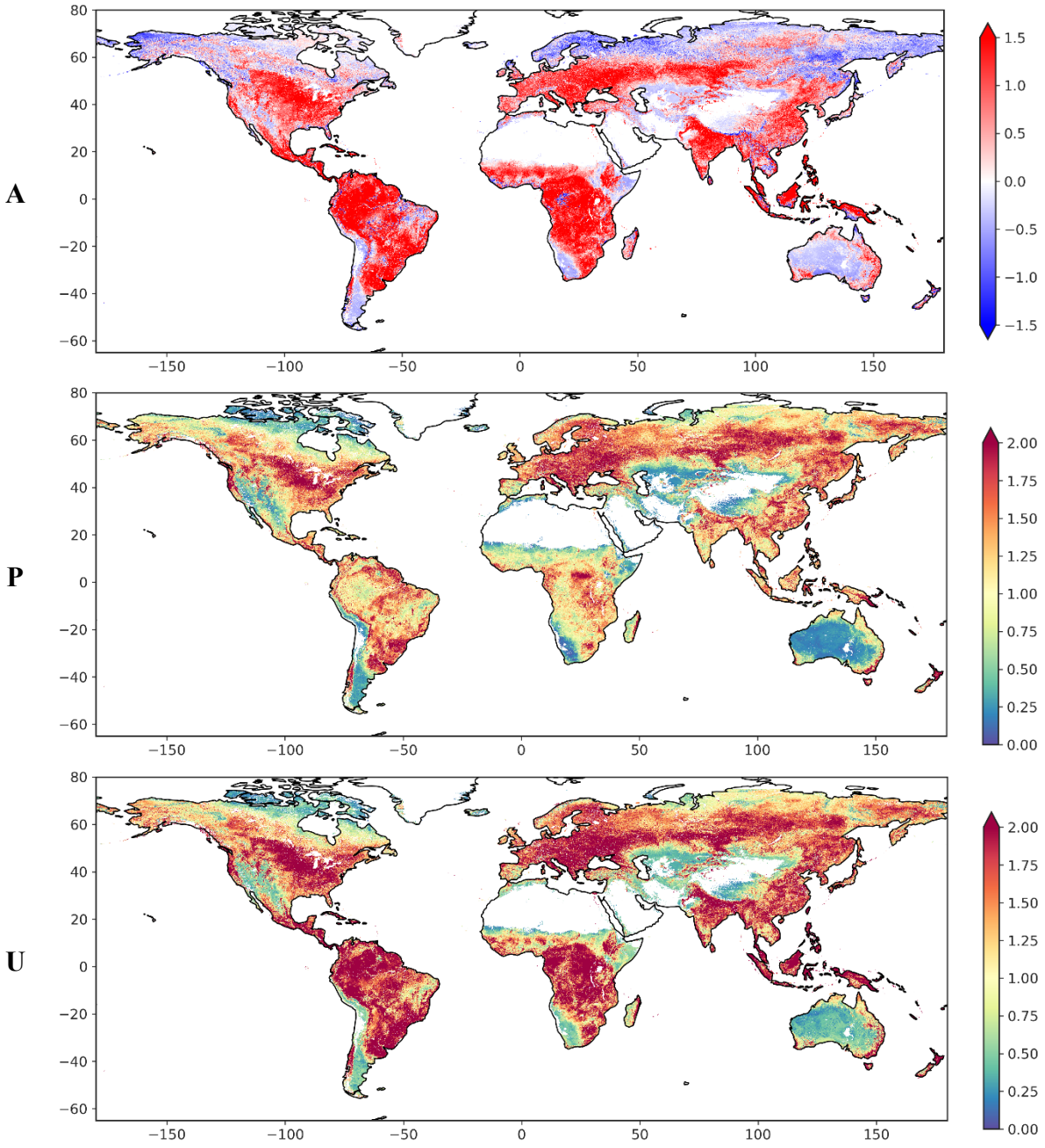


Figure 47: Spatial pattern of pixel wise APU between GPP 300m V2 Sentinel-3/OLCI and MOD17 V6.1 GPP for all pairwise valid GPP values for the year 2019

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6.8.4 Conclusion

What is the spatial and statistical consistency of DMP/GDMP/NPP/GPP 300m V2 in comparison to MOD17 V6.1 GPP?

There is a large positive bias between GPP 300m V2 and MOD17 V6.1 (mean bias = 0.6 gC/m^2/day and median bias = 0.2 gC/m^2/day). These differences are related to differences in sensor, in FAPAR computation, and most importantly in the computation of GPP. For MOD17 V6.1, a drought-stress related factor is incorporated in the GPP algorithm while this is not present in the CLMS GPP computation. The effect is that MOD17 V6.1 GPP values are in general lower than GPP 300m V2 values.

6.9 Q9: COMPARISON WITH IN-SITU FLUXNET DATA

6.9.1 Introduction

The scientific questions to be answered are the following: what is the statistical consistency of the GPP 300m V2 product compared to in-situ FLUXNET GPP data? Is the uncertainty between GPP 300m V2 and FLUXNET within the requirement range?

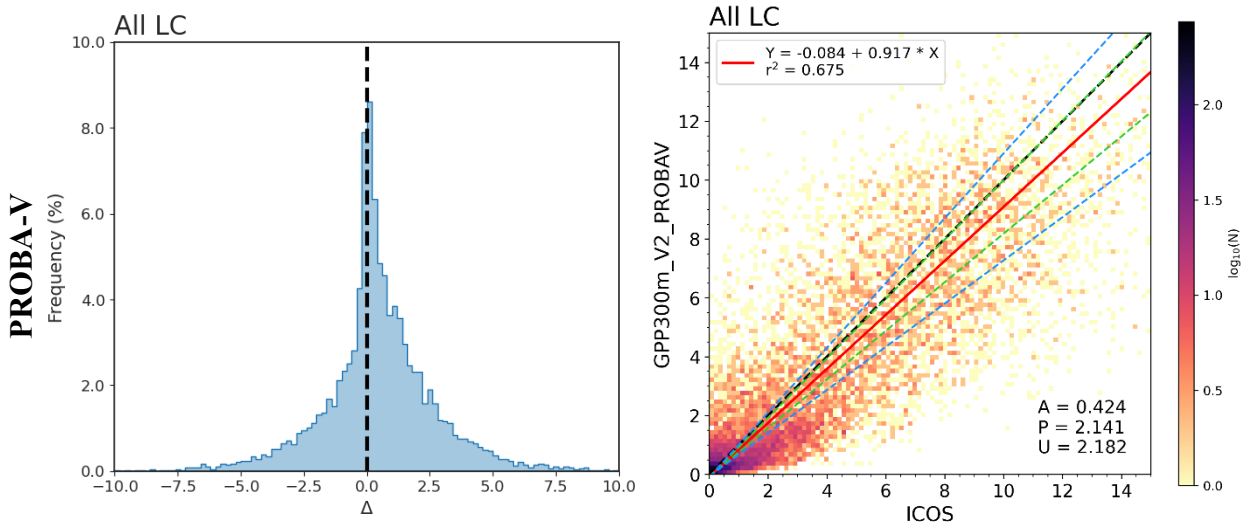
The analyses are performed over the full time series (2014 – 2024) and are based on FLUXNET extractions (see §5.2.3).

The analyses include:

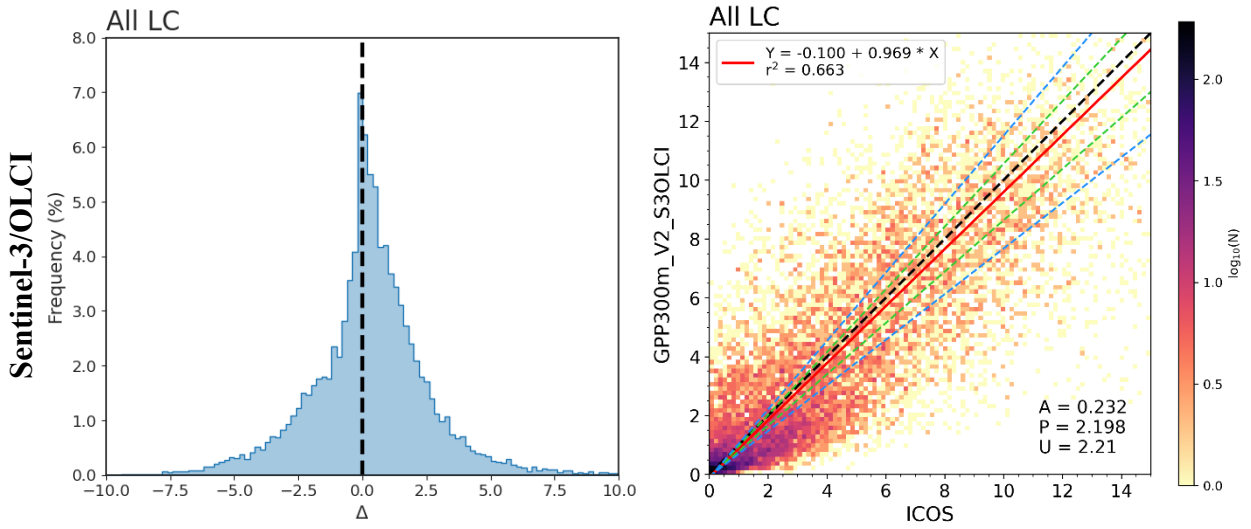
- Statistical consistency analysis:
  - Histograms of overall bias
  - GM regression analysis
  - Uncertainty requirement
- Temporal consistency analysis:
  - Temporal profiles

6.9.2 Statistical consistency

The statistical consistency with FLUXNET data is assessed for the GPP 300m V2 dataset of both PROBA-V and Sentinel-3/OLCI over the period 2014 – 2024 The FLUXNET data are split into the dataset from ICOS and from Ameriflux. Frequency histograms of the bias and scatterplots of all four combinations (PROBA-V and Sentinel-3/OLCI vs ICOS and Ameriflux) are shown in Figure 48 and Figure 49. The bias histograms are all centred around zero but with a large spread. There is some positive skewedness for comparison with ICOS data, and a larger tail of negative values for Ameriflux data, but the mean bias is in all four cases very low. The scatterplots confirm these results: the GPP 300m V2 are slightly lower than ICOS GPP values resulting in GM regression line with slope below one but intercept very close to zero, the GPP 300m V2 on the other hand are higher than Ameriflux values causing the slope of the GM regression to be larger than one. Considering the different spatial footprints and instruments of the datasets, the regression lines are still surprisingly close to the 1:1 line.



Mean bias: 0.5 gC/m^2/day
Median bias: 0.3 gC/m^2/day



Mean bias: 0.3 gC/m^2/day
Median bias: 0.3 gC/m^2/day

Figure 48: Frequency histograms of the overall bias (left) and scatter plots (right) between ICOS GPP and GPP 300m V2 PROBA-V (top) and GPP 300m V2 Sentinel-3/OLCI (bottom) for 2014 – 2024.

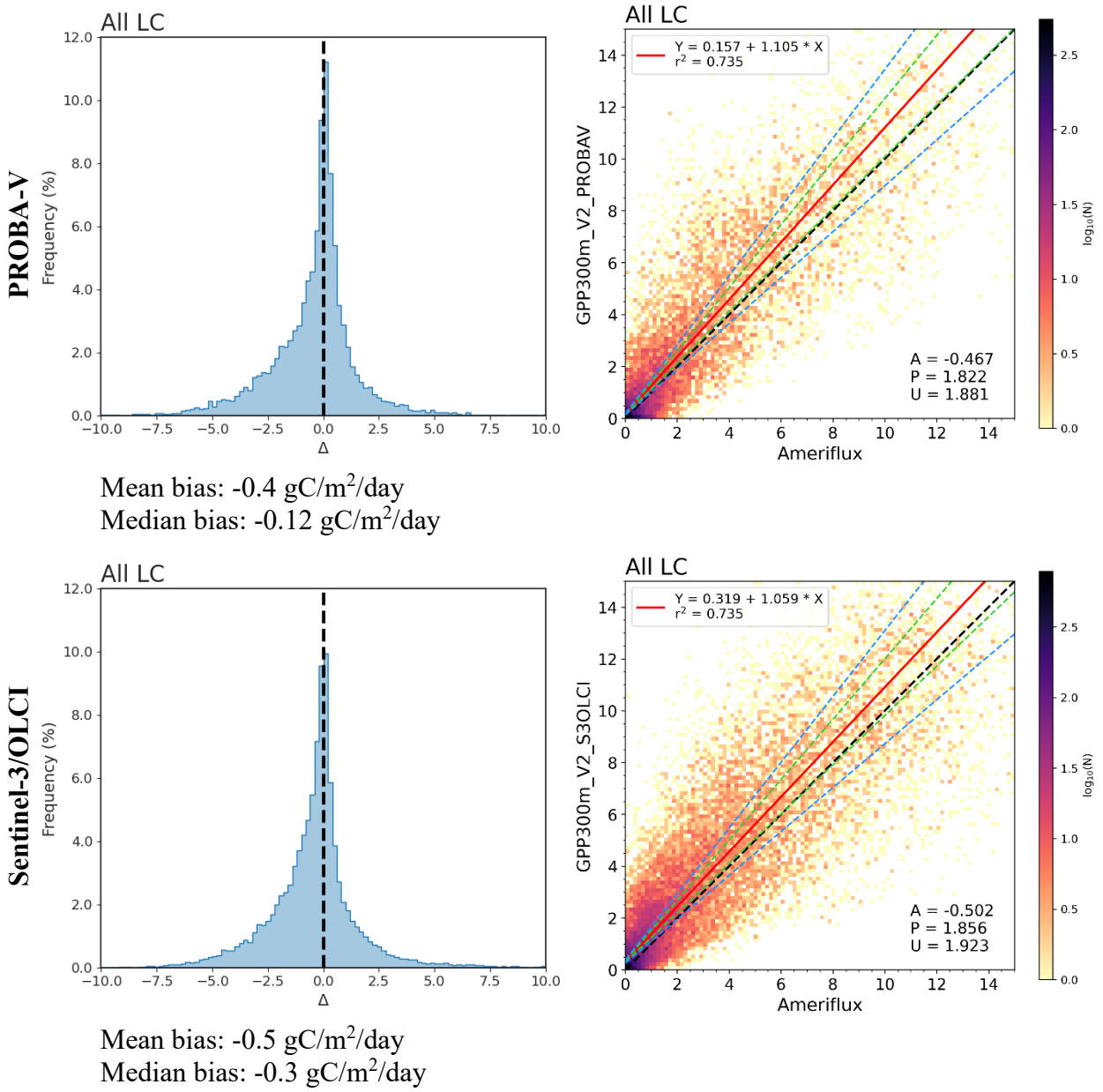


Figure 49: Frequency histograms of the overall bias (left) and scatter plots (right) between Ameriflux GPP and GPP 300m V2 PROBA-V (top) and GPP 300m V2 Sentinel-3/OLCI (bottom) for 2014 – 2024.

The statistics associated to the various scatterplots are given in Table 16, along with the percentage of pixels within the goal and threshold uncertainty. These values are computed for all land cover types combined. About a quarter of the sample shows comparison statistics within the threshold range, related to the high scatter observed in Figure 49. It should be noted that the good correspondence for low GPP values (GMR equation offset close to zero) is not reflected in this statistic because of its relative character. The correlation coefficients have values above 0.8, which is still relatively high.

For all dataset pairs, the (relative) uncertainty (RMSD) and precision (STD) metrics are high, indicating relatively large unsystematic bias between both datasets under comparison.

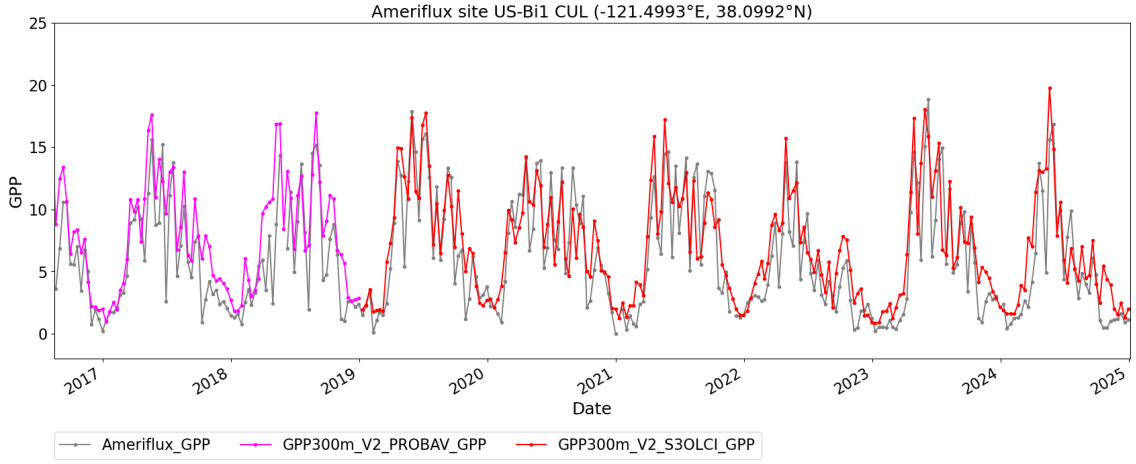
Table 16: Statistics of comparison between GPP 300m V2 PROBA-V or Sentinel-3/OLCI and GPP from FLUXNET over all ICOS and Ameriflux sites for the years 2014-2024. The absolute values of the statistics are given in gC/m²/day. The last column contains the percentage of pixels with uncertainty within the goal (10%) and threshold (20%) range.

Dataset	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset / slope	% Goal / Thres.
GPP 300m V2 PROBA-V vs ICOS	9495	0.82	0.4 (12%)	0.3 (8%)	2 (58%)	1.1 (29%)	2 (59%)	-0.08 / 0.91	12 / 25
GPP 300m V2 Sentinel-3/OLCI vs ICOS	12229	0.81	0.2 (6%)	0.2 (6%)	2 (54%)	1.1 (28%)	2 (54%)	-0.10 / 0.97	13 / 26
GPP 300m V2 PROBA-V vs Ameriflux	10906	0.86	-0.5 (-14%)	-0.13 (-4%)	1.8 (53%)	0.7 (21%)	1.9 (55%)	0.16 / 1.1	13 / 26
GPP 300m V2 Sentinel-3/OLCI vs Ameriflux	18446	0.86	-0.5 (-14%)	-0.3 (-7%)	1.9 (52%)	0.8 (22%)	1.9 (53%)	0.3 / 1.1	14 / 27

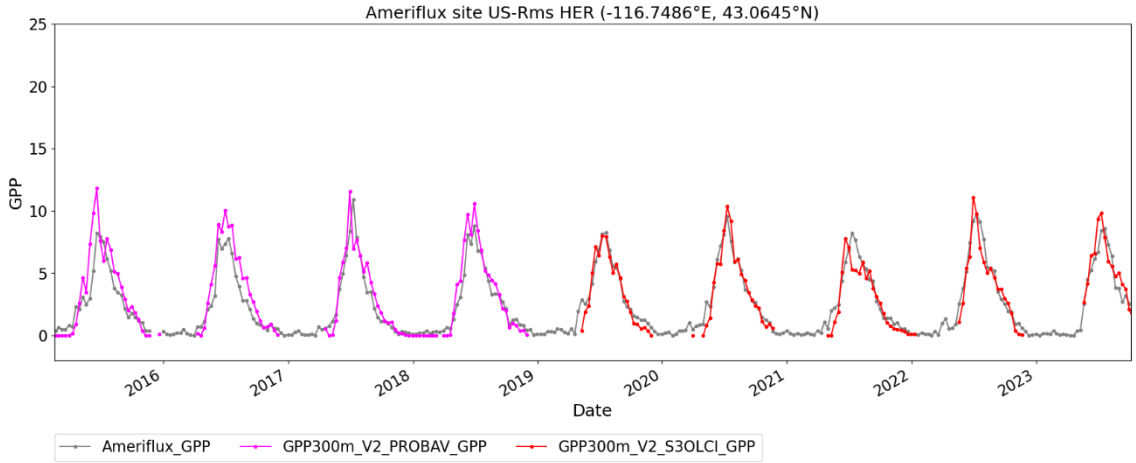
6.9.3 Temporal profiles

Figure 50 shows temporal profiles over several ICOS and Ameriflux sites with different biome types for GPP 300m V2 PROBA-V, GPP 300m V2 Sentinel-3/OLCI, and ICOS and Ameriflux GPP. Even though there are quite some differences (mostly high frequency fluctuations) between the in-situ data and the remote sensing observations, the overall temporal patterns and seasonal trends are quite similar. An exact agreement is also not expected because of differences in observation techniques, spatial footprints, GPP computation, etc.

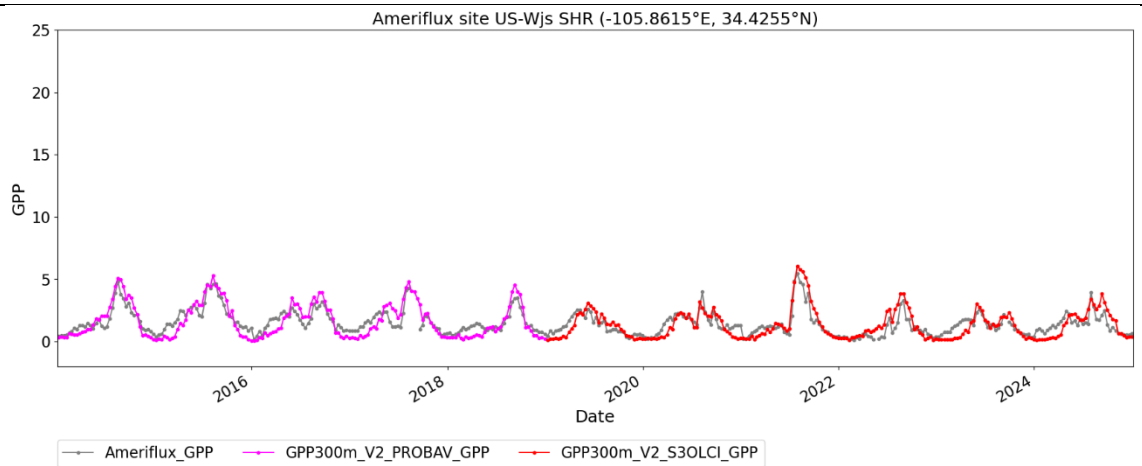
Ameriflux CRO



Ameriflux HER



Ameriflux SHR



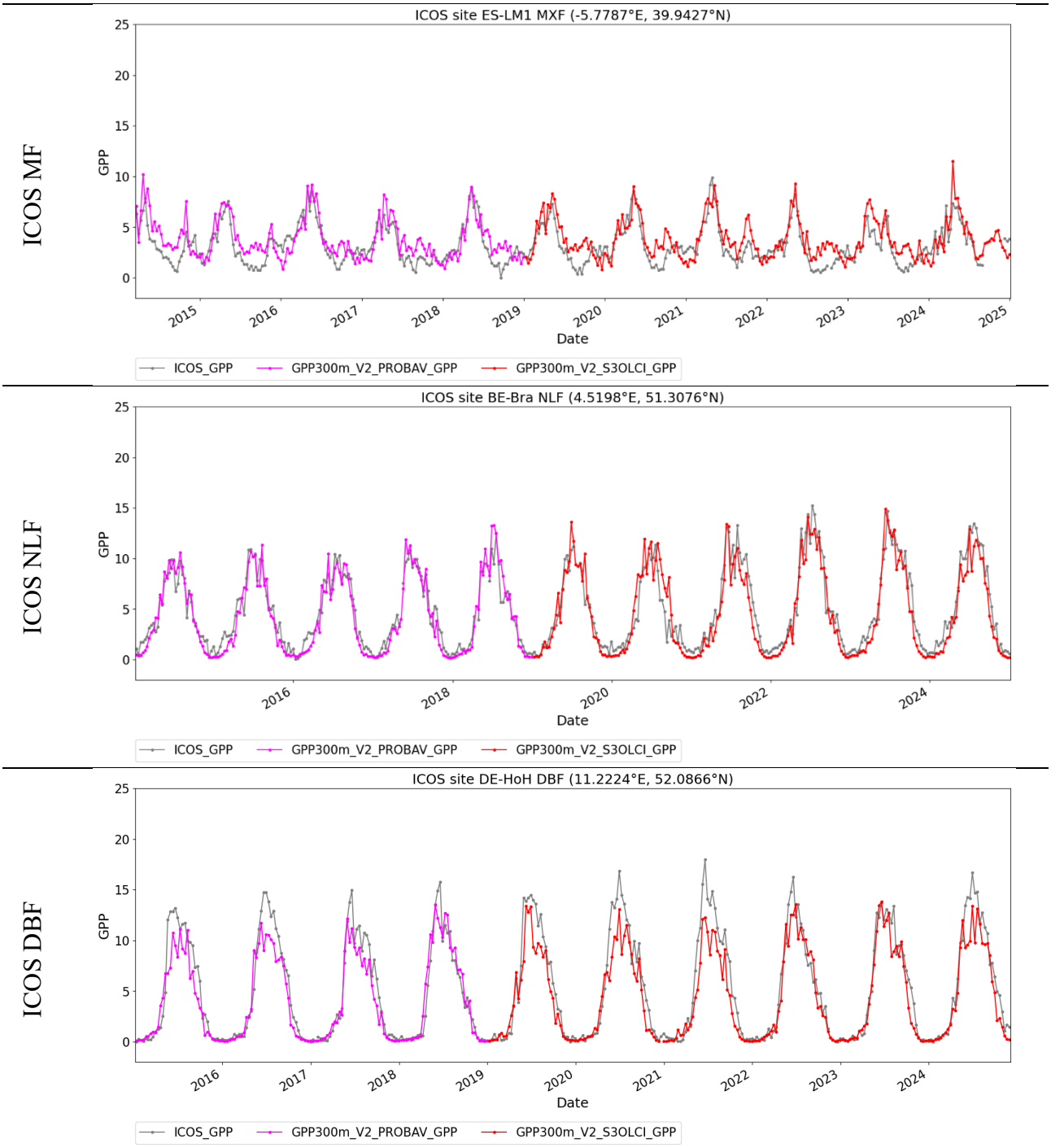


Figure 50: Temporal profiles over a selection of FLUXNET sites of Ameriflux/ICOS GPP (grey), GPP 300m V2 PROBA-V (pink), and GPP 300m V2 Sentinel-3/OLCI (red).

6.9.4 Conclusion

What is the statistical consistency of the DMP/GDMP/NPP/GPP 300m V2 product compared to in-situ FLUXNET GPP data?

The GPP 300m V2 product agrees relatively good with the ICOS and Ameriflux GPP data. Although the APU statistics only indicate moderate agreement, the time series show that both datasets follow the same seasonal patterns albeit with slightly different amplitudes for some data points.

Is the uncertainty between GPP 300m V2 and FLUXNET within the requirement range?

Around 13% and 26% of the FLUXNET sites comply with the goal and threshold uncertainty requirement, respectively, when comparing GPP 300m V2 with ICOS and Ameriflux. These percentages are quite low which is due to differences between in-situ and remote sensing observations, their footprints and GPP computation. Additionally, the sample size is not very big and mainly covers parts of Europe and North America.

6.10 Q10: TEMPORAL CONSISTENCY OF DMP/GDMP/NPP/GPP 300M V2

6.10.1 Introduction

The scientific questions to be answered are the following: what is the temporal consistency of DMP/GDMP/NPP/GPP 300m V2 in comparison to MOD17 V6.1 GPP? How do the results based on GPP 300m V2 compare to those of DMP/GDMP 300m V1.0, DMP/GDMP 300m V1.1, and DMP/GDMP 1km V2?

The analysis of the temporal smoothness is performed over two one-year periods (2019 and 2023), while the other analyses use the full time series (January 2014 – December 2024). The temporal pixel profiles and temporal smoothness analysis are performed on extractions of valid GPP values for the LANDVAL V2 sites (see §5.3.5). The spatio-temporal evolution of the mean GPP and of validation metrics of the comparison with an external reference is performed on a global systematic subsample (see §5.3.3) created with 1km resolution extractions.

The analyses include:

- Temporal pixel profiles over LANDVAL V2 sites
- Temporal smoothness
- Stability analysis
- Spatio-temporal evolution of validation metrics (Hovmöller plots) of the comparison with an external reference

6.10.2 Temporal pixel profiles

Figure 51 shows temporal pixel profiles for GPP 300m V2 and all reference datasets for a selection of sites in different biomes. Temporal profiles for all LANDVAL V2 sites can be found in [Annex 1: Digital annex].

The GPP 300m V2 dataset shows similar temporal behaviour as the reference datasets and is more stable over time than GPP 300m V1.0. Especially, the continuity over the whole time series is better in V2 than in a combination of V1.0 and V1.1 (see for example the EBF site 416 and MXF site 103 in Figure 51). Note that V1.1 mostly coincides with V2 which is plotted on top.

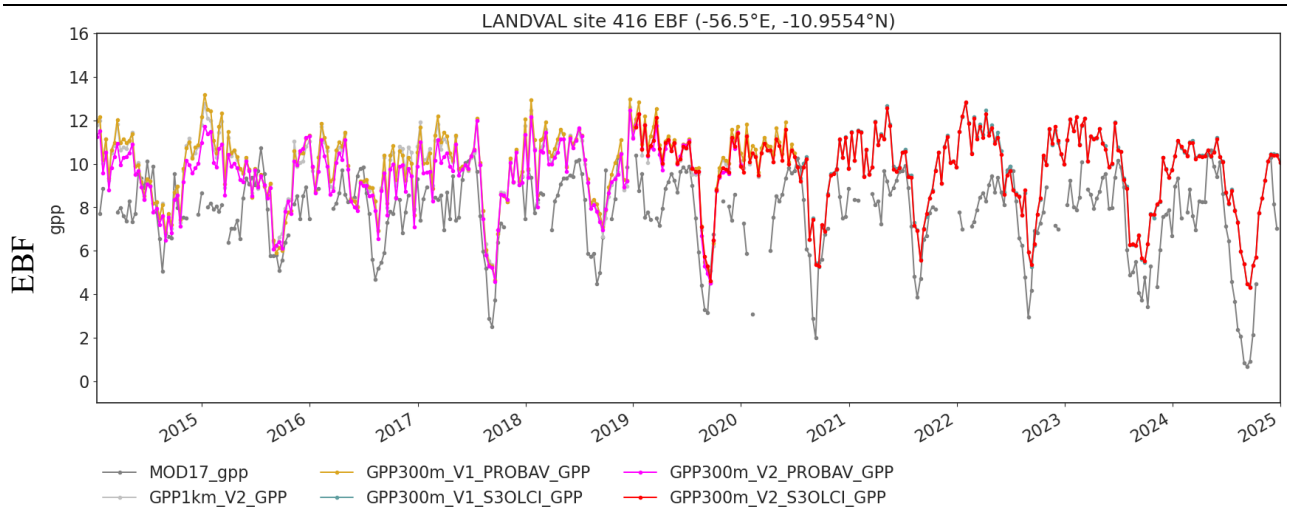
For some sites (e.g. EBF site 416, MXF site 103, CRO site 184, BA site 596, NLF site 24, and HER site 565), there are quite some differences visible between GPP 300m V2 on the one hand and GPP 300m V1.0 and GPP 1km V2 on the other hand. This is a result of the updated PROBA-V C2 TOC data used in V2. Additionally, a spectral conversion was applied to the PROBA-V TOC reflectances for the 1km dataset to better align them with the SPOT/VGT reflectances. Therefore, an offset with the 300m GPP values is to be expected and can be seen for example in some parts of the NLF and HER temporal profiles of Figure 51.

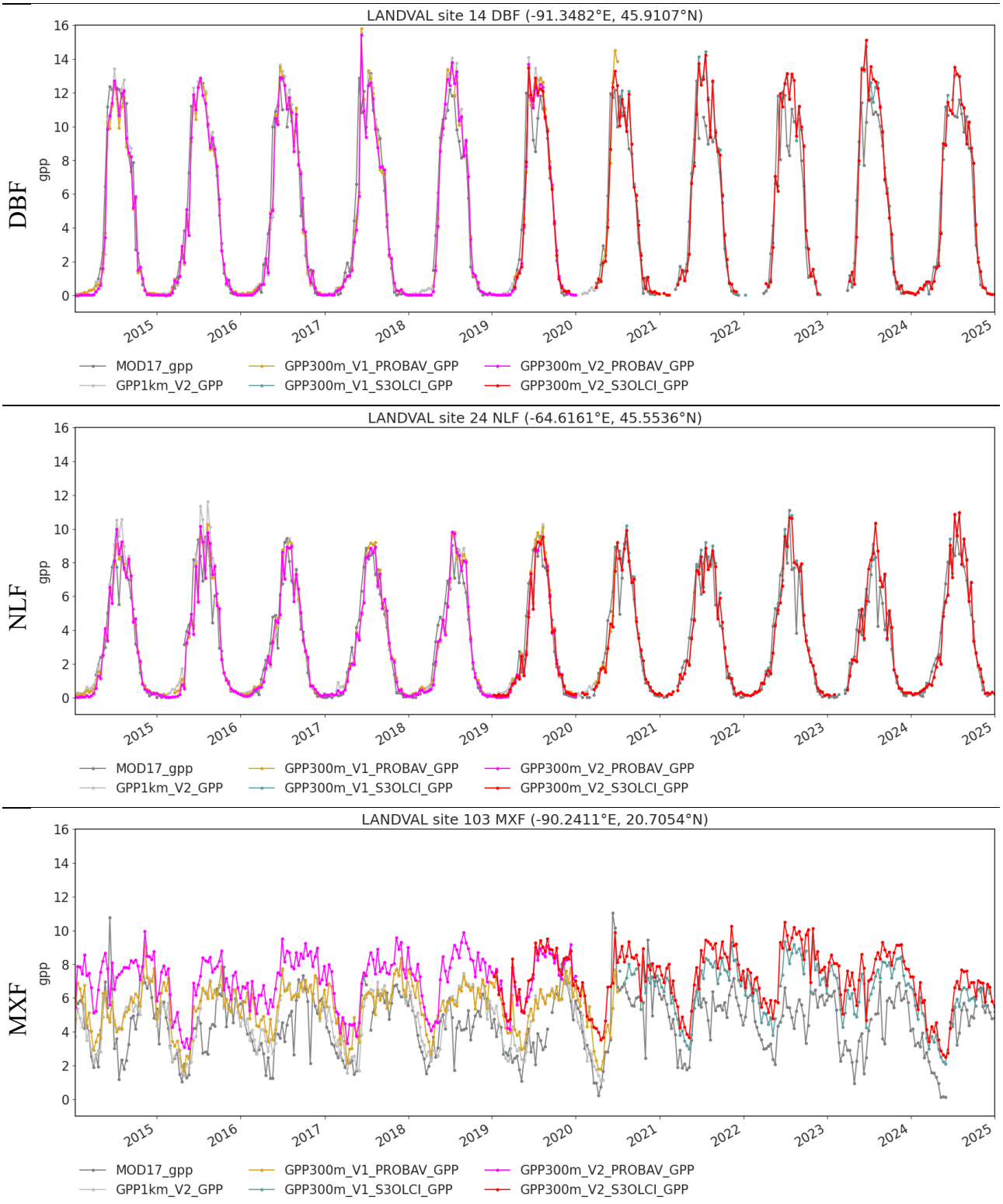
In most of the temporal profiles of Figure 51, the MOD17 V6.1 GPP patterns are shifted to lower values with respect to the other datasets, and are also noisier (see for example HER site 565). This is caused by the inclusion of a drought-stress factor in the MOD17 V6.1 dataset. The effect is most drastic for the EBF and MXF site but is also present in the temporal patterns of the DBF, CRO, and HER site, more specifically during the growing season peak.

Another remarkable difference is the offset between GPP 300m V1 and GPP 300m V2 for certain EBF and MXF sites such as MXF site 103. Both the V1.0 and V1.1 values are lower than V2 GPP. The sites for which this offset is present, are always located close to the equator in the Tropics. It is hence likely that these differences in GPP values are caused by different input TOC versions (C1 vs C2 for PROBA-V and V1 vs V2.3 for Sentinel-3/OLCI), since TOC versions vary mostly in regions affected by persistent cloud cover.

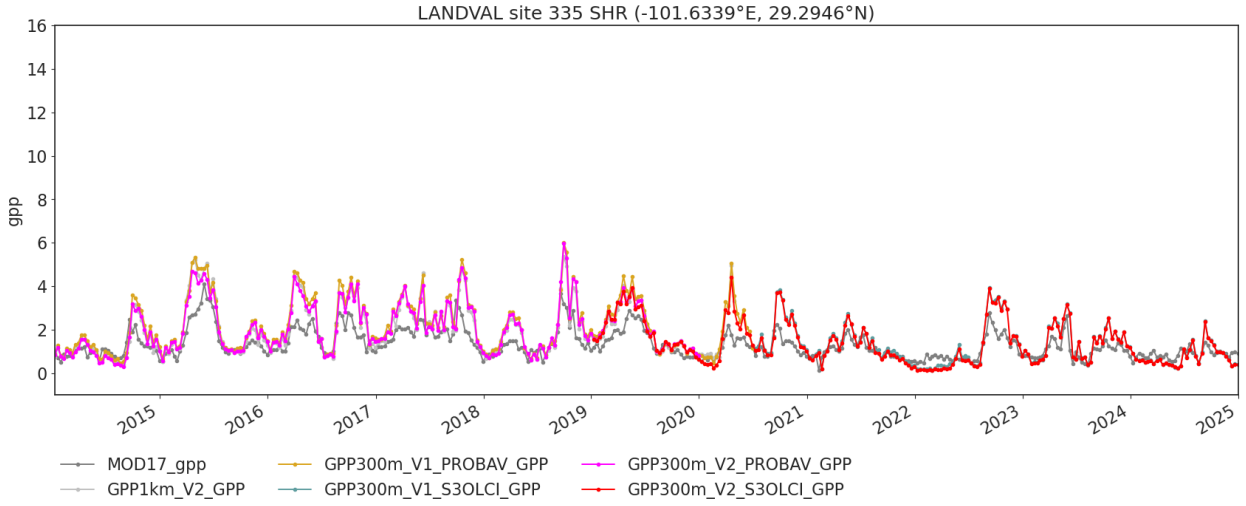
For some sites, such as CRO site 184, GPP 300m V1.1 shows higher GPP values than GPP 300m V2 for the period 2020 – 2022. This is probably due to the geolocation issue present in TOC V1.0 data on which the GPP 300m V1.1 dataset is based up until 30/06/2022. After this date, the temporal patterns of the two datasets are more in agreement.

The temporal patterns of many BA sites are close or equal to zero for the whole GPP 300m V2 time series. This is different to what is observed from the other datasets, and resulting from the correction for low FAPAR values that sets all values below 0.05 equal to 0 and scales values from 0.05 to 0.15 linearly between 0 and 0.15.

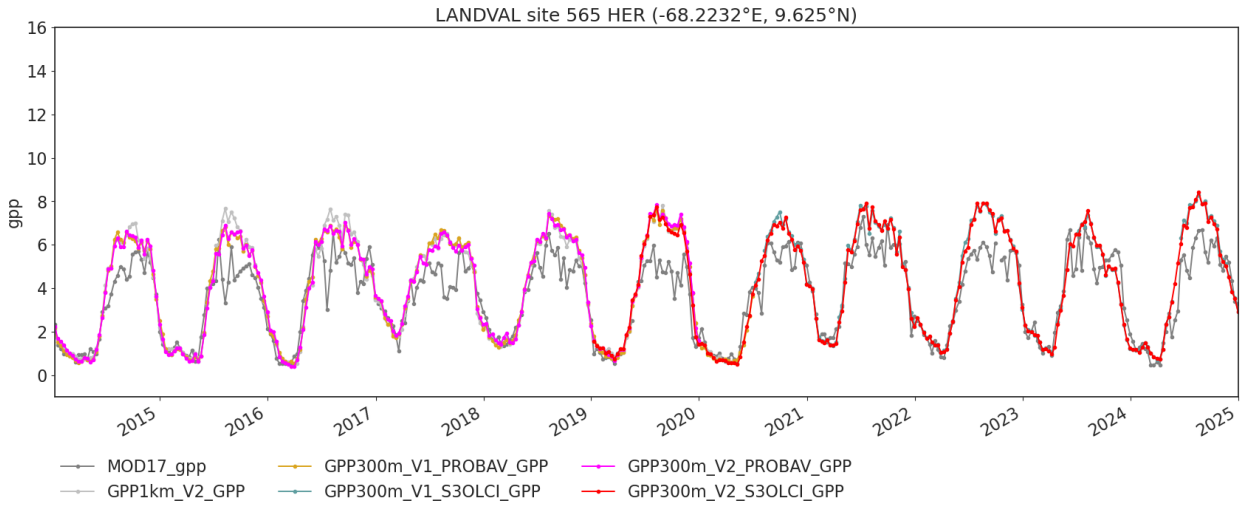



LANDVAL site 14 DBF (-91.3482°E, 45.9107°N)
DBF
gpp
LANDVAL site 24 NLF (-64.6161°E, 45.5536°N)
NLF
gpp
LANDVAL site 103 MXF (-90.2411°E, 20.7054°N)
MXF
gpp
MOD17_gpp
GPP1km_V2_GPP
GPP300m_V1_PROBAV_GPP
GPP300m_V1_S3OLCI_GPP
GPP300m_V2_PROBAV_GPP
GPP300m_V2_S3OLCI_GPP

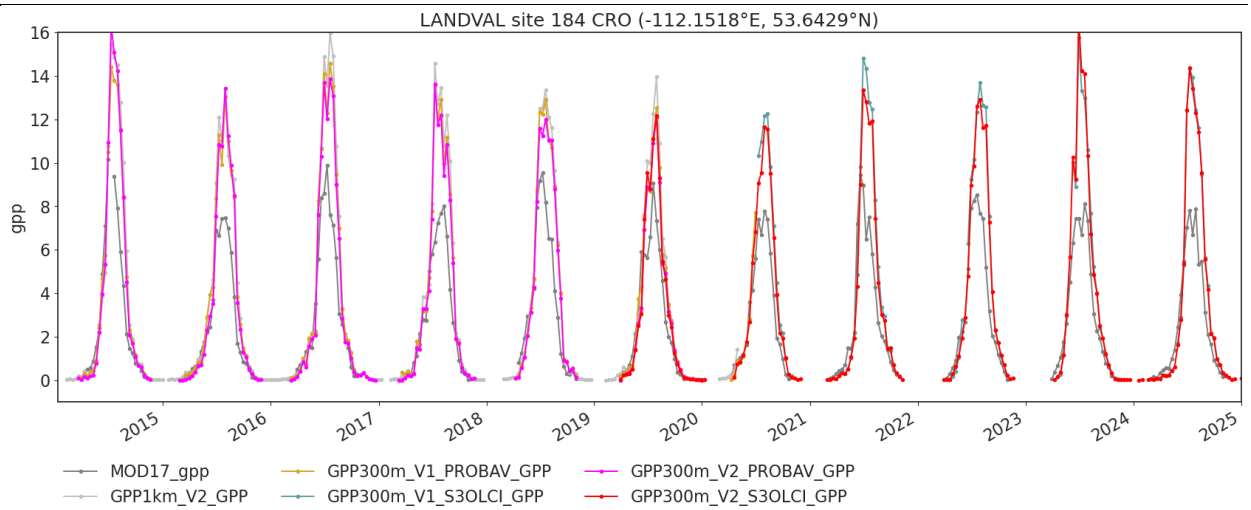
SHR


LANDVAL site 335 SHR (-101.6339°E, 29.2946°N)
gpp
MOD17_gpp
GPP1km_V2_GPP
GPP300m_V1_PROBAV_GPP
GPP300m_V1_S3OLCI_GPP
GPP300m_V2_PROBAV_GPP
GPP300m_V2_S3OLCI_GPP

HER


LANDVAL site 565 HER (-68.2232°E, 9.625°N)
gpp
MOD17_gpp
GPP1km_V2_GPP
GPP300m_V1_PROBAV_GPP
GPP300m_V1_S3OLCI_GPP
GPP300m_V2_PROBAV_GPP
GPP300m_V2_S3OLCI_GPP

CRO


LANDVAL site 184 CRO (-112.1518°E, 53.6429°N)
gpp
MOD17_gpp
GPP1km_V2_GPP
GPP300m_V1_PROBAV_GPP
GPP300m_V1_S3OLCI_GPP
GPP300m_V2_PROBAV_GPP
GPP300m_V2_S3OLCI_GPP

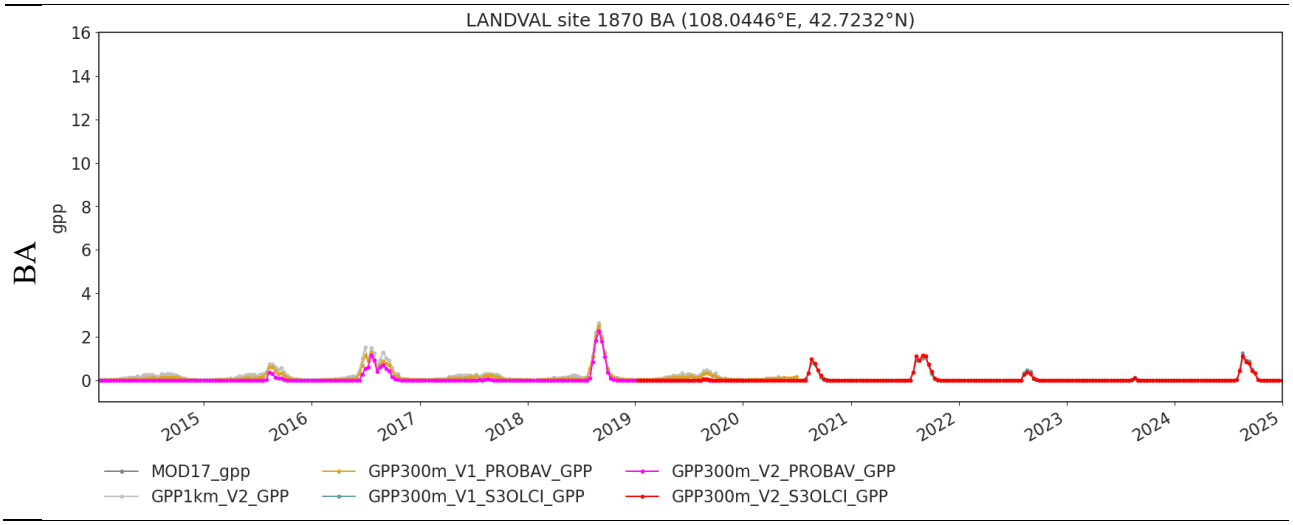
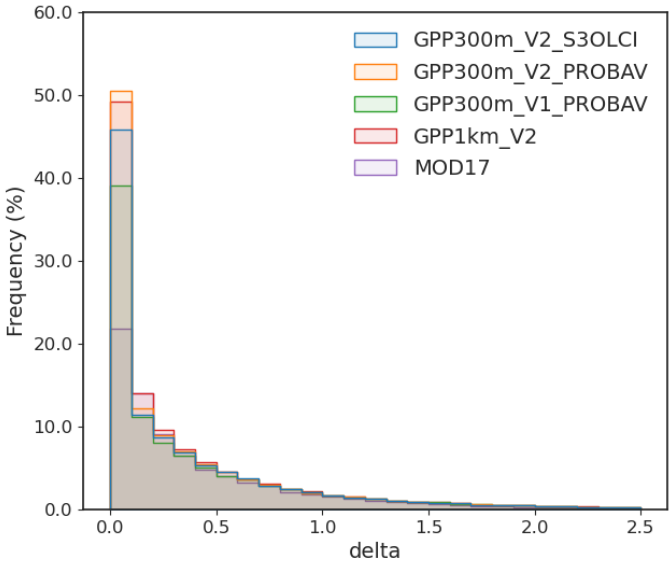


Figure 51: Temporal profiles over a selection of LANDVAL sites of GPP 300m V2 Sentinel-3/OLCI (red), GPP 300m V2 PROBA-V (pink), GPP 300m V1.0 (yellow), GPP 300m V1.1 (blue), DMP 1km V2 (light grey), MODIS MOD17 V6.1 (grey). Temporal profiles for all LANDVAL V2 sites can be found in Annex 1: Digital annex.

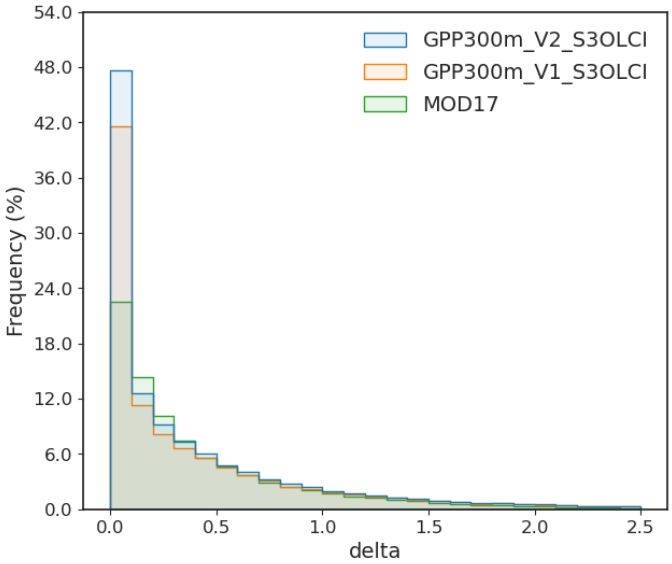
6.10.3 Temporal smoothness

The frequency histograms of the temporal smoothness (also called the intra-annual precision, or δ) of the various GPP datasets are shown in Figure 52 for the year 2019 and in Figure 53 for the year 2023. Median values of δ and the time series relative noise are provided as an indication of the overall intra-annual precision of the products. The GPP 300m V2 dataset is a bit smoother than the previous version GPP 300m V1.0 and GPP 300m V1.1, i.e. the median smoothness statistic is lower for V2, although the results for all datasets are very close to each other. The relative noise of the MOD17 V6.1 dataset is slightly lower than that of GPP 300m V2. This is due to the higher aggregation time (MODIS data are aggregated over 16 days, while the GPP 300m V2 input data are 10-daily aggregations), and the fact that MODIS has two overpasses per day from the Terra and Aqua satellites. The median δ of the GPP 1km V2 dataset is low, because the input FAPAR makes use of a climatology for gap filling.



	Median δ	Relative noise
GPP 300m V2 Sentinel-3/OLCI	0.16	21.9
GPP 300m V2 PROBA-V	0.14	22.4
GPP 300m V1.0	0.18	21.4
GPP 1km V2	0.15	22.9
MOD17 V6.1 GPP	0.24	20.2

Figure 52: Frequency histograms of temporal smoothness (left), and median and time series relative noise per dataset (right) for the year 2019, for GPP 300m V2 Sentinel-3/OLCI (blue), GPP 300m V2 PROBA-V (orange), GPP 300m V1.0 (green), GPP 1km V2 (red), and MOD17 V6.1 GPP (purple).



	Median δ	Relative noise
GPP 300m V2 Sentinel-3/OLCI	0.18	23.0
GPP 300m V1.1	0.19	23.3
MOD17 V6.1 GPP	0.26	20.8

Figure 53: Frequency histograms of temporal smoothness (left), and median and time series relative noise per dataset (right) for the year 2023, for GPP 300m V2 Sentinel-3/OLCI (blue), GPP 300m V1.1 (orange), and MOD17 V6.1 GPP (green).

6.10.4 Stability analysis

To evaluate the general stability of the GPP 300m V2 product, a Hovmöller plot depicting the spatio-temporal evolution of the mean GPP is shown in Figure 54. Seasonal patterns are visible, as expected, with higher GPP in the Northern hemisphere during summertime and an almost constant GPP around

the equator. The transition from PROBA-V to Sentinel-3/OLCI at the start of 2019 results in minor spatiotemporal effects. For PROBA-V, values above 70° North are cut off, while they are still present in Sentinel-3/OLCI. For Sentinel-3/OLCI, the GPP values at the equator are slightly higher, and there also seems to be a trend towards higher GPP values over the years. A similar trend has been observed in the FAPAR 300m V2 dataset [CGLOPS1_VR_FAPAR300m-V2].

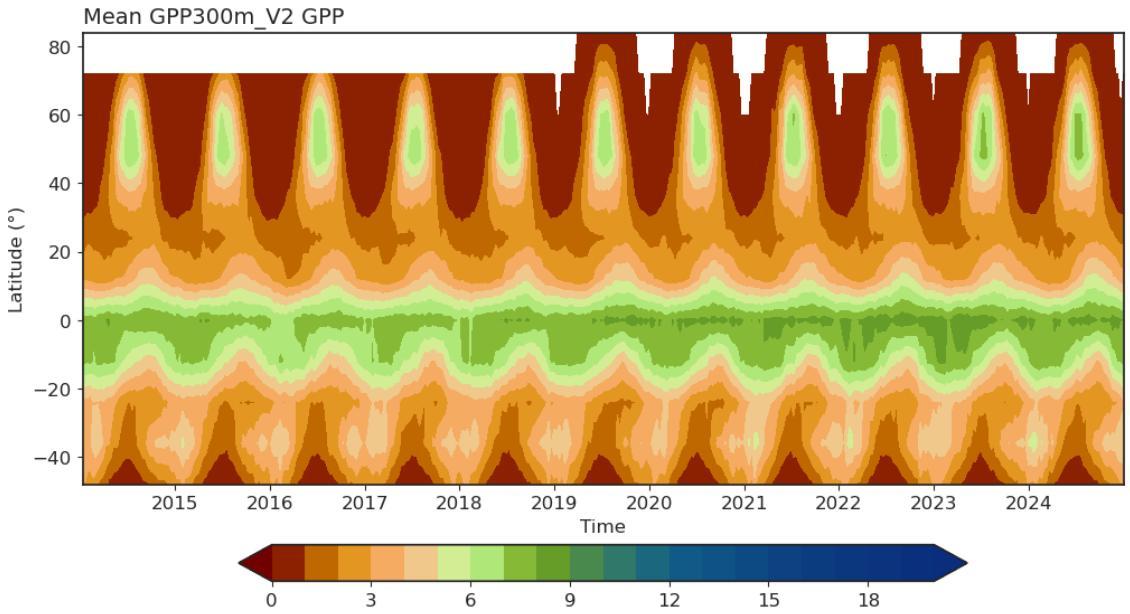


Figure 54: Spatio-temporal evolution of mean GPP of the GPP 300m V2 dataset for the period 2014 – 2024

The inter-annual precision (i.e., dispersion of GPP 300m V2 values from year to year) is assessed over all LANDVAL V2 sites with completeness above 90%.. Sites with completeness below 90% are excluded because gaps also cause changes in the stability statistic, while this is not related to changes in the GPP values themselves. For every quarter, the year up to the end of that quarter is compared to the year before that, using anomalies of the 5th and 95th percentile. Thus, the first precision value that can be computed for the GPP 300m V2 dataset is the last quarter of 2015, using the year 2015 as current dataset and the year 2014 as reference dataset.

The stability of the GPP 300m V2 product is determined as the slope of the linear regression of all quarterly precision values over time. It is expressed as percentage change per decade (10 years). This is illustrated in Figure 55. Overall, the stability of the product is very good, with a value of 0.02 (0.6%), well within the goal requirement of 10% (see §2.1).

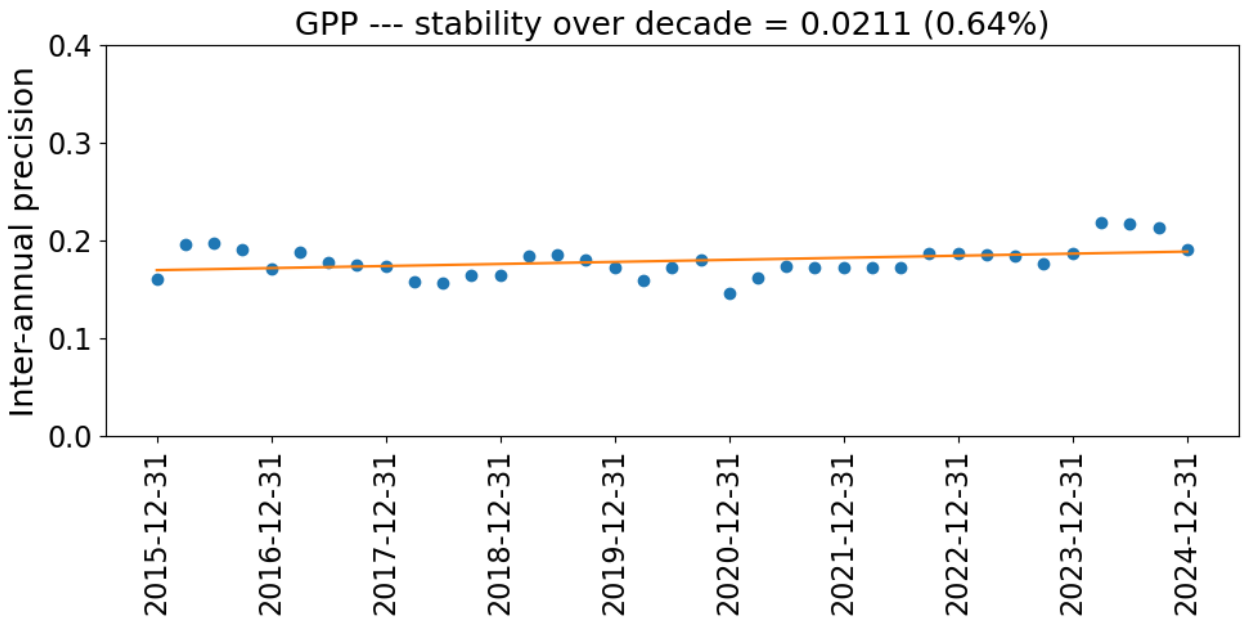


Figure 55: Temporal evolution of inter-annual precision of the GPP 300m V2 product.

6.10.5 Spatio-temporal agreement with MOD17 V6.1

The spatio-temporal variation of the Accuracy metric (MBE) and the Uncertainty metric (RMSD) is illustrated in the Hovmöller plots of Figure 56. This figure shows the comparison of resp. GPP 300m V2, a combination of GPP 300m V1.0 and GPP 300m V1.1, and a combination of GPP 1km V2 and GPP 300m V2 with MOD17 V6.1 over the full time series. Both the MBE and RMSD show yearly patterns, related to seasonal variations in vegetation growth. In general, differences are largest around the equator (i.e. wet tropics, regions with persistent cloud cover) and at high latitudes (i.e. regions with bad illumination conditions and a limited number of good quality observations in winter).

The switch between PROBA-V and Sentinel-3/OLCI is slightly visible both in the top and middle plots of Figure 56, but the differences with MOD17 V6.1 are somewhat lower for the new GPP 300m V2 dataset. This indicates that the temporal consistency is improved in GPP 300m V2. Even though the switch is visible, the agreement between PROBA-V and Sentinel-3/OLCI is very high.

For some applications, a time series over a long period is needed and therefore the 1km (available since 1999) and 300m GPP datasets can be combined. The bottom plots in Figure 56 show the combination of GPP 1km V2 and GPP 300m V2 versus MOD17 V6.1 to investigate the temporal consistency when combining these two datasets. The switch between the 1km and 300m dataset at the start of 2019 is almost invisible. Therefore, the consistency between both datasets is very high, and the GPP 1km V2 dataset can be used to extend the GPP 300 V2 dataset to the past.

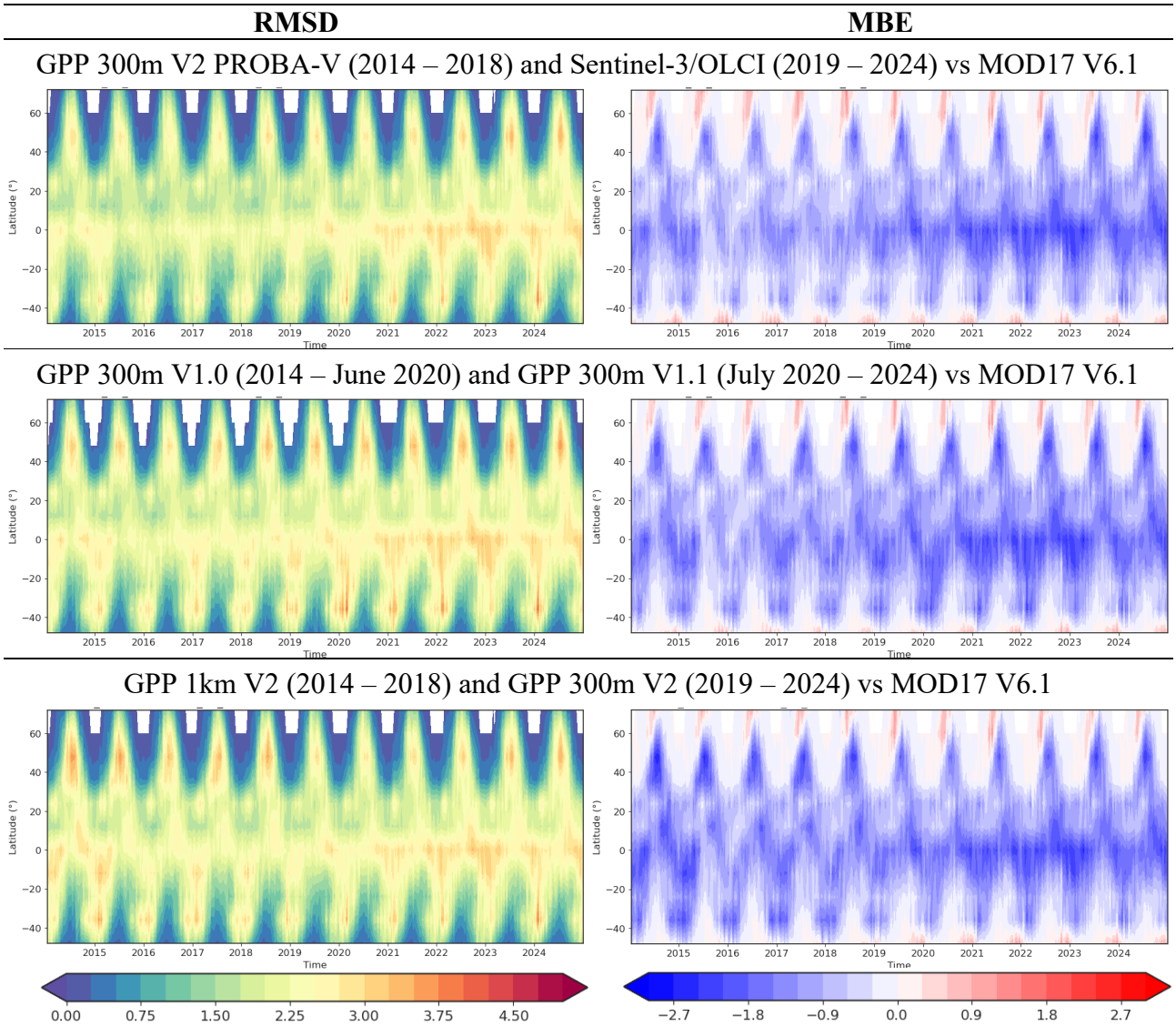


Figure 56: Hovmöller plots of the RMSD (left) and MBE (right) between GPP 300m V2 and MOD17 V6.1 for the period 2014 – 2024 with switch between PROBA-V and Sentinel-3/OLCI at the start of 2019 (top), between GPP 300m V1.0 (2014 – June 2020) + GPP 300m V1.1 (July 2020 – 2024) and MOD17 V6.1 (middle), and between GPP 1km V2 (2014 – 2018) + GPP 300m V2 (2019 – 2024) and MOD17 V6.1 (bottom)

6.10.6 Conclusion

What is the temporal consistency of DMP/GDMP/NPP/GPP 300m V2 in comparison to MOD17 V6.1 GPP?

The temporal profiles of GPP 300m V2 over all LANDVAL V2 sites are in good agreement with the temporal profiles of the MOD17 V6.1 dataset, albeit with in some cases a shift in absolute GPP values due to the inclusion of a drought-stress related factor in MOD17 V6.1. The smoothness of both datasets is comparable, although that of MOD17 V6.1 is a bit higher due to a longer aggregation

period. The transition between PROBA-V and Sentinel-3/OLCI for GPP 300m V2 does not lead to discontinuities in the time series.

How do the results based on DMP/GDMP 300m V2 compare to those of DMP/GDMP 300m V1.0, DMP/GDMP 300m V1.1, and DMP/GDMP 1km V2?

The temporal behaviour of GPP 300m V2 is in general similar to that of the other CLMS GPP datasets. The GPP 300m V2 time series are smoother than the combination of GPP 300m V1.0 and GPP 300m V1.1. Differences between the CLMS temporal profiles can be traced back to differences in the input TOC versions such as the geolocation issue in TOC reflectance V1.1 for the GPP 300m V1.1 dataset and the alignment with SPOT/VGT for the GPP 1km V2 dataset.

Hovmöller plots show that the GPP 300m V2 dataset has overall better temporal consistency than the combination of GPP 300m V1.0 and GPP 300m V1.1. They also indicate that the GPP 1km V2 dataset is suitable to extend the GPP 300m V2 dataset to the past.

What is the stability of the GPP 300m V2 product?

The analysis of the inter-annual precision indicates that the stability of the GPP 300m V2 product is high, with stability criterion computed over 2014 – 2024 equal to 0.02 (0.6%), well below the goal requirement of 10%. The inter-annual precision does not change significantly over the switch between PROBA-V and Sentinel-3/OLCI.

7 CONCLUSIONS AND RECOMMENDATIONS

7.1 CONCLUSIONS

This validation report for the DMP/GDMP/NPP/GPP 300m V2 dataset focussed on the questions listed in §4.1 to evaluate the quality of the product. A short summary of the results per question is given below.

1. *What are the occurrences of quality flags in DMP/GDMP/NPP/GPP 300m V2 in terms of temporal and spatial distribution? Which quality flags are recommended for users and are applied in the subsequent analysis?*

   The only quality flags specific to the DMP/GDMP/NPP/GPP product are flags indicating whether FAPAR data is missing, whether meteorological data is missing, or if low FAPAR values are rescaled. The first two flags are never set for valid DMP/GDMP/NPP/GPP values and can hence be ignored. The rescaled low FAPAR flag is higher during Northern hemisphere winter-time and lower in summer. It is also higher for the Sentinel-3/OLCI dataset which has a lower completeness than PROBA-V. It is recommended to include all DMP/GDMP/NPP/GPP 300m V2 pixels, and to not filter on the QFLAG layer.

2. *What are the temporal variation and spatial distribution of the completeness of DMP/GDMP/NPP/GPP 300m V2? How do the results compare to those of DMP/GDMP 300m V1.0, DMP/GDMP 300m V1.1, DMP/GDMP 1km V2, and MOD17 V6.1 GPP?*

   The product completeness of DMP/GDMP/NPP/GPP 300m V2 is higher than that of previous versions V1.0 and V1.1. This is related to the difference in input TOC data that has been used in each version and the corresponding cloud masking that has been applied. Especially for regions in the Tropics with EBF biome type, the completeness is improved in the latest version.

3. What are the *differences between RT6 (final consolidation mode) and the NRT modes (RT2, RT1, and RT0) in DMP/GDMP/NPP/GPP 300m V2?*

   Differences between the four consolidations modes are very small, with largest deviations present between RT0 and RT6. The RT0 mode contains some artefacts which, due to the short consolidation window, are not removed by the outlier detection algorithm. It is hence advised to use higher RT modes, and preferably RT6, instead.

4. *How do the DMP/GDMP/NPP/GPP 300m V2 PROBA-V and Sentinel-3/OLCI datasets compare visually to the DMP/GDMP 300m V1.0, DMP/GDMP 300m V1.1, and MOD17 V6.1 GPP datasets?*

   GPP 300m V2 quicklooks show similar spatial patterns and sharpness as the reference products. Differences in completeness between the CLMS datasets are caused by differences in completeness of the FAPAR datasets, including the input TOC reflectance datasets, cloud masking algorithms, and the specifics of the FAPAR algorithm for each version.

5. *What is the statistical and spatial consistency between PROBA-V and Sentinel-3/OLCI based DMP/GDMP/NPP/GPP 300m V2?*

   Overall, there is good statistical and spatial consistency between PROBA-V and Sentinel-3/OLCI for DMP/GDMP/NPP/GPP 300m V2. Differences are largest for the EBF biome for which higher Sentinel-3/OLCI FAPAR values are found compared to PROBA-V.

6. *What are the differences between DMP/GDMP/NPP/GPP 300m V2, DMP/GDMP 300m V1.0, and DMP/GDMP 300m V1.1 in terms of statistical and spatial consistency?*

   The consistency between DMP/GDMP/NPP/GPP 300m V2 and previous versions is generally good. There is a small negative bias with DMP/GDMP 300m V1.0 which is largest for the EBF biome due to changes in FAPAR processing. The consistency with DMP/GDMP 300m V1.1 is much better since the FAPAR algorithm is quasi the same for V1.1 and V2. Any difference is linked to differences in input TOC reflectance versions.

7. *What is the statistical and spatial consistency with DMP/GDMP 1km V2?*

   There is good statistical and spatial consistency between DMP/GDMP/NPP/GPP 300m V2 and DMP/GDMP 1km V2, although the 1km dataset provides in general somewhat higher values than the 300m data. The largest differences are again found for the EBF biome, which benefit of gap-filling implemented in the 1km V2 algorithm.

8. *What is the spatial and statistical consistency of DMP/GDMP/NPP/GPP 300m V2 in comparison to MOD17 V6.1 GPP?*

   There is a large positive bias between GPP 300m V2 and MOD17 V6.1 GPP, related to the inclusion of a drought-stress related factor in the latter. The effects are largest for forest biomes and crop lands.

9. *What is the statistical consistency of the DMP/GDMP/NPP/GPP 300m V2 product compared to in-situ FLUXNET GPP data? Is the uncertainty between GPP 300m V2 and FLUXNET within the requirement range?*

   The GPP 300m V2 product is in relatively good agreement with ICOS and Ameriflux GPP data, especially when looking at the temporal patterns that display similar seasonality. For 13% and 26% of the data, the uncertainty between GPP 300m V2 and ICOS/Ameriflux meets the goal and threshold criteria, respectively.

10. *What is the temporal consistency of DMP/GDMP/NPP/GPP 300m V2 in comparison to MOD17 V6.1 GPP? How do the results based on DMP/GDMP 300m V2 compare to those of DMP/GDMP 300m V1.0, DMP/GDMP 300m V1.1, and DMP/GDMP 1km V2?*

    The temporal profiles of DMP/GDMP/NPP/GPP 300m V2 over LANDVAL V2 sites indicate similar temporal behaviour compared to the reference datasets. The temporal profiles of V2 are more stable than for DMP/GDMP 300m V1.0 and have a higher completeness. Hovmöller plots show that the temporal consistency between PROBA-V and Sentinel-3/OLCI is improved in DMP/GDMP/NPP/GPP 300m V2 compared to the combination of DMP/GDMP 300m V1.0 and DMP/GDMP 300m V1.1. They also indicate that the DMP/GDMP 1km V2 dataset is suitable to extend the DMP/GDMP/NPP/GPP 300m V2 dataset to the past in order to have a longer time series. The stability of the DMP/GDMP/NPP/GPP 300m V2 product is within the requirement ranges.

It can be concluded that the GPP 300m V2 product has high product completeness, good stability, and large statistical, spatial, and temporal agreement with reference datasets for the 2014 – 2024 period. Some differences are expected due to differences in TOC reflectance data and accompanying cloud masking, in the FAPAR algorithm versions, and the inclusion/exclusion of a drought-stress factor for GPP calculation. Most discrepancies are minimal and unsystematic. Since the FAPAR algorithm has been updated on the processing of EBF sites, slightly larger differences (both systematic and unsystematic) are observed over this biome (see also [CGLOPS1_VR_FAPAR300m-V2]). From the analysis of the various consolidation modes, it is apparent that some artefacts are present in RT0 which are reduced towards higher consolidation modes (RT1 → RT2 → RT6), with the best results obtained for RT6. The transition between PROBA-V and Sentinel-3/OLCI is minimally visible in the data, well below the stability requirement. The GPP 300m V2 dataset is thus homogeneous over the time series from 2014 to the present.

7.2 RECOMMENDATIONS

From the conclusions of this validation exercise, the following user recommendations can be formulated:

- Users are recommended to include all valid DMP/GDMP/NPP/GPP 300m V2 pixels, and to not filter additionally based on the QFLAG layer. For expert users, the QFLAG layer provides additional information on the FAPAR algorithm (e.g. EBF case or gap filled), and – in case a pixel is not filled – on the reason (e.g. due to missing FAPAR or missing meteorological data).
- Users are recommended to use the RT6 mode for historical analysis and the largest available RT mode available for near real time analyses, since the short consolidation period for the near-real time RT0 mode is sometimes insufficient to remove all cloud contaminated observations.
- If users want to extend their time series analysis backwards up to 1999, this can be done by combining the DMP 300m V2 with the DMP/GDMP 1km V2 dataset (1999 – June 2020), since both datasets are largely statistically and spatially consistent. However, some discrepancies might be observed due to differences in the FAPAR products that were used as input (gap filling applied on all biomes in the FAPAR 1km V2 product, while only on the EBF pixels in the FAPAR 300m V2 product).

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ANNEX 1: DIGITAL ANNEX

The Digital Annex can be downloaded from the CLMS website:
https://land.copernicus.eu/en/technical-library

It contains the following folders:

- QUICKLOOKS:

  Maps over Northern America (NAME), Western Europe (WEUR), South America (SAMI), Sahel (SAHE), and West Africa (WAFR) 10° x 10° tiles, and maps over Brussels (BRU) and Ivory Coast (IVO) 1° x 1° tiles.
  Subfolders per tile (as specified in §5.3.4).
  Filenames: "<DATASET>_<YYYYMMDD>_<TILE>_GPP.png"

- TEMPORAL_PROFILES:

  Temporal profiles for the LANDVAL sites.
  Subfolders per biome (as specified in §5.3.2).
  Filenames: "site<SITE_ID>.png"

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ANNEX 2: ICOS AND AMERIFLUX SITES

Table 17 and Table 18 list the ICOS and Ameriflux sites used in the analysis of §6.9. The ICOS data are taken from the Ecosystem final quality (L2) product in ETC-Archive format – release 2025-1 (https://meta.icos-cp.eu/collections/1-HA2r4l5QUjAgQr5CCEfJe3) and from the FLUXNET2015 dataset (Pastorello et al., 2020). The Ameriflux data has been taken from the FLUXNET2015 archive only.

Table 17: Overview of ICOS sites. IGBP is the International Geosphere-Biosphere Programme vegetation classification and LC300 is the classification according to the CGLOPS Global Land Cover resampled to 300m (see §5.3.2).

Site	IGBP	LC300	Reference
AT-Neu	GRA	HER	https://doi.org/10.18140/FLX/1440121
BE-Bra	MF	NLF	https://doi.org/10.18140/FLX/1440128 https://hdl.handle.net/11676/LkArZFhE0oxjbMSf29M0zr-v
BE-Dor	GRA	HER	https://hdl.handle.net/11676/geqYNPk-fvkZDGtnJI-sq1Je https://hdl.handle.net/11676/fQH-ofhiDWgNY20tcC6X92tK
BE-Lcr	DBF	CRO	https://hdl.handle.net/11676/oc2rvgUbNz-kUIeMsNmvNv8X
BE-Lon	CRO	CRO	https://doi.org/10.18140/FLX/1440129 https://hdl.handle.net/11676/pXMqHUqaOhH2HBBphZfPyuy1
BE-Maa	CSH	MXF	https://hdl.handle.net/11676/ZG_uAsLMIT7yUsnXnJSB5Ufb https://hdl.handle.net/11676/alvFQlxKMyvv0EGMLln5pkzy
BE-Vie	MF	MXF	https://doi.org/10.18140/FLX/1440130 https://hdl.handle.net/11676/wSydovEhV_Qqz-1oppmNgOzR
CH-Aws	GRA	MXF	https://hdl.handle.net/11676/quYooZm1POt6BTN6jvcIeZXb
CH-Cha	GRA	CRO	https://doi.org/10.18140/FLX/1440131
CH-Dav	ENF	NLF	https://doi.org/10.18140/FLX/1440132 https://hdl.handle.net/11676/C9GvCZEmctcfy3ylhxwgdBZw
CH-Frk	GRA	HER	https://hdl.handle.net/11676/5P16NBXaIqCXQWi9gZfYH3Lo
CH-Fru	GRA	CRO	https://doi.org/10.18140/FLX/1440133
CH-Lae	MF	DBF	https://doi.org/10.18140/FLX/1440134
CH-Oe2	CRO	CRO	https://doi.org/10.18140/FLX/1440136
CZ-BK1	ENF	NLF	https://doi.org/10.18140/FLX/1440143 https://hdl.handle.net/11676/89KKJ3j2R37t0BNVSOJ5720y
CZ-KrP	CRO	CRO	https://hdl.handle.net/11676/nMA34HofEPJou2VKmDMTBo_9
CZ-Lnz	MF	DBF	https://hdl.handle.net/11676/B2BjRQCUXetA0_vGI3x7WFUJ https://hdl.handle.net/11676/USap29V42bW-AdF0j5ONfAOy
CZ-RAJ	ENF	NLF	https://hdl.handle.net/11676/LlmTy3u9548FEMXEW59RiWpY
CZ-Stn	DBF	DBF	https://hdl.handle.net/11676/CM9ZMyBcfqBoKkmLPUDwiyM
CZ-wet	WET	CRO	https://doi.org/10.18140/FLX/1440145
DE-Akm	WET	HERW	https://doi.org/10.18140/FLX/1440213
DE-Geb	CRO	CRO	https://doi.org/10.18140/FLX/1440146 https://hdl.handle.net/11676/xz0tNSuk4HX-SjZAlGJUtdwU

DE-Gri	GRA	HER	https://doi.org/10.18140/FLX/1440147
DE-Hai	DBF	DBF	https://doi.org/10.18140/FLX/1440148
DE-Har	MF	NLF	https://hdl.handle.net/11676/RMAJ-gFOgcGACadN_gZG_GqU
DE-Hdn	CRO	CRO	https://hdl.handle.net/11676/gIBXGnpsVQgebVi_Cmd3Hocb
DE-HoH	DBF	DBF	https://hdl.handle.net/11676/-aRYvkkX6D6wd2ijWclesXaK https://hdl.handle.net/11676/f87nZjEcPrK929geFDhBFFsE
DE-Hte	WET	HERW	https://doi.org/10.18140/FLX/1669634
DE-Hzd	DBF	MXF	https://hdl.handle.net/11676/uq6kULT1EqxHKftJz-CqeaJa
DE-Kli	CRO	CRO	https://doi.org/10.18140/FLX/1440149
DE-Msr	ENF	NLF	https://hdl.handle.net/11676/FYr5igpqgstIKkMu0YL-eKyI
DE-Obe	ENF	NLF	https://doi.org/10.18140/FLX/1440151
DE-Rns	CRO	CRO	https://hdl.handle.net/11676/T08mrGq_-E_H2EIzKejmRpFA
DE-RuC	OSH	NLF	https://hdl.handle.net/11676/MKtmPmXNZnCQc-p38RBQGATH
DE-RuR	GRA	HER	https://doi.org/10.18140/FLX/1440215
DE-RuS	CRO	CRO	https://doi.org/10.18140/FLX/1440216 https://hdl.handle.net/11676/_7e5aWDof-DlwljHAggWitni
DE-RuW	ENF	NLF	https://hdl.handle.net/11676/BytRjI_RJ0pRtLQCYfnb2Sd1
DE-SfN	WET	NLF	https://doi.org/10.18140/FLX/1440219
DE-SfS	WET	NLF	https://hdl.handle.net/11676/7LvrgFY84v0co_E4Nfm_PpEK
DE-Tha	ENF	NLF	https://doi.org/10.18140/FLX/1440152 https://hdl.handle.net/11676/qNKn4AM0QueV6jGDUulrduSs
DK-Gds	ENF	NLF	https://hdl.handle.net/11676/vDQDn0hIxnz9gRAIc3duWbUE
DK-Skj	WET	HERW	https://hdl.handle.net/11676/JPkhq8xhlXF20Cijd7pHUd-T
DK-Sor	DBF	DBF	https://doi.org/10.18140/FLX/1440155 https://hdl.handle.net/11676/V4jef_y7u1KnUWY5BA5Xaeph
DK-Vng	CRO	CRO	https://hdl.handle.net/11676/8P6u6WR0FOT-TowVtUqDo0VH
EE-Pts	WET	MXF	https://hdl.handle.net/11676/TvJugTmkPEuepYIbK_ERtfG3
ES-Abr	SAV	MXF	https://doi.org/10.18160/BWYS-J050
ES-LJu	OSH	HER	https://doi.org/10.18140/FLX/1440157
ES-LM1	SAV	MXF	https://hdl.handle.net/11676/9c9e--DRFTTyhtl6c0lcA3O6
ES-LM2	SAV	MXF	https://hdl.handle.net/11676/CgWe1iCQaO3oYYxgSMEe1BSq
ES-LMa	SAV	MXF	https://hdl.handle.net/11676/MSnF-sfvP507KrXcu5yl6LAx
ES-Pdu	WET	CRO	https://hdl.handle.net/11676/Irf83eIco13mKrjYN7jGr4Wf
FI-Hyy	ENF	NLF	https://doi.org/10.18140/FLX/1440158 https://hdl.handle.net/11676/oL1SNLmzDQBfxh2fmyPmgSkO
FI-Ken	ENF	NLF	https://hdl.handle.net/11676/BhlqiN95ECtoSXoxmf0Q4V35
FI-Let	ENF	NLF	https://doi.org/10.18140/FLX/1440227
FI-Qvd	CRO	CRO	https://hdl.handle.net/11676/UuvYrldK3v4VzDgfH6nC0ed7
FI-Sii	WET	MXF	https://hdl.handle.net/11676/mkbfyrZKqg3Djz0tW1rRap1c https://hdl.handle.net/11676/D_8tpErz001XgFe_nbjAC2LI
FI-Sod	ENF	NLF	https://hdl.handle.net/11676/i08hsgs8ttOf9hCCWlBHKU-B
FI-Var	ENF	NLF	https://hdl.handle.net/11676/Ve8588DeHh2Y-t-dtZGWggl6
FR-Aur	CRO	CRO	https://hdl.handle.net/11676/I1_Yl1wLiTSnkW_qrpbFbFw0
FR-Bil	ENF	NLF	https://hdl.handle.net/11676/Qa701KsgkCkpYT4vev--kpOk https://hdl.handle.net/11676/6Gtaz-S_00vHaRILxxf9KBwr
FR-CLt	GRA	HER	https://hdl.handle.net/11676/mbWVxsxcn7lbSZmVE08_nIUR

FR-EM2	CRO	CRO	https://hdl.handle.net/11676/XylBXyJlRDxIhbqaNsGdapw2
FR-FBn	ENF	NLF	https://hdl.handle.net/11676/dw5JqZoWiNVw0JQS-QXykPY6 https://hdl.handle.net/11676/dkNJiwTDeMFJSSlQ4zuEvyhY
FR-Fon	DBF	DBF	https://hdl.handle.net/11676/FRqdGz1Hn6ANHW8o2e_hsv4c
FR-Gri	CRO	CRO	https://doi.org/10.18140/FLX/1440162 https://hdl.handle.net/11676/gEj9nfPY-ImTFfWmZmjCmxg4
FR-Hes	DBF	DBF	https://hdl.handle.net/11676/762ZAgt8A5AGDMT1ry74n13K https://hdl.handle.net/11676/NfVYkVAdGTpqHfumpcqNU_iP
FR-Lam	CRO	CRO	https://hdl.handle.net/11676/JhEP9E7ZyJoy59VmdApHBGzY
FR-Lqu	GRA	CRO	https://hdl.handle.net/11676/hOru2oNW9dquS4oULLggOov2
FR-Lus	GRA	CRO	https://hdl.handle.net/11676/S_NN3abNF1zutV6hnPOuCupl https://hdl.handle.net/11676/e3uQrM8lf2ga3osm2zPnj4g_
FR-Mej	GRA	CRO	https://hdl.handle.net/11676/89dJgrylR97XwCZTAHP5qw5a
FR-Pue	EBF	MXF	https://doi.org/10.18140/FLX/1440164 https://hdl.handle.net/11676/fiekQRTqiAcpQ-uXQSbrfdjw
FR-Tou	GRA	CRO	https://hdl.handle.net/11676/LZSDmUp2MspQPIyTfDwNXYhi
IE-Cra	WET	HER	https://doi.org/10.18160/4EEY-EZSZ
IT-BFt	DBF	DBF	https://hdl.handle.net/11676/IUOlBIndAhz4kdSgYJUDyq-D
IT-BsB	DBF	MXF	https://hdl.handle.net/11676/m7FE4F9G5kn2XaEsWLMCtomo
IT-Bsn	GRA	MXF	https://hdl.handle.net/11676/wlhfJOFvQ_5kotb8D5m_9LK-
IT-Cp2	EBF	MXF	https://doi.org/10.18140/FLX/1440233 https://hdl.handle.net/11676/17IOTr6UHN-InqB626N47BPR
IT-Lav	ENF	NLF	https://doi.org/10.18140/FLX/1440169
IT-Lsn	OSH	CRO	https://hdl.handle.net/11676/wjxZGQl8gUxBPVKPVlu4-g5L
IT-MBo	GRA	HER	https://doi.org/10.18140/FLX/1440170 https://hdl.handle.net/11676/JCNDTBWwUE2geoWtCsSA3JIH
IT-MtM	GRA	MXF	https://hdl.handle.net/11676/wapiBlojkz-8ILRNxAYCMhIk
IT-MtP	GRA	NLF	https://hdl.handle.net/11676/E7nUZjcz96QTa5eNkPaTVFaB
IT-Niv	GRA	HER	https://hdl.handle.net/11676/KEdPYVMu2ozICBTFUa7oRkYX
IT-Noe	CSH	MXF	https://doi.org/10.18140/FLX/1440171
IT-Ren	ENF	NLF	https://doi.org/10.18140/FLX/1440173 https://hdl.handle.net/11676/9y39dDLebENU-sWH4qzvXC5S
IT-SR2	ENF	NLF	https://doi.org/10.18140/FLX/1440236 https://hdl.handle.net/11676/BwK-6iDNKSg7642Ll55MOO69
IT-Tor	GRA	HER	https://doi.org/10.18140/FLX/1440237
NL-Loo	ENF	NLF	https://doi.org/10.18140/FLX/1440178 https://hdl.handle.net/11676/RBbGtSvPuyjgktNpRDUq5fpw
SE-Deg	WET	SHR	https://hdl.handle.net/11676/UaxomkcCXf3DQVhI2WV2AjRz
SE-Htm	ENF	NLF	https://hdl.handle.net/11676/8hMY0lY7kERIfokXkEpev-03 https://hdl.handle.net/11676/KUlZl_YDg5Dkyb0OKxqDiDj2
SE-Lnn	CRO	CRO	https://doi.org/10.18160/5GZQ-S6Z0
SE-Nor	ENF	NLF	https://hdl.handle.net/11676/TQJ13O19n72qJ3MoGHnXCdRB https://hdl.handle.net/11676/qHDB5MwRh-MpbuzvavKUhCa6
SE-Ros	ENF	NLF	https://doi.org/10.18160/RJ1S-CC2Z
SE-St1	WET	HERW	https://doi.org/10.18140/FLX/1440187
SE-Svb	ENF	NLF	https://hdl.handle.net/11676/SxJ5MBCJuCmHLPK4C1fgwGCc

			https://hdl.handle.net/11676/qoFATco_TlPNn2AICLlWhRy3
UK-AMo	WET	HER	https://hdl.handle.net/11676/bUNge2BcADhYDR1h9NHDXbem https://hdl.handle.net/11676/3K_uD1PW4dpFIuri-A05WkxE

Table 18: Overview of Ameriflux sites. IGBP is the International Geosphere-Biosphere Programme vegetation classification and LC300 is the classification according to the CGLOPS Global Land Cover resampled to 300m (see §5.3.2).

Site	IGBP	LC300	Reference
AR-CCg	GRA	HER	https://doi.org/10.17190/AMF/2469434
AR-TF1	WET	DBF	https://doi.org/10.17190/AMF/1818370
AR-TF2	WET	HER	https://doi.org/10.17190/AMF/2571120
BR-Npw	WSA	MXF	https://doi.org/10.17190/AMF/1881563
CA-ARB	WET	HER	https://doi.org/10.17190/AMF/1902821
CA-ARF	WET	SHR	https://doi.org/10.17190/AMF/1902822
CA-BOU	WET	MXF	https://doi.org/10.17190/AMF/2469435
CA-Cbo	DBF	MXF	https://doi.org/10.17190/AMF/1854365
CA-DB2	WET	MXF	https://doi.org/10.17190/AMF/1881564
CA-DBB	WET	MXF	https://doi.org/10.17190/AMF/1881565
CA-EM1	WET	HER	https://doi.org/10.17190/AMF/2469436
CA-EM2	WET	HER	https://doi.org/10.17190/AMF/2571141
CA-ER1	CRO	CRO	https://doi.org/10.17190/AMF/1832154
CA-HPC	ENF	SHR	https://doi.org/10.17190/AMF/2469437
CA-LP1	ENF	NLF	https://doi.org/10.17190/AMF/1832155
CA-Mer	WET	MXF	https://doi.org/10.17190/AMF/2571102
CA-SCB	WET	SHR	https://doi.org/10.17190/AMF/2469439
CA-SCC	ENF	SHR	https://doi.org/10.17190/AMF/2469440
CA-SMC	ENF	NLF	https://doi.org/10.17190/AMF/2571111
CA-TP1	ENF	CRO	https://doi.org/10.17190/AMF/2006962
CA-TP3	ENF	NLF	https://doi.org/10.17190/AMF/1881566
CA-TPD	DBF	DBF	https://doi.org/10.17190/AMF/1881567
CL-ACF	ENF	ENF	https://doi.org/10.17190/AMF/2571129
CL-SDP	WET	HER	https://doi.org/10.17190/AMF/2571108
US-A32	GRA	CRO	https://doi.org/10.17190/AMF/1881568
US-A37	CVM	CRO	https://doi.org/10.17190/AMF/2571154
US-A39	CRO	HER	https://doi.org/10.17190/AMF/2571116
US-A74	CRO	CRO	https://doi.org/10.17190/AMF/2006963
US-Akn	MF	NLF	https://doi.org/10.17190/AMF/2469442
US-ALQ	WET	MXF	https://doi.org/10.17190/AMF/2006964
US-ARM	CRO	CRO	https://doi.org/10.17190/AMF/1854366
US-Bar	DBF	MXF	https://doi.org/10.17190/AMF/2006969
US-Bi1	CRO	CRO	https://doi.org/10.17190/AMF/1871134
US-Bi2	CRO	CRO	https://doi.org/10.17190/AMF/1871135
US-BRG	GRA	MXF	https://doi.org/10.17190/AMF/2006970

US-BZB	WET	MXF	https://doi.org/10.17190/AMF/1881569
US-BZF	WET	SHR	https://doi.org/10.17190/AMF/1881570
US-BZo	WET	MXF	https://doi.org/10.17190/AMF/1881571
US-BZS	ENF	MXF	https://doi.org/10.17190/AMF/1881572
US-CdM	WSA	SHR	https://doi.org/10.17190/AMF/2469445
US-CF1	CRO	CRO	https://doi.org/10.17190/AMF/1832158
US-CF2	CRO	CRO	https://doi.org/10.17190/AMF/1881573
US-CF3	CRO	CRO	https://doi.org/10.17190/AMF/1881574
US-CF4	CRO	CRO	https://doi.org/10.17190/AMF/1881575
US-CGG	GRA	HER	https://doi.org/10.17190/AMF/2469446
US-CRK	ENF	NLF	https://doi.org/10.17190/AMF/2571140
US-CS1	CRO	CRO	https://doi.org/10.17190/AMF/1881576
US-CS2	ENF	NLF	https://doi.org/10.17190/AMF/1881577
US-CS3	CRO	CRO	https://doi.org/10.17190/AMF/1881578
US-CS4	CRO	CRO	https://doi.org/10.17190/AMF/1881579
US-CS5	CRO	CRO	https://doi.org/10.17190/AMF/2469447
US-CS6	CRO	CRO	https://doi.org/10.17190/AMF/2469448
US-CS8	CRO	CRO	https://doi.org/10.17190/AMF/2469449
US-Cst	ENF	NLF	https://doi.org/10.17190/AMF/2469450
US-DFC	CRO	CRO	https://doi.org/10.17190/AMF/2006975
US-Dmg	WET	CRO	https://doi.org/10.17190/AMF/2469451
US-DS1	CRO	CRO	https://doi.org/10.17190/AMF/2571132
US-DS2	CRO	CRO	https://doi.org/10.17190/AMF/2571148
US-DS3	CRO	CRO	https://doi.org/10.17190/AMF/2229376
US-EA5	CSH	MXF	https://doi.org/10.17190/AMF/2571138
US-EA6	WSA	MXF	https://doi.org/10.17190/AMF/2571142
US-EDN	WET	HERW	https://doi.org/10.17190/AMF/1832159
US-Elm	WET	HERW	https://doi.org/10.17190/AMF/2007169
US-EML	OSH	HER	https://doi.org/10.17190/AMF/2007170
US-EvM	WET	HEW	https://doi.org/10.17190/AMF/2571136
US-Fcr	OSH	MXF	https://doi.org/10.17190/AMF/2007172
US-GLE	ENF	NLF	https://doi.org/10.17190/AMF/1871136
US-HB1	WET	HERW	https://doi.org/10.17190/AMF/1832160
US-HB2	ENF	NLF	https://doi.org/10.17190/AMF/2229377
US-HB3	ENF	NLF	https://doi.org/10.17190/AMF/2229378
US-HB4	WET	HERW	https://doi.org/10.17190/AMF/2571130
US-Hn2	GRA	HERW	https://doi.org/10.17190/AMF/1902832
US-Hn3	OSH	HER	https://doi.org/10.17190/AMF/1881580
US-Ho1	ENF	NLF	https://doi.org/10.17190/AMF/2469453
US-Ho2	ENF	NLF	https://doi.org/10.17190/AMF/1881581
US-Hsm	WET	MXF	https://doi.org/10.17190/AMF/2571131
US-HWB	CVM	MXF	https://doi.org/10.17190/AMF/1881582
US-Ich	OSH	HER	https://doi.org/10.17190/AMF/2007175

US-ICs	WET	HER	https://doi.org/10.17190/AMF/1871138
US-ICt	OSH	HER	https://doi.org/10.17190/AMF/1881583
US-Jo1	OSH	SHR	https://doi.org/10.17190/AMF/1902833
US-Jo2	OSH	SHR	https://doi.org/10.17190/AMF/1881584
US-KFS	GRA	HER	https://doi.org/10.17190/AMF/1881585
US-KLS	GRA	HER	https://doi.org/10.17190/AMF/1854367
US-Kon	GRA	CRO	https://doi.org/10.17190/AMF/2316062
US-KLP	WET	NLF	https://doi.org/10.17190/AMF/2571133
US-KS3	WET	HERW	https://doi.org/10.17190/AMF/1881586
US-LA3	WET	HERW	https://doi.org/10.17190/AMF/2571117
US-Me2	ENF	NLF	https://doi.org/10.17190/AMF/1854368
US-Me6	ENF	NLF	https://doi.org/10.17190/AMF/2204871
US-MN1	CRO	CRO	https://doi.org/10.17190/AMF/2571145
US-MN2	CRO	CRO	https://doi.org/10.17190/AMF/2571146
US-MN3	CRO	CRO	https://doi.org/10.17190/AMF/2571147
US-Mo1	CRO	CRO	https://doi.org/10.17190/AMF/2229382
US-Mo2	GRA	CRO	https://doi.org/10.17190/AMF/2229383
US-Mo3	CRO	CRO	https://doi.org/10.17190/AMF/2229384
US-MOz	DBF	DBF	https://doi.org/10.17190/AMF/1854370
US-Mpj	OSH	NLF	https://doi.org/10.17190/AMF/1832161
US-Myb	WET	CRO	https://doi.org/10.17190/AMF/1871139
US-NC3	ENF	MXF	https://doi.org/10.17190/AMF/2204872
US-NC4	WET	NLF	https://doi.org/10.17190/AMF/1902837
US-NGB	SNO	HERW	https://doi.org/10.17190/AMF/1832162
US-NGC	GRA	SHR	https://doi.org/10.17190/AMF/1902838
US-NR1	ENF	NLF	https://doi.org/10.17190/AMF/1871141
US-ONA	GRA	HER	https://doi.org/10.17190/AMF/1832163
US-Orv	WET	MXF	https://doi.org/10.17190/AMF/1832164
US-OWC	WET	MXF	https://doi.org/10.17190/AMF/1871142
US-PAS	GRA	MXF	https://doi.org/10.17190/AMF/2571135
US-RGA	CRO	CRO	https://doi.org/10.17190/AMF/2204873
US-RGB	CRO	CRO	https://doi.org/10.17190/AMF/2204874
US-RGF	CRO	CRO	https://doi.org/10.17190/AMF/2469466
US-RGo	CRO	CRO	https://doi.org/10.17190/AMF/2204875
US-RGW	CRO	CRO	https://doi.org/10.17190/AMF/2469467
US-Rms	CSH	HER	https://doi.org/10.17190/AMF/1881587
US-Ro1	CRO	CRO	https://doi.org/10.17190/AMF/1881588
US-Ro2	CRO	CRO	https://doi.org/10.17190/AMF/2204876
US-Ro4	GRA	CRO	https://doi.org/10.17190/AMF/1881589
US-Ro5	CRO	CRO	https://doi.org/10.17190/AMF/1818371
US-Ro6	CRO	CRO	https://doi.org/10.17190/AMF/1881590
US-Rpf	DBF	SHR	https://doi.org/10.17190/AMF/2229388
US-RRC	WET	MXF	https://doi.org/10.17190/AMF/2469469

US-Rwf	CSH	HER	https://doi.org/10.17190/AMF/1881591
US-Rws	OSH	HER	https://doi.org/10.17190/AMF/1881592
US-Seg	GRA	HER	https://doi.org/10.17190/AMF/1984572
US-Ses	OSH	SHR	https://doi.org/10.17190/AMF/1984573
US-SHC	ENF	NLF	https://doi.org/10.17190/AMF/2571110
US-Slt	DBF	MXF	https://doi.org/10.17190/AMF/2571104
US-Sne	GRA	MXF	https://doi.org/10.17190/AMF/1871144
US-Snf	GRA	CRO	https://doi.org/10.17190/AMF/1854371
US-SP1	ENF	NLF	https://doi.org/10.17190/AMF/2469470
US-SRC	OSH	SHR	https://doi.org/10.17190/AMF/1871145
US-SRG	GRA	SHR	https://doi.org/10.17190/AMF/2204877
US-SRM	WSA	SHR	https://doi.org/10.17190/AMF/2475756
US-Srr	WET	MXF	https://doi.org/10.17190/AMF/2229389
US-SRS	WSA	SHR	https://doi.org/10.17190/AMF/2204878
US-SSH	DBF	DBF	https://doi.org/10.17190/AMF/2469471
US-StJ	WET	HER	https://doi.org/10.17190/AMF/2229390
US-Syv	MF	DBF	https://doi.org/10.17190/AMF/2204879
US-Ton	WSA	MXF	https://doi.org/10.17190/AMF/2204880
US-Tw1	WET	CRO	https://doi.org/10.17190/AMF/1832165
US-Tw3	CRO	CRO	https://doi.org/10.17190/AMF/1881594
US-Tw5	WET	CRO	https://doi.org/10.17190/AMF/1881595
US-UC1	CVM	CRO	https://doi.org/10.17190/AMF/2469472
US-UC2	CVM	CRO	https://doi.org/10.17190/AMF/2469473
US-UMB	DBF	DBF	https://doi.org/10.17190/AMF/2204882
US-UMd	DBF	DBF	https://doi.org/10.17190/AMF/1881597
US-UTB	BSV	HERW	https://doi.org/10.17190/AMF/2469474
US-UTD	CRO	HER	https://doi.org/10.17190/AMF/2571139
US-UTV	CRO	SHR	https://doi.org/10.17190/AMF/2571143
US-Var	GRA	HER	https://doi.org/10.17190/AMF/1993904
US-Vcm	ENF	HER	https://doi.org/10.17190/AMF/2229391
US-Vcp	ENF	NLF	https://doi.org/10.17190/AMF/2229392
US-Whs	OSH	SHR	https://doi.org/10.17190/AMF/1984574
US-Wjs	SAV	SHR	https://doi.org/10.17190/AMF/1871146
US-Wkg	GRA	SHR	https://doi.org/10.17190/AMF/1984575
US-xAE	GRA	HER	https://doi.org/10.17190/AMF/1985434
US-xBL	DBF	CRO	https://doi.org/10.17190/AMF/2229405
US-xBN	ENF	MXF	https://doi.org/10.17190/AMF/2229406
US-xBR	DBF	DBF	https://doi.org/10.17190/AMF/1881598
US-xCL	GRA	MXF	https://doi.org/10.17190/AMF/1985435
US-xCP	GRA	CRO	https://doi.org/10.17190/AMF/1985436
US-xDC	GRA	HER	https://doi.org/10.17190/AMF/1985437
US-xDJ	ENF	NLF	https://doi.org/10.17190/AMF/2229407
US-xDL	MF	MXF	https://doi.org/10.17190/AMF/1985438

US-xDS	CVM	HER	https://doi.org/10.17190/AMF/1985439
US-xGR	DBF	DBF	https://doi.org/10.17190/AMF/1985440
US-xHA	DBF	DBF	https://doi.org/10.17190/AMF/1985441
US-xHE	OSH	HER	https://doi.org/10.17190/AMF/1985442
US-xJE	ENF	MXF	https://doi.org/10.17190/AMF/1985443
US-xJR	OSH	HER	https://doi.org/10.17190/AMF/2229408
US-xKA	GRA	CRO	https://doi.org/10.17190/AMF/1985444
US-xKZ	GRA	CRO	https://doi.org/10.17190/AMF/1985445
US-xMB	OSH	HER	https://doi.org/10.17190/AMF/1985446
US-xML	DBF	DBF	https://doi.org/10.17190/AMF/1985447
US-xNG	GRA	HER	https://doi.org/10.17190/AMF/1985448
US-xNQ	OSH	HER	https://doi.org/10.17190/AMF/1985449
US-xNW	ENF	HER	https://doi.org/10.17190/AMF/2571112
US-xRM	ENF	NLF	https://doi.org/10.17190/AMF/1985450
US-xSB	ENF	HER	https://doi.org/10.17190/AMF/1985451
US-xSC	DBF	DBF	https://doi.org/10.17190/AMF/2229409
US-xSE	DBF	DBF	https://doi.org/10.17190/AMF/1985452
US-xSJ	SAV	MXF	https://doi.org/10.17190/AMF/2229410
US-xSL	CRO	HER	https://doi.org/10.17190/AMF/2229411
US-xSR	OSH	SHR	https://doi.org/10.17190/AMF/1985453
US-xST	DBF	DBF	https://doi.org/10.17190/AMF/1985454
US-xTA	ENF	NLF	https://doi.org/10.17190/AMF/1985455
US-xTL	WET	HER	https://doi.org/10.17190/AMF/2469475
US-xTR	DBF	DBF	https://doi.org/10.17190/AMF/1985456
US-xUK	DBF	DBF	https://doi.org/10.17190/AMF/1985457
US-xUN	MF	DBF	https://doi.org/10.17190/AMF/1985458
US-xWD	GRA	HER	https://doi.org/10.17190/AMF/2229412
US-xWR	ENF	NLF	https://doi.org/10.17190/AMF/2571118
US-xYE	ENF	HER	https://doi.org/10.17190/AMF/1985459
US-YK2	WET	HER	https://doi.org/10.17190/AMF/2469476

ANNEX 3: SCIENTIFIC QUALITY MONITORING 2025

OBJECTIVE

This annex summarises the results of the scientific quality monitoring (SQM) of the DMP/GDMP/NPP/GPP 300m V2 product for the period January – December 2025. The SQM is the annual scientific validation of the most recent products to check the quality and stability of the product over time. The year under evaluation (i.e. 2025) is compared to a short-term statistic (STS) which serves as reference dataset. The results of this comparison are evaluated with respect to the results of the overall quality assessment in the main document.

METHODOLOGY

A subset of the procedures as applied in the main VR is repeated for the GPP 300m V2 products covering the period January – December 2025. The same sampling strategy is used as in the main VR (global subsamples and LANDVAL V2 extractions), but instead of comparing to other datasets, the products of 2025 are compared to an STS dataset. The STS is created for all 36 dekads in a year and is computed as the median per-pixel value of the GPP 300m V2 product over the years 2020 to 2024. The median is chosen to minimise the influence of extreme outliers on the STS. The period 2020 – 2024 is chosen to have a reference dataset of the most recent five years, representing the average state of GPP at present, that has been fully validated. We did not include all 11 years (2014 – 2024) that were validated in the main VR, because GPP is expected to change over time due to changes in climate and land cover/land use. The dataset under evaluation, i.e. the period of January – December 2025, is also referred to as the 'ACT' period.

Because DMP, GDMP, NPP, and GPP are related by a constant factor (see equations in section 3), the validation results for one of the products is also valid for the other three products. Therefore, the SQM is only performed for GPP, as in the main VAR.

RESULTS

Quality flags

The spatial and temporal distribution of the low FAPAR correction flag (QFLAG bit 6) of the GPP 300m V2 product for the period January – December 2025 is shown in Figure 57. Both the spatial and the temporal patterns are similar to the Sentinel-3/OLCI patterns for 2019 in the main VR (see right hand side of Figure 9).

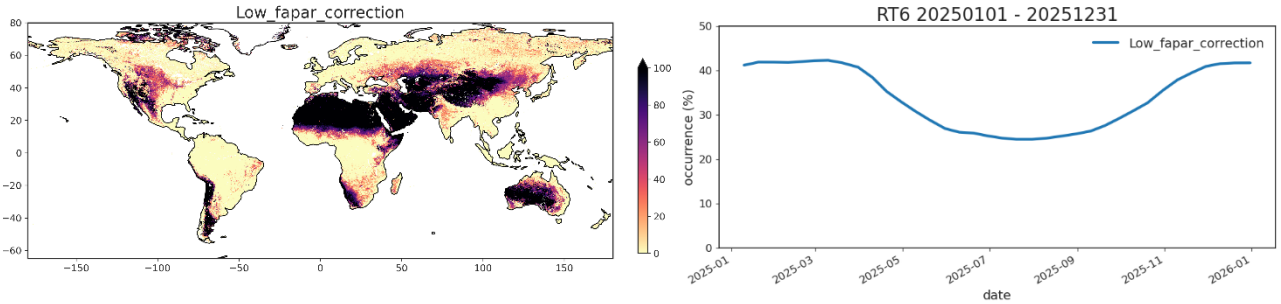


Figure 57: Spatial (*left*) and temporal (*right*) distribution of number of low FAPAR correction (QFLAG bit 6) occurrences over land pixels for GPP 300m V2 for the year 2025.

Product completeness

Figure 58 shows the spatial distribution of the percentage of missing observations of the GPP 300m V2 product during January – December 2025 (left) and the temporal evolution of the percentage of valid observations over land during this period and for the STS (right). The spatial data completeness is similar to that of the Sentinel-3/OLCI dataset and other reference datasets in the main VR (see Figure 10 and Figure 11). The temporal distribution is also similar to that of Figure 12 of the main VR. The same phenological cycle is visible in both figures with increasing GPP from January to northern hemisphere summer-time and decreasing GPP in the second part of the year. The number of valid observations is lower for the ACT dataset (i.e. the GPP 300m V2 dataset for the year 2025) than for the STS, for which each pixel is more likely to have a valid observation since data from five years are combined.

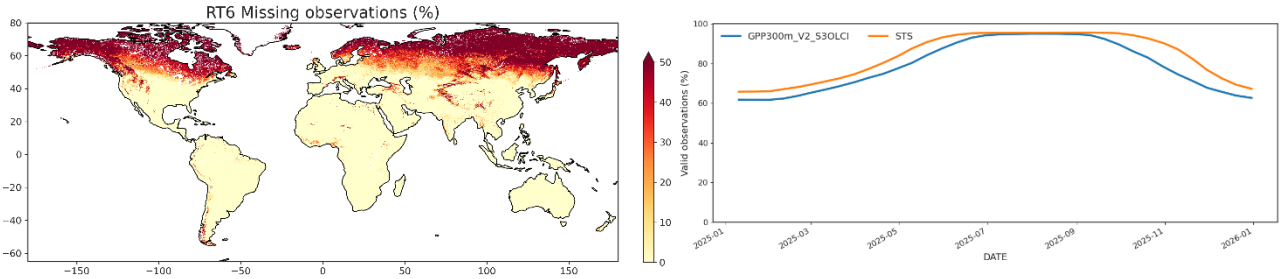
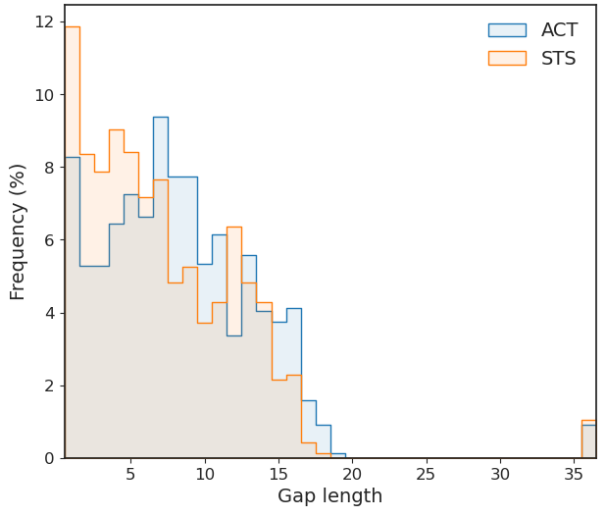


Figure 58: Spatial distribution of the percentage of missing observations (*left*) and temporal evolution of the percentage of valid observations over land (*right*) based on a global subsample of the GPP 300m V2 product for the year 2025 (ACT, blue) and the STS (orange).

The frequency distribution of gap lengths (i.e. the number of consecutive dekads with missing data points) for the ACT and STS datasets is shown in Figure 59. The January – December 2025 dataset has a similar distribution of gap length compared to the STS and to the reference datasets in the main VR (see Figure 13 and Figure 14), i.e. most gaps have a length between one and 17 dekads. The STS gap lengths are in general a bit shorter than for the ACT dataset which can be explained by the fact that the STS is the median over five years of data.



Dataset	Mean gap length (dekads)
ACT (2025)	8.3
STS	7.1

Figure 59: Histograms of gap length over all 2000 LANDVAL V2 sites (*left*) and mean gap length (*right*) for the GPP 300m V2 January – December 2025 dataset (ACT, blue) and the STS (orange).

Statistical consistency

The frequency histogram of the overall bias and the GM regression over all LANDVAL V2 sites between the GPP 300m V2 January – December 2025 dataset and the STS are shown in Figure 60. The bias histogram is centred around zero with mean bias = 0.16 gC/m^2/day and median bias = 0 gC/m^2/day. The histogram is slightly skewed towards positive values, indicating that GPP 300m V2 for 2025 is, in general, somewhat higher than the STS. The scatterplot shows the same bias, although the GM regression line is close to the 1:1 line and has a slope of 0.96. Overall, there is good agreement between the two datasets. Some differences are expected because different periods are compared for which variation in vegetation growth and production is natural.

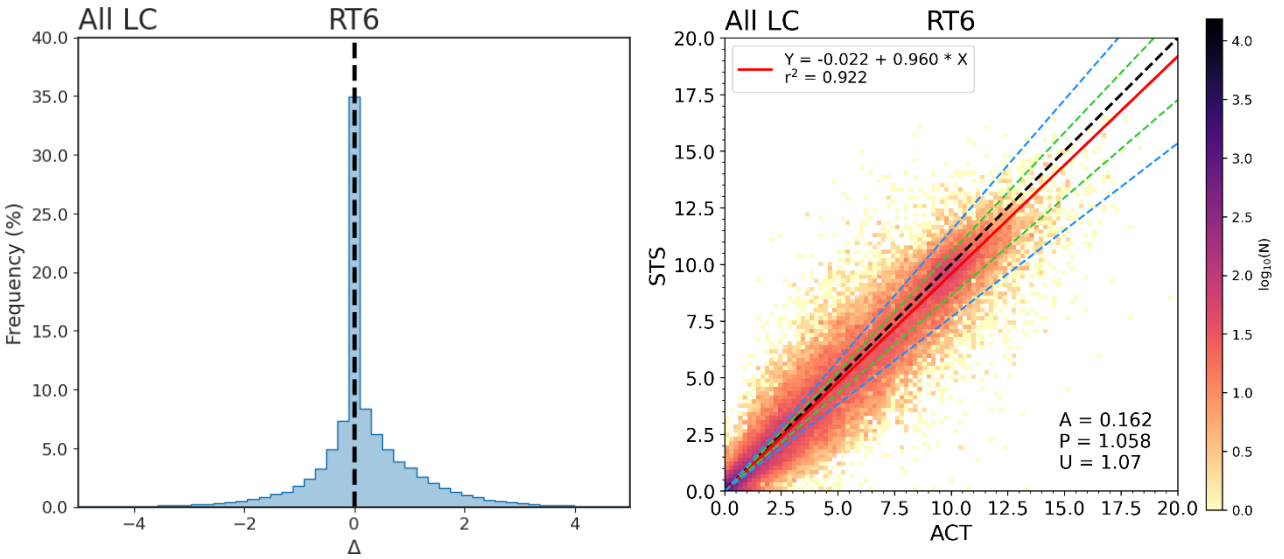


Figure 60: Frequency histogram of the overall bias (*left*) and scatter density plot (*right*) between the GPP 300m V2 January – December 2025 dataset and the STS for all pairwise samples.

Table 19 contains the statistics related to the GM regression over all land cover types. The last column are the percentages of LANDVAL V2 sites with uncertainty within the goal and threshold requirements. The percentages are relatively low because two different periods are compared to each other.

Table 19: Statistics of comparison between GPP 300m V2 January – December 2025 and the STS over all LANDVAL V2 sites. The last column contains the percentage of pixels with uncertainty within the goal (10%) and threshold (20%) range. The absolute values of the statistics are given in gC/m²/day.

N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset / slope	% Goal / Threshold
58267	0.96	0.16 (5%)	0.0 (0%)	1.1 (32%)	0.3 (9%)	1.1 (32%)	-0.02 / 0.96	25 / 40

Spatial consistency

The spatial consistency between GPP 300m V2 January – December 2025 and the STS is analysed with histograms over different biomes (Figure 61). The histograms show good agreement between the two datasets. There are small discrepancies for forest biome sites (EBF, MXF, DBF, and NLF) for which natural variation in vegetation production is expected between different years.

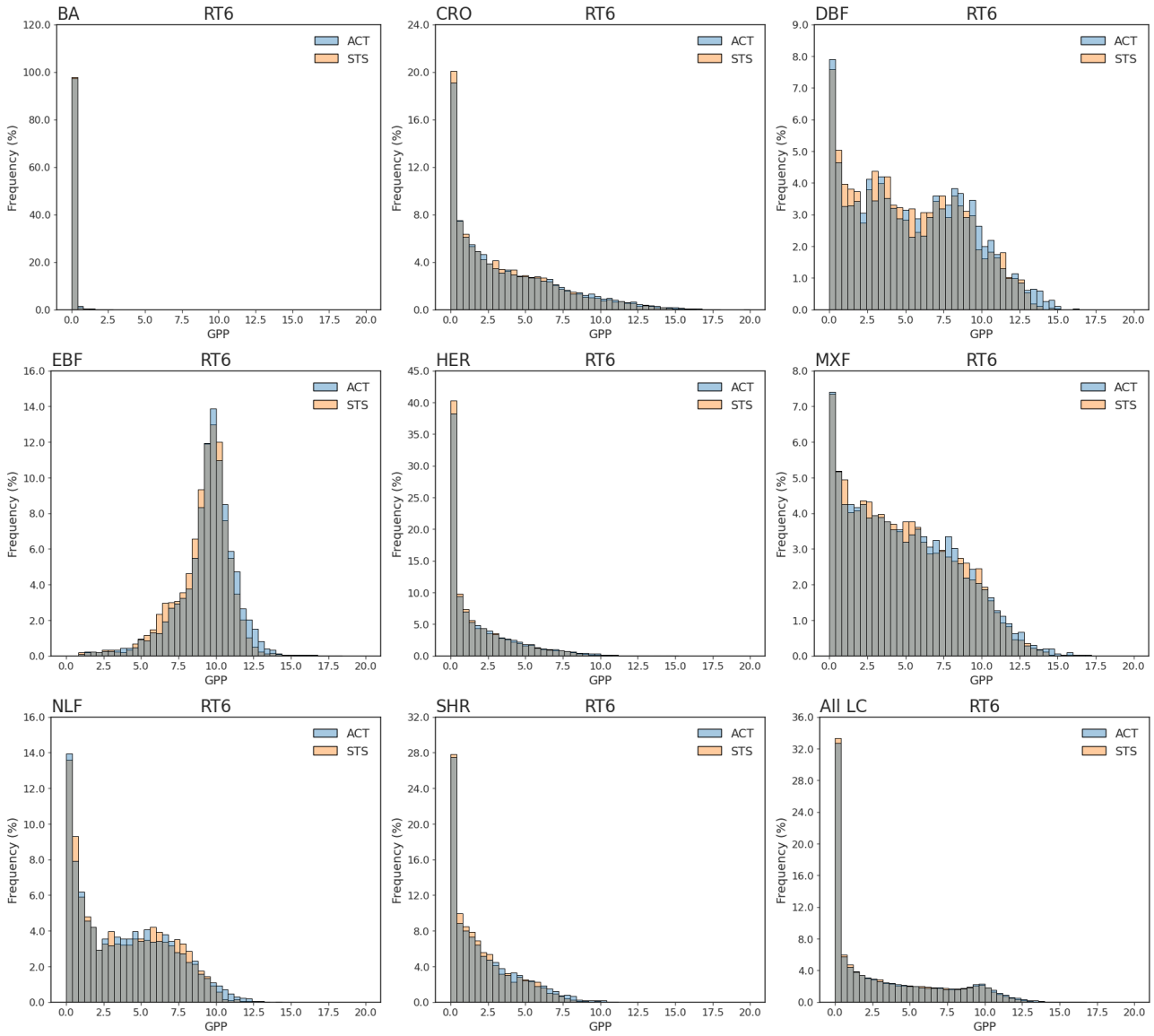


Figure 61: Frequency histograms per biome and for all land cover types comparing the GPP 300m V2 dataset for the year 2025 (ACT, blue) with the STS (orange). Grey colour indicates where both histograms overlap.

Figure 62 shows the spatial patterns of pixel wise APU validation metrics between GPP 300m V2 January – December 2025 and the STS. The validation metrics are closest to zero for deserts and other bare areas (e.g. the Sahara, the Arabian Peninsula, and part of Australia). The regions with the highest precision and uncertainty metric values (i.e. South America, South Africa, Northern Europe, and Russia) are known to have experienced severe droughts during 2025. For these regions, a larger difference with the STS is expected.

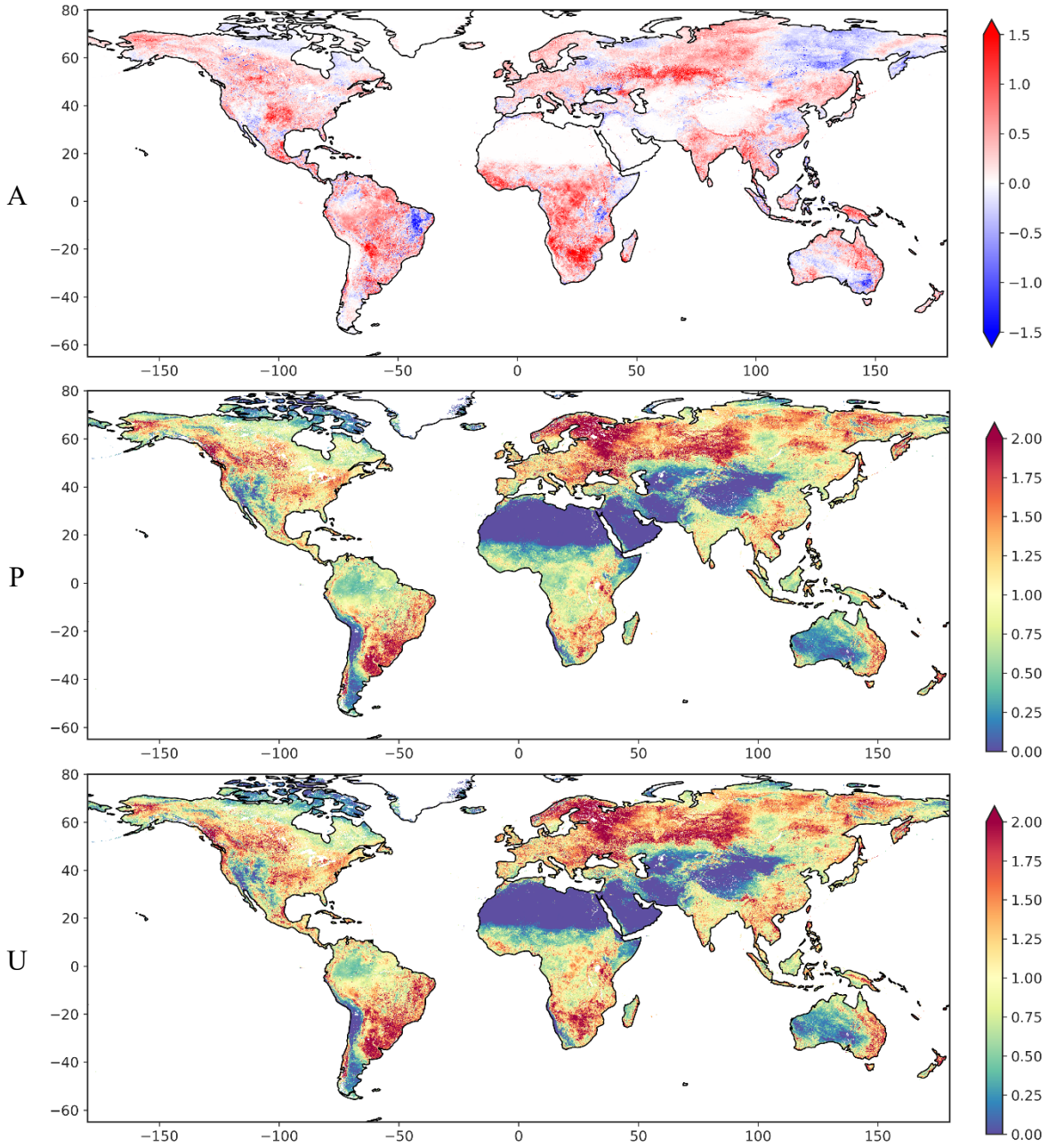


Figure 62: Spatial patterns of pixel-wise APU between the 2025 GPP 300m V2 product and the STS for all pairwise valid GPP values.

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Temporal consistency

Figure 63 and Figure 64 show the temporal evolution of percentiles (P5, P25, P50, P75, and P95) for the GPP 300m V2 January – December 2025 dataset and the STS over all LANDVAL V2 sites for all landcover types and per biome, respectively. Pairwise masking is applied in such a way that the percentiles for each dekad are based on the same set of pixel extractions in both datasets. The temporal evolution of 2025 is in good agreement with that of the STS. As always, the P5 percentile for the ACT dataset overlaps exactly with that of the STS for the BA, SHR, and HER biomes, because of the GPP equal to zero values for these biome types. For the BA class, very little datapoints deviate from zero and therefore only the P95 percentile has a temporal pattern that changes over time.

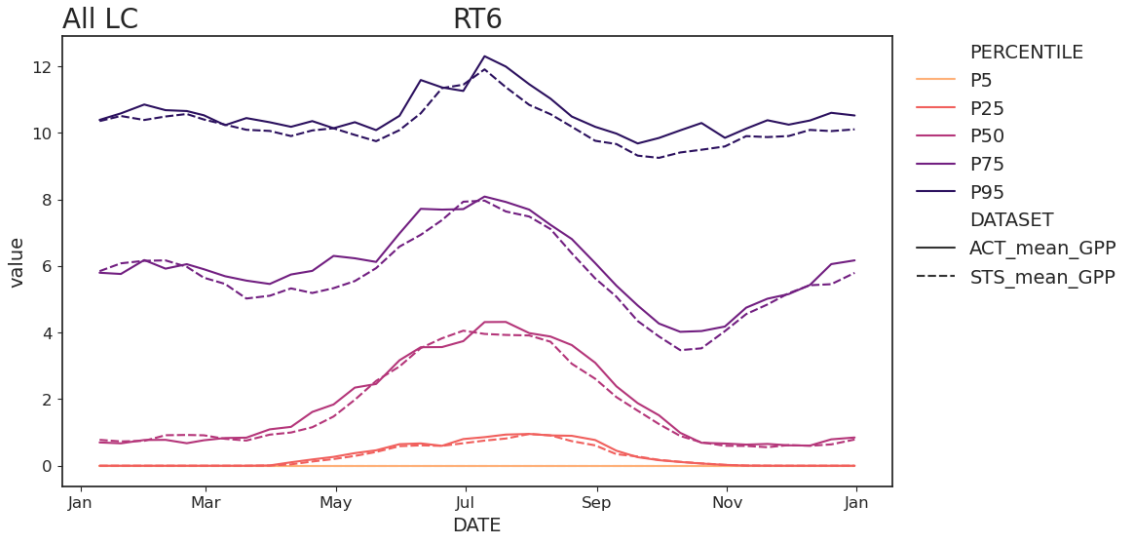


Figure 63: Temporal evolution of percentiles (P5, P25, P50, P75, and P95) for GPP 300m V2 January – December 2025 (ACT, solid lines) and the STS (dashed lines) over all biome types. Pairwise masking between ACT and STS is applied per dekad.

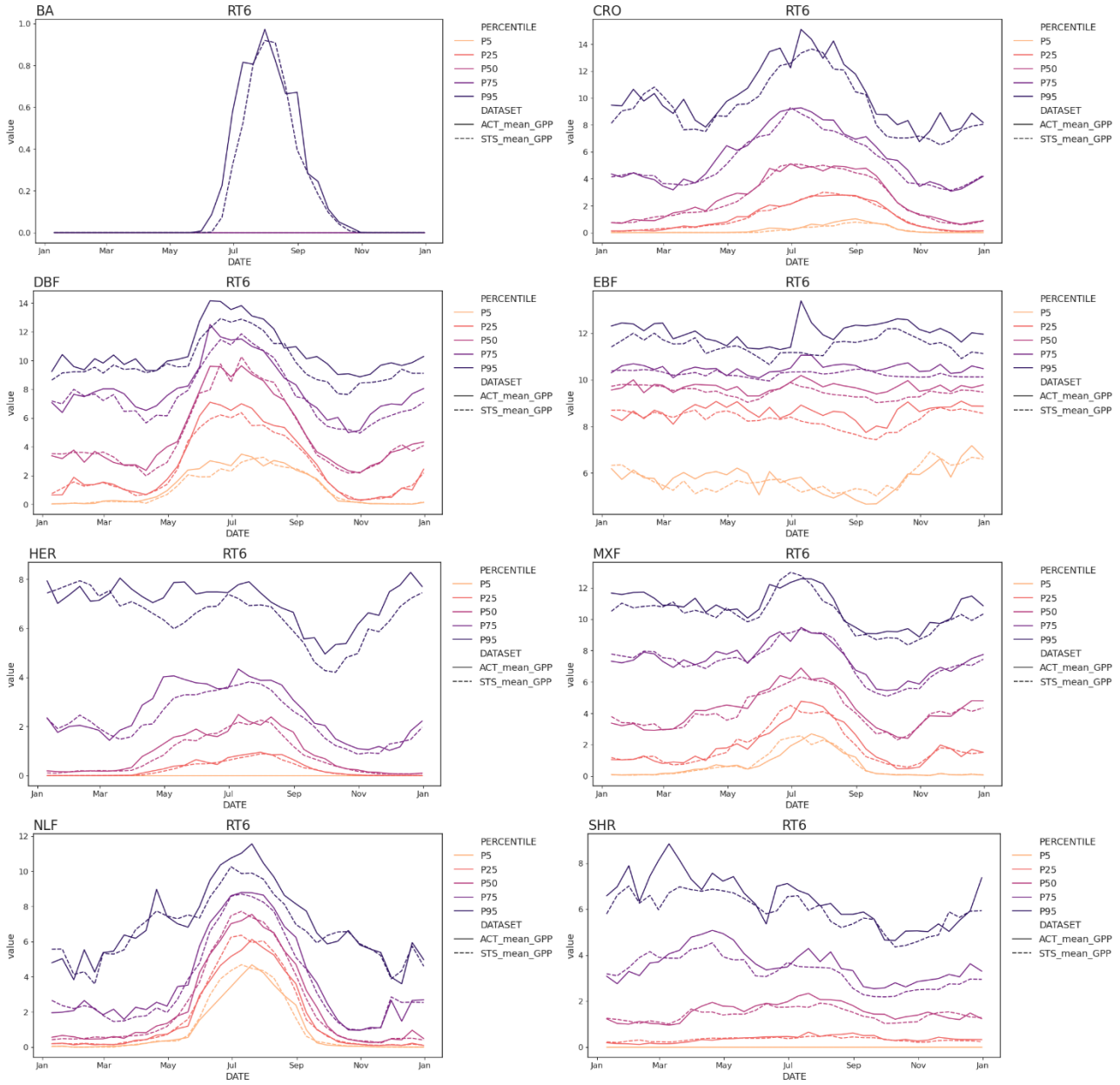
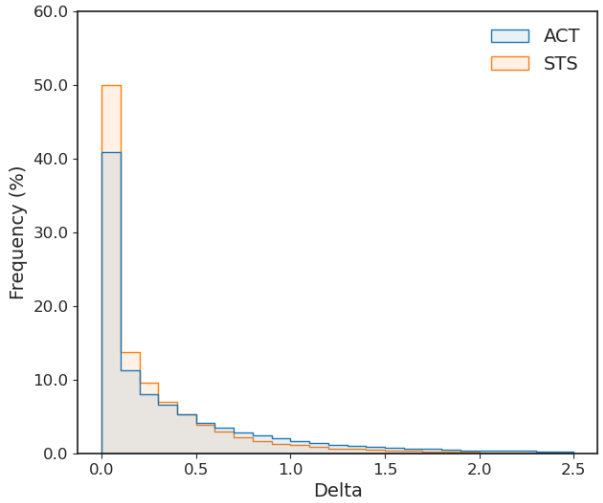


Figure 64: Temporal evolution of percentiles (P5, P25, P50, P75, and P95) for GPP 300m V2 January – December 2025 (ACT, solid lines) and the STS (dashed lines) per biome type.

The frequency histograms of the temporal smoothness (or delta, δ) for the GPP 300m V2 January – December 2025 dataset and the STS dataset are presented in Figure 65. The STS dataset has a higher smoothness because observations are averaged over five years. The values of the median temporal smoothness and the relative noise of ACT are comparable to those of the OLCI reference 2019 and 2023 datasets in §6.10.3.



	Median δ	Relative noise
ACT (2025)	0.17	21
STS	0.11	15

Figure 65: Frequency histogram of temporal smoothness (δ) for GPP 300m V2 January – December 2025 (ACT, blue) and the STS (orange) on the left, and median temporal smoothness and relative noise on the right.

The stability analysis from §6.10.4 is extended to include the period from January – December 2025. The temporal evolution of inter-annual precision is shown in Figure 66. The stability over the entire time frame is equal to 0.02 (0.7%), well within the goal requirement of 10% (see §2.1).

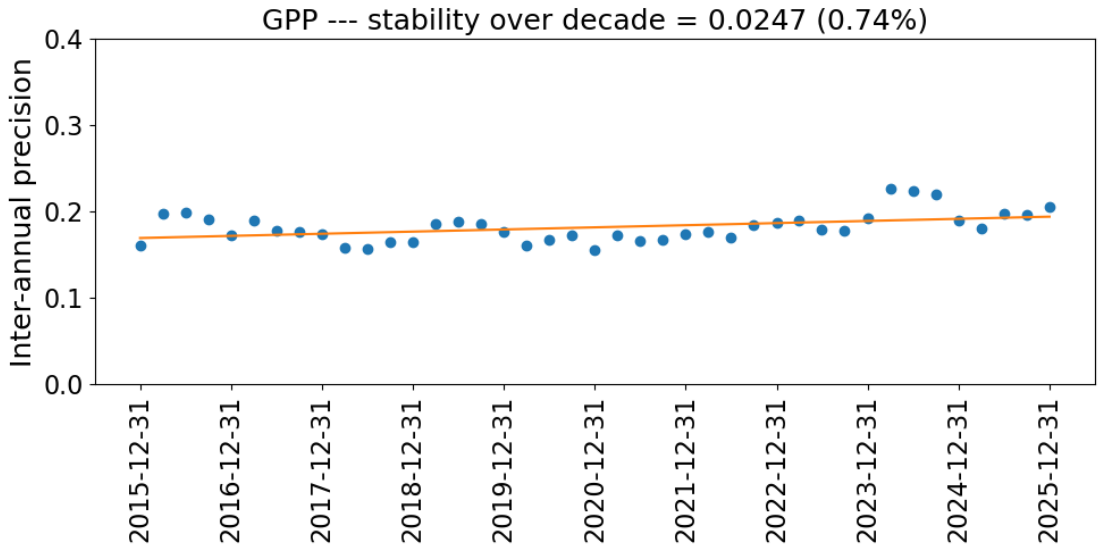


Figure 66: Temporal evolution of the inter-annual precision of the GPP 300m V2 product from 2014 to 2025.

Comparison with in-situ FLUXNET data

The accuracy of the GPP 300m V2 January – December 2025 product is assessed by comparing the GPP estimates with FLUXNET measurements. The analysis of §6.9 is repeated for the new products and therefore additional FLUXNET data for the year 2025 were downloaded and prepared. In total, data for 33 ICOS sites and for 21 Ameriflux sites were available. The sites are listed in Table 20, detailed site characteristics are in Table 17 and Table 18.

Table 20: Overview of FLUXNET sites used in the SQM of 2025.

Dataset	Sites used in SQM 2025 analysis
ICOS	BE-Bra, BE-Dor, BE-Lon, BE-Maa, BE-Vie, CH-Dav, CZ-BK1, CZ-Lnz, DE-Geb, DE-HoH, DE-RuS, DE-Tha, DK-Sor, FI-Hyy, FI-Sii, FI-Sod, FR-Bil, FR-FBn, FR-Fon, FR-Gri, FR-Hes, FR-Lqu, FR-Lus, FR-Pue, IT-Cp2, IT-MBo, IT-Ren, IT-SR2, NL-Loo, SE-Deg, SE-Htm, SE-Svb, UK-AMo
Ameriflux	US-BZB, US-BZo, US-CRK, US-Dmg, US-Hsm, US-ICh, US-ICs, US-Me6, US-Mpj, US-Myb, US-Ro4, US-Seg, US-Ses, US-SRG, US-SRM, US-Ton, US-UMd, US-UTE[11]*, US-UTJ[12]*, US-Vcp, US-Wjs * New sites for the year 2025

Frequency histograms of the bias and scatterplots of the GPP 300m V2 January – December 2025 dataset compared to FLUXNET data are shown in Figure 67. The relation between GPP 300m V2 on one hand and ICOS or Ameriflux on the other hand is similar to what was found in §6.9.2. The ICOS data is, on average, slightly higher than the GPP 300m V2 product, while the Ameriflux data is, on average, a bit lower. For both FLUXNET datasets, a large spread is present, but the bias histograms are centred around zero.

Similar conclusions can be drawn from the statistics values in Table 21. The APU values are slightly worse for the 2025 period because a much smaller sample size is available compared to the analysis in the main VR. That is also the reason why the percentages of pixels with uncertainty within the goal or threshold range is lower than before.

[11] https://ameriflux.lbl.gov/sites/siteinfo/US-UTE
[12] https://ameriflux.lbl.gov/sites/siteinfo/US-UTJ

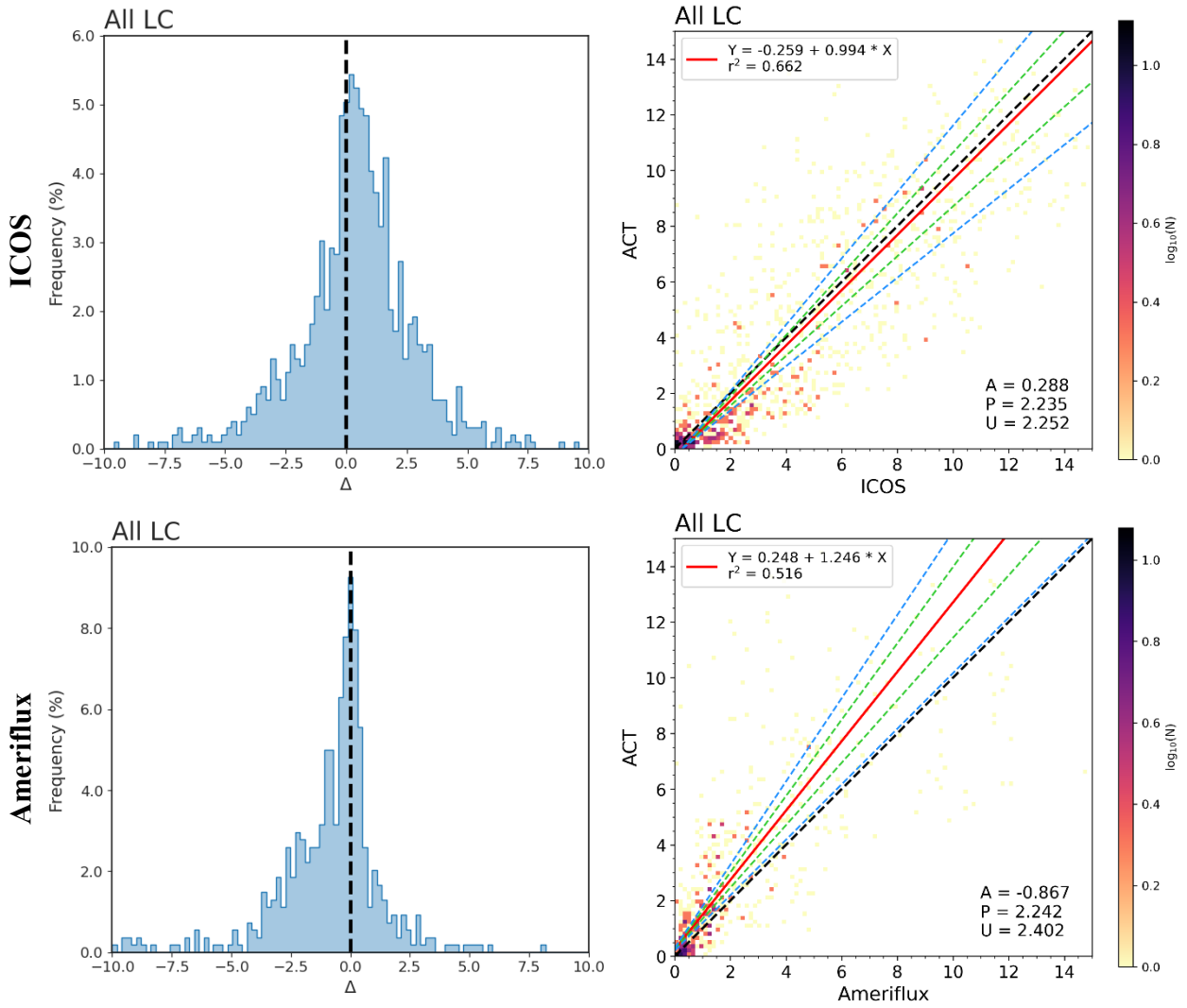
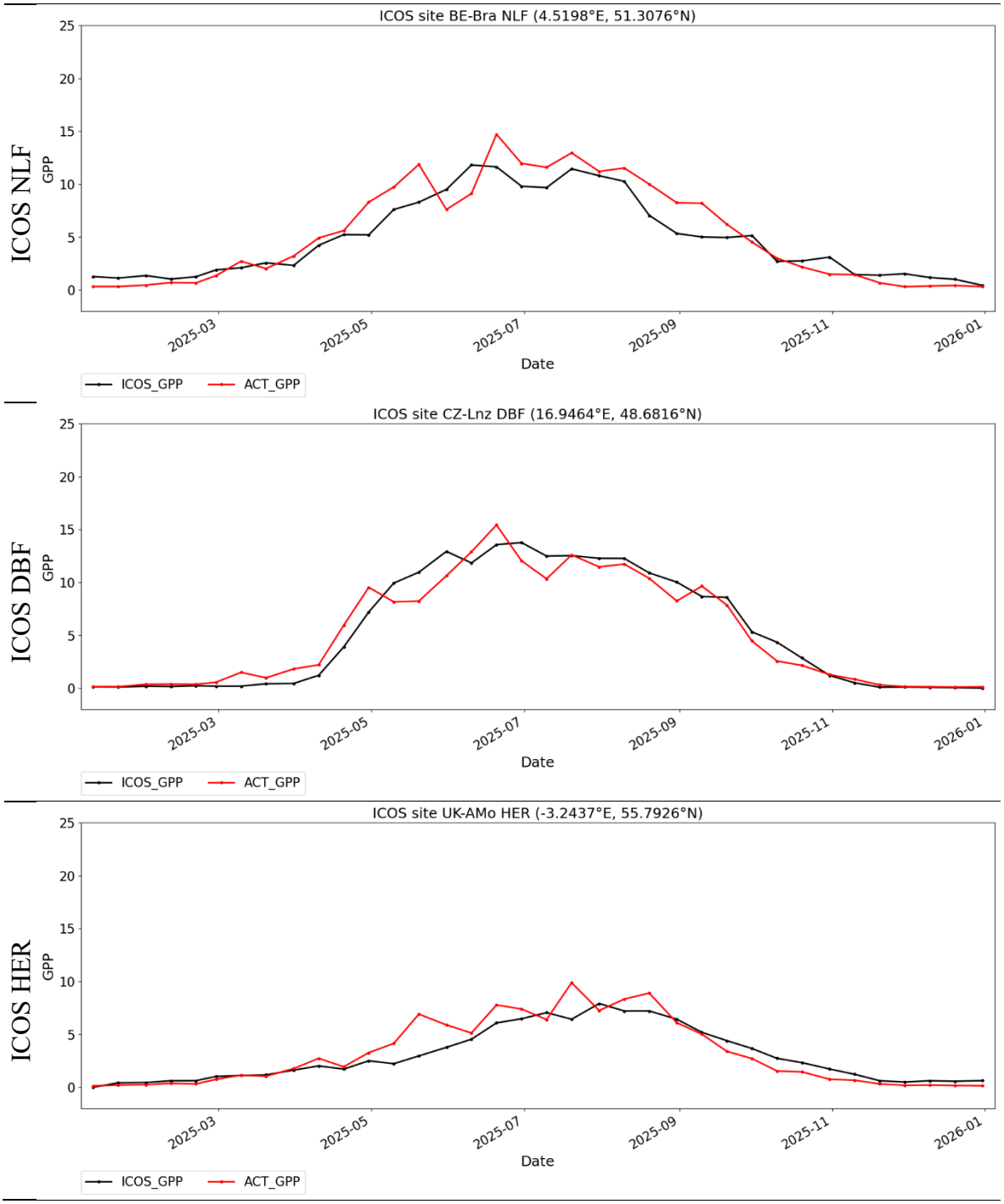


Figure 67: Frequency histogram of the overall bias (*left*) and scatter plot (*right*) between ICOS GPP (*top*) or Ameriflux GPP (*bottom*) and GPP 300m V2 for the year 2025.

Table 21: Statistics of comparison between GPP 300m V2 and FLUXNET GPP over ICOS and Ameriflux sites for the year 2025. The absolute values of the statistics are given in gC/m²/day. The last column contains the percentage of pixels with uncertainty within the goal (10%) and threshold (20%) range.

Dataset	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset / slope	% Goal / Thres.
GPP 300m V2 vs ICOS	986	0.81	0.3 (7%)	0.4 (9%)	2 (53%)	1.2 (29%)	2 (53%)	-0.3 / 0.99	11 / 24
GPP 300m V2 vs Ameriflux	541	0.72	-0.9 (-26%)	-0.4 (-13%)	2 (66%)	1.0 (28%)	2 (71%)	0.2 / 1.2	13 / 24

Figure 68 shows temporal profiles over a selection of ICOS and Ameriflux sites with different biome types for the GPP 300m V2 January – December 2025 dataset and the ICOS/Ameriflux GPP data. There are some differences between the two datasets, but phenological temporal patterns are the same for both. Considering that the in-situ data and the GPP 300m V2 data have a different spatial footprint and are processed with a different method, the GPP values agree fairly well.



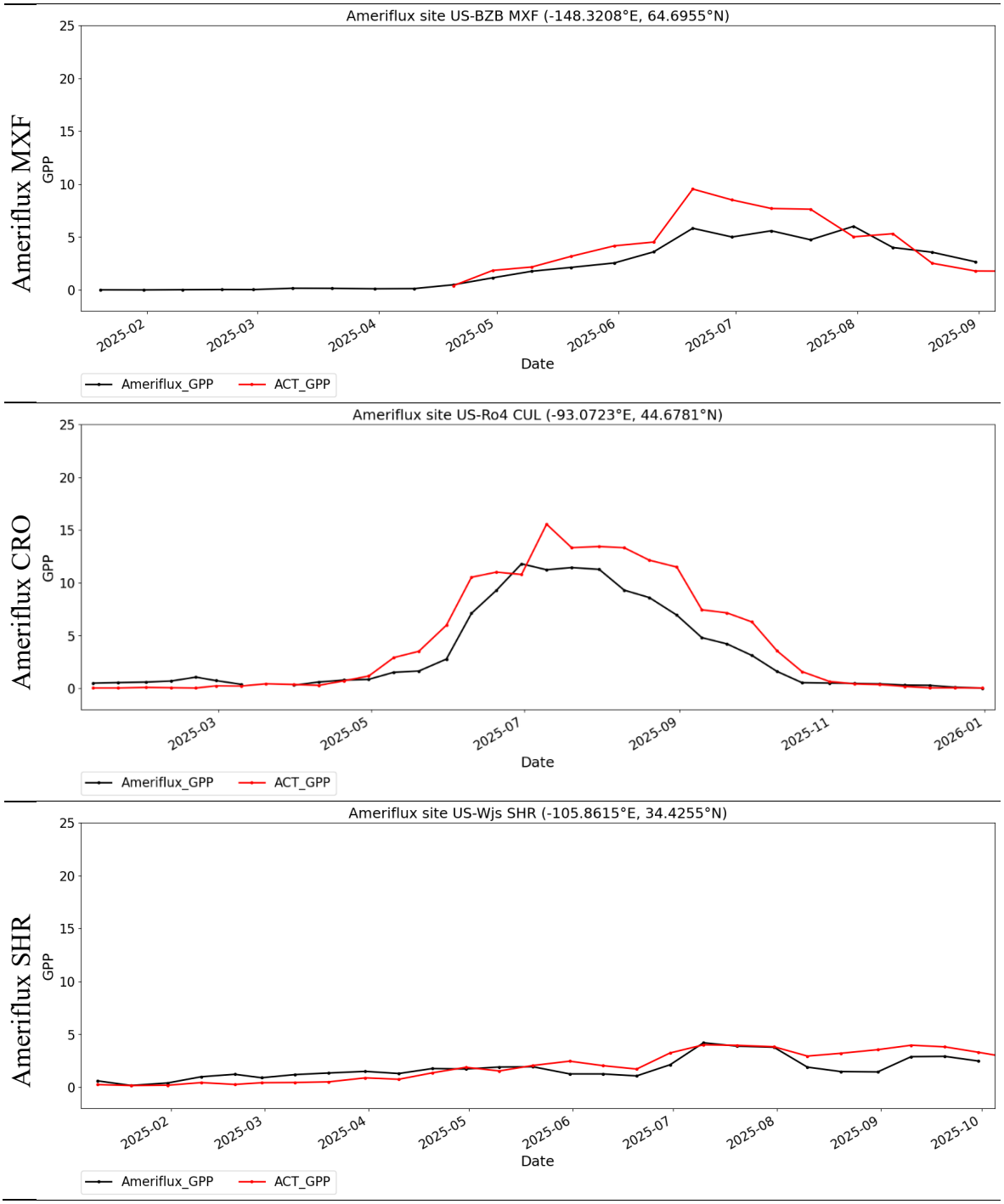


Figure 68: Temporal profiles over a selection of FLUXNET sites of ICOS/Ameriflux GPP (black) and GPP 300m V2 (red).

CONCLUSION

The results of the SQM analysis for the GPP 300m V2 product of January – December 2025 conclude that there is overall good consistency between the 2025 period and the STS. Some differences between both datasets are present, but they are acceptable since different periods are compared for which natural variation in vegetation is expected. The stability of the product is still high with stability metric equal to 0.7%, well within the goal requirement of 10%.